MODELLING OF THE COVID-19 PANDEMIC

SUN YINXIAOHE

(MSc, NUS)

A THESIS SUBMITTED

FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

SAW SWEE HOCK SCHOOL OF PUBLIC HEALTH

NATIONAL UNIVERSITY OF SINGAPORE

2023

Thesis Advisors:

Associate Professor Alex Richard Cook

Examiners:

Assistant Professor Hannah Eleanor Clapham

Assistant Professor Wang Yi

DECLARATION

I hereby declare that this thesis is my original work, and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in the thesis.

This thesis has also not been submitted for any degree in any university previously.



Sun Yinxiaohe

01 January 2023

ACKNOWLEDGEMENT

I am truly indebted to everyone who has helped me and made my PhD journey possible.

First and foremost, I am very grateful to my supervisor for his patience, guidance, support and encouragement. *A/Prof Alex R Cook*, has been a fantastic mentor to me, the best mentor I could have imagined. He generously shared his passion and knowledge, not only in daily research but also in life experience.

I would also like to express my special thanks to my thesis committee members – *Dr Clarence Tam* and *A/Prof Angela Chow*; to my examiners – P*rof Hsu Liyang, Dr. Ong Twee Hee and Dr Yi Siyan*; to my previous workplace, Centre for Infectious Disease Epidemiology and Research (CIDER) and my previous director *Dr Vincent Pang*; and all other professors, researchers, teachers and colleagues who have given me valuable comments and suggestions on the work I have done.

Many thanks to my lovely friends, *Tang Me*i and Dr *Zhao Qitong*, for listening to me and encouraging me when I felt any depression during my PhD journey. Also, I owe a special thanks to my parents for supporting every decision I have made and providing unconditional love; to my cat, *Fu Xing*, for her accompanying me on many sleepiest nights of thesis writing.

Finally, but most importantly, a special thanks to my husband *Peng Yuming*, for his selfless support, patience, and endless understanding during the entire process.

Let there be peace, let peace never cease. — Ahmed Elnawawy

TABLE OF CONTENTS

DECLARATION ..... *i*

ACKNOWLEDGEMENT ..... *i*

TABLE OF CONTENTS ..... *ii*

LIST OF TABLES ..... *vii*

LIST OF FIGURES ..... *viii*

LIST OF ABBREVIATIONS ..... *xi*

LIST OF PUBLICATIONS *0*

CHAPTER 1: INTRODUCTION *1*

1.1 Chronology of the COVID-19 pandemic **1**

1.2 Epidemiology, clinical manifestations, the transmission of COVID-19 **8**

1.3 Public health interventions and vaccination programmes **9**

1.3.1 Public health interventions 9

1.3.2 Vaccination programmes 11

1.3.3 Planning and selection of interventions 12

1.4 Overview of thesis **13**

1.4.1 Knowledge gaps and motivation. 13

CHAPTER 2: ESTIMATE THE PERSONAL PROTECTIVE EQUIPMENT FOR THE COMING COURSE OF A PANDEMIC: A MULTIPLE SCENARIOS SIMULATION STUDY *17*

2.1 Abstract **17**

2.2 Introduction **18**

2.3 Methods **20**

2.3.1 Epidemic Scenarios and model structure 20

2.3.2 N95 respirators Policies 27

2.3.3 Ratio of Confirmed and Suspected but Negative Cases 30

2.4 Results **31**

2.4.1 Epidemiological features in simulated outbreaks under scenarios 31

2.4.2 Final N95 respirators usage comparison according to different scenarios 32

2.4.3 N95 respirators usage with implementation of rapid testing 36

2.5 Discussion **38**

2.6 Motivation for Next Chapter **42**

CHAPTER 3. ESTIMATING THE NUMBER OF COVID-19 CASES BY INFECTION ORIGIN IN SINGAPORE *43*

3.1 Abstract **43**

3.2 Introduction **44**

3.3 Materials and Methods **46**

3.3.1 Data Sources 46

3.3.2 Modelling 47

3.3.3 Model Fitting 52

3.4 Results 56
3.5 Discussion 63
3.6 Motivation for Next Chapter 66

CHAPTER 4. USE OF BLUETOOTH CONTACT TRACING TECHNOLOGY TO MODEL COVID-19 QUARANTINE POLICIES IN HIGH RISK CLOSED POPULATIONS....67

4.1 Abstract 67
4.2 Introduction....68
4.3 Materials and Methods....70
4.4 Results 81
4.5 Discussion 94

CHAPTER 5. CONCLUSION AND FUTURE WORK....98

5.1 Summary of major findings....98
5.2 Public health contributions and implications....100
5.3 Limitations....102
5.4 Future studies....103
5.5 Conclusion 104

Reference....105

SUMMARY

On 11 March 2020, coronavirus disease 2019 (COVID-19) was declared as a pandemic by WHO at 118,319 cases and 4,292 deaths reported globally. As of 31 December 2021， 278 million cases have been reported across the world, and among all reported cases, there have been more than 44,000,000 (16.0% of all reported cases) cases in the WHO's South-east Asia region, 11,000,000 (4% of all reported cases) in the Western Pacific, and 2,177,701 (0.8% of all reported cases) cases in Singapore. COVID-19 posed severe challenges to healthcare systems in the early phase of the pandemic, resulting in a shortage of healthcare resources including personal protective equipment (PPE), intensive care unit (ICU) beds and inpatient beds, which not only left frontline healthcare workers (HCWs) at grave risk, but also brought challenges for policymakers and society as a whole.

Studies showed that N95 respirators (N95s) offer good protection against viral respiratory pathogens, especially during aerosol generating procedures or when a patient's COVID-19 status is positive or unknown. Accordingly, N95s are in high demand for HCWs compared with other PPE categories due to the limited production capacity. To address the shortage of N95s, health care planners suggested the extensive-use or re-use of N95s. A study which screened 27 countries or regions revealed that only 5 countries (19% of 27 countries or regions) allowed extended use; 2 countries (7% of 27 countries or regions) mentioned reuse; and 3 countries (11% of 27 countries or regions) recommended both strategies for rationing N95s. Therefore, soon after of the pandemic, optimizing N95s requirements based on different strategies to maximize the protection of HCWs safety has been an urgent issue.

While timely, stringent, and effective intervention policies, as implemented in many countries in early 2020, was demonstrated to have mitigated deteriorating COVID-19 outbreaks, they were not viable long-term strategies as it was impractical to implement strict *lockdowns* or border control policies for the months or years for mass vaccination occurred. Obviously, COVID-19 infection has setting-specific heterogeneity due to differences in living and/or health conditions, such as the significant differences between general communities and nursing homes, migrant workers, homeless and travellers. In addition, outbreaks in army and prison settings were reported. Hence, it is essential to update responsive interventions promptly according to the type of infection by real-time forecasting of COVID-19. Such forecasting can inform the need to escalate or reduce the degree of interventions, which has proven to be one of the approaches taken by the Singapore government to avoid rapid resurgence following lockdowns.

Singapore began lockdown measures for the COVID-19 outbreak on 6 April 2020, enforcing social distancing and widespread closure of schools, workplaces, and commercial buildings. Exiting lockdown on 1 June 2020, two significant heterogeneous disease states emerged in Singapore: a low endemic community state and a high dormitory outbreak state (over 95% of cases belong to the latter). Consequently, policy responses differed between sites. In dormitory settings, operators were expected to limit inter-mixing of workers from different blocks or sites, performed movement tracking and regular 14-day swabbing of uninfected workers. We seek to understand the spread between each dormitory to improve the interventions strategies to minimise the risk of new disease clusters emerging.

This thesis aims to use statistical and mathematical models to predict and evaluate the COVID-19 pandemic in a timely manner, and provide reliable evidence for public health officials, health care providers and individuals to better prepare and respond to the disease.

Specifically, Chapter 1 provides an overview of the development of COVID-19 pandemic, including the chronological spread events and epidemiological characteristics of COVID-19, the implementation of public health control measures, the development of the COVID-19 vaccine, and the importance of modelling the spread of the pandemic and modelling control measures to evaluate their effect on the spread. Chapter 2 estimates the demands of N95s required for the healthcare system in the early outbreak of the pandemic, among three epidemic levels under different strategies, including single-use, extend-use, and prolong-use policies, to run with the maximized possible protection for HCWs. Chapter 3 proposes to develop a system to estimate COVID-19 cases of different infection types in Singapore and forecasts of ICU needs evolving from all infections while multiple epidemics were concurrently ongoing. Chapter 4 explores the contact network in dormitories in the period once transmission levels had fallen, and further develops a simulation model to evaluate multiple control measures via the empirical contact patterns.

LIST OF TABLES

Table 1. Parameters used across different scenarios.....26
Table 2. Summary of the average demand of N95s per patient per day in the ICU where 39 sets are required under the extended-use policy and 6 sets under the prolonged-use policy.....30
Table 3. Average number of N95s being used in the isolation wards. 30
Table 4. Total N95s consumption (millions) for different scenarios.....33
Table 5. N95s consumption (millions) in the peak month for different scenarios.....35
Table 6. Overview of dataset 74
Table 7. Epidemiological parameters 77
Table 8. Scenarios settings in our simulation studies. 80
Table 9. Characteristics of the dormitory residents within Dorm A and Dorm C dormitories. 81
Table 10. Summary of regular and transient contacts in a day.....86

LIST OF FIGURES

Figure 1. Flow chart of the model structure.....21
Figure 2. Exponential distribution versus the observed data presentation. 25
Figure 3 Schematic of N95s utilisation under three protocols in ICU (A and B represent 2 patients, separately). Schematic of N95s utilisation under three protocols in ICU (A and B represent 2 patients, separately). Each symbol represents an interaction with a patient by a healthcare worker. Closed symbols represent new N95s being donned before that interaction with the patient; open symbols represent extensive use of the N95s for that interaction.....28
Figure 4. Schematic of N95 utilisation under three protocols in respiratory isolation wards. Each symbol represents an interaction with a patient by a healthcare worker. Closed symbols represent new N95s being donned before that interaction with the patient; open symbols represent extensive use of the N95s for that interaction. 29
Figure 5. The epidemic curves in different scenarios (A) Hospitalized cases (B) ICU admitted cases. The colours indicate the scenarios. Black: Severe scenario, orange: moderate scenario, and green: mild scenario. 32
Figure 6. The cumulative N95s consumption for selected ratios of confirmed cases versus suspected cases. The simulated scenarios in plots A-C are severe, D-F moderate, and G-I mild. The ratios of confirmed cases versus suspected cases for: A, D and G is 1:0; B, E and F is 1:2; C, F and I is 1:10. The line colours describe policies of N95s usage where the red line shows single use, blue line shows extended use, and green line shows prolonged use.....34
Figure 7. Cumulative N95s consumption when considering confirmed cases versus suspected cases as 2:1, 1:1 and 1:5. Plots A-C reflect the severe epidemic scenario, D-F the moderate scenario, and G-I the mild scenario. The policies for N95s usage are represented in colour (red for single use, blue for extended use, and green for prolonged use).....37
Figure 8. N95s use across time until the switch to rapid testing in different epidemic scenarios where (A) is severe, (B) is moderate and (C) is mild. The orange line series show single use, blue the extended-use and green the prolonged-use policies. Each bar is stratified by the levels of health care setting. Darker colours describe ICU cases, medium shading the cases in regular beds and lighter shades the suspected COVID -19 cases.....38
Figure 9. Model structure. Cases originating from the general community, importation and foreign worker dormitories transfer to ICU with risk rC, ri, and rFW, respectively. Google mobility data and daily situation reported cases were used to estimate and project the general community cases; daily situation reported cases was used for estimation and projections of imported cases and foreign worker cases. And SEIR model was implemented to assess the epidemic curve in foreign worker subgroup.....48

Figure 10. Estimations and projections and projections of 6 types of mobility changes from baseline. The grey polygons represent the 95% CI in the estimation period and the pink polygons are the equivalent for the projection period. The black points-line represent the observed google mobility changes. 54
Figure 11. Description of COVID-19 cases in Singapore as of November 13. A) imported and general community cases; B) foreign workers cases. Note the difference in scales; C) the overall trend of google mobility changes from 15 February 2020 to 31 December 2020. ...57
Figure 12. Evaluation of projections with 30-day rolling windows. The panels show A) general community cases, B) imported cases and C) foreign worker cases. Panel D) shows the active number of ICU cases. In each panel, the redlines represent model-based projections, and the red polygons represent corresponding 95% credible intervals (95% CI). Each segment of the projected data in red polygons represents the estimates from one 30-day forecast made at a previous point in time. 60
Figure 13. Quantification of projection error. The projection accuracy across time by aggregation of daily projection models. 60
Figure 14. 90-day hindcast from 31 December 2020. Daily number of cases by the origin of infection where the panels show A) general community cases, B) imported cases, and C) foreign worker cases. Panel D) shows the active number of ICU cases. The grey polygons represent the 95% CI in the estimation period and the red polygons are the equivalent for the projection period. The points represent the observed case data.....61
Figure 15. 90-day hindcast of cumulative cases from 31 December 2020. The panels show total number of A) The total number of community cases would be increased but should be under control within first 90-days in 2021, according to current projection of mobility changes. B) the number of imported cases would be increased to 1,850 in total under border control policies of December 2020, and C) the total number of foreign worker cases would be stabilised at ~54,300 cases in the first three months of 2021. The grey polygons represent the 95% CI in the estimation period and the red polygons are the equivalent for the projection period. The points represent the observed case data. 62
Figure 16. Definition of regular and transient contacts 73
Figure 17. Model Structure. Here, G and $\mathcal{G}$ represent the collected contact network, and the simulated contact network, respectively. With the simulated contact network, we used an SIR-style branching process model to simulate daily cases (Λt) from day 1 to day 120 under different infection rates ($pinfect$), since the first wave of the outbreak in Singapore dormitories was about 4 months. We then looked for infection rates by individually matching cumulative cases simulated for different infection rates with reported prevalence. Thereafter, we simulated the epidemic scale of multiple intervention policies under these two scenarios, including a smaller outbreak (matching what was observed in Dorm A) and a larger outbreak (Dorm C).....75

Figure 18. Transmission scenarios from infected i.....76
Figure 19. Demographic characterises comparison by different contact types. (a) Contact types in age groups. (b) Contact types in the original countries. The red colour indicated regular contact, and the blue colour indicated transient contact. 83
Figure 20. Overview of interaction frequency of regular contacts and transient contacts. The red colour indicated regular contact, and the blue colour indicated transient contact. 84
Figure 21. a) The average number of interactions that occurred per hour across the day of the week; b) the average number of interactions that occurred per hour in a day across regular and transient events. 85
Figure 22 Interaction duration between each pair of MWs within a day. We found that the mean interaction duration of regular contacts was longer than transient contacts. We did however observe that there were some long interaction events between transient contacts, which occurred when MWs had multiple non-consecutive transient contacts across certain days. 86
Figure 23. Simulated contact network. Regular (a) and Transient contacts (b) for Dorm A (dark orange) and Dorm C (steel blue).....87
Figure 24. (a) The overall proportions of infected MWs at different infection probabilities within four months. The red solid line represents the median infection rate in Dorms A and C based on 33% and 66% being infected, and the red dashed line is the 95CIs; (b) The overall proportions of infected at different infection probabilities within six months. The red solid line represents the median infection rate in Dorm A and Dorm C based on 33% and 66% being infected, and the red dashed line is the 95CIs. Orange and blue lines indicate Dorm A and Dorm C respectively. 88
Figure 25. Simulation of epidemic size under multiple intervention scenarios. (a) and (d) Scenario S1: Isolate cases with moderate delay (i.e., 3 days); (b) and (e) Scenario S2: Isolate cases and quarantine all contacts with reasonable delay; (c) and (f) Scenario S3: Isolate cases and quarantine regular contacts with a moderate delay. 90
Figure 26. Simulation of epidemic size under multiple NPIs under smaller and larger outbreaks (i.e., with θ of 0.4 and 0.87). (a) and (f) Scenario 1: Immediate isolation of cases in the baseline scenario; (b) and (g) Scenario 2: Immediate isolation of cases and quarantining for all contacts; (c) and (h) Scenario 3: Immediate isolation and quarantining of regular contacts; (d) and (i) Scenario 4: No isolation of cases or quarantining of contacts; (e) and (j) Scenario 5: baseline scenario under a 30% reduction in contacts. Daily cases of the epidemic can be found in Figure 27. 92
Figure 27. Simulation of epidemic under multiple NPIs under smaller and larger outbreaks (i.e., with θ of 0.4 and 0.87). (a) and (f) Scenario 1: Immediate isolation of cases in the baseline scenario; (b) and (g) Scenario 2: Immediate isolation of cases and quarantining for all contacts; (c) and (h) Scenario 3: Immediate isolation and quarantining of regular contacts; (d) and (i) Scenario 4: No isolation of cases or quarantining of contacts; (e) and (j) Scenario 5: baseline scenario under a 30% reduction in contacts 93

LIST OF ABBREVIATIONS

ART tests	Antigen-based rapid diagnostic tests
BP	*BluePass* device
COVID-19	Coronavirus disease
CDC	Centre for Disease Control and Prevention
CI	Credible Interval
CMP	Construction, marine shipyard, and processing industries
CMR	Community Mobility Reports
COVAX	COVID-19 vaccines global access
DCT	Digital contact tracing
DZIF	German Centre for Infection Research
FDA	Food and Drug Administration
HCWs	Healthcare workers
ICU	Intensive care unit
MCMC	Markov Chain Monte Carlo
MERS-CoV	Middle East respiratory syndrome coronavirus
MHRA	Medicines and Healthcare products Regulatory Agency
MWs	Migrant workers
NPIs	Non-pharmaceutical interventions
NUH	National University Hospital
N95s	N95 respirators
PCR	Polymerase chain reaction
PPE	Personal protective equipment
RRT	Rostered Routine Testing
SARS-CoV	Severe acute respiratory syndrome
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
SEIR	Susceptible-Exposed-Infectious-Recovered model

SIR	Susceptible-Infectious-Recovery model
WHO	World Health Organisation

LIST OF PUBLICATIONS

- Yinxiaohe Sun, Hirono Otomaru, Sharon Esi Duoduwa Quaye, Jyoti Somani, Natasha Bagdasarian, Darius LL Beh, Dale A Fisher, Alex R Cook, Borame L Dickens. Scenarios to Manage the Demand for N95 Respirators for Healthcare Workers During the COVID-19 Pandemic. Risk Management and Healthcare Policy. Volume 13:2489-2496

CHAPTER 1: INTRODUCTION

1.1 Chronology of the COVID-19 pandemic

Coronavirus Disease (COVID-19), caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) (1–3), was one of the three global coronavirus outbreaks in the world over the past two decades (4). The other two were severe acute respiratory syndrome (SARS-CoV) and Middle East respiratory syndrome coronavirus (MERS-CoV) in 2003 and 2012, with mortality rates of 10% and 37%, respectively (5–9). Although SARS-CoV-2 has a lower mortality rate (10), it is overwhelmingly more contagious and can spread rapidly between individuals and countries because of its low severity. As of this writing, the COVID-19 outbreak has been ongoing.

DECEMBER 2019 — The first record of pneumonia-like symptoms for COVID-19 was on 1 December 2019, in Wuhan, the capital of Hubei Province, China (11), while some phylogenetic analyses showed that the SARS-CoV-2 has first arisen in October or November 2019 (12–14). Subsequently, some sources quoted more patients presenting the same symptoms on December 8, the first day of the COVID-19 outbreak (15). On December 27, the Hubei Provincial Hospital of Integrated Chinese & Western Medicine reported three new coronavirus cases causing abnormal pneumonia to the China Centre for Disease Control and Prevention (CDC) (16). Confirmed with an investigation, China CDC officially reported to the World Health Organisation (WHO) that there was a cluster of 27 pneumonia-like cases with an unclear source of viral transmission, of which seven were in critical condition on 31 December 2019 (15).

JANUARY 2020 — The outbreak in China has started. On January 8, SARS-CoV-2 was officially announced as the causative pathogen of the outbreak by WHO (17,18). The day after, the first death (a 61-year older man with severe underlying conditions) among 41 confirmed cases was reported in China (19). WHO shared preliminary results of whole-genome sequencing with the public on January 12 (20). The results show that the novel coronavirus has similarities with the SARS-CoV. Since then, cases have been detected nationwide in China, starting from Shenzhen and Guangdong (21,22). Cases reported in Shenzhen were in the same household, indicating human-to-human transmission of SARS-CoV-2 (21).

A large-scale outbreak was observed in Wuhan, and the virus continued to spread for two main reasons. One is that the novel coronavirus hit during the Chinese New Year, an annual reunion that culminates in the peak traffic movement between cities throughout China. Another one is that limited knowledge of SARS-CoV-2 prevented the rapid identification of cases (23–25). After that, the Beijing government isolated Wuhan, the epicentre. The lockdown policy was first proved to be a successful and effective intervention in controlling the virus's spread in the early pandemic outbreak (26,27).

Transmission of SARS-CoV-2 outside China began in early January 2020, starting in Thailand, a 61-year-old Chinese traveller from Wuhan reported on January 8 (28–30). A week later, on January 16, Japan confirmed its first case (31). In the second half of January, South Korea (32), Singapore (33), Vietnam (34), Nepal (35), Malaysia (36), Cambodia (37), Sri Lanka (38), India (39) and the Philippines (40) reported their first cases in succession. On January 21, the first case was identified in Washington, US (41), indicating the beginning of

COVID-19 outside Asia. Following that, France reported its first cases on January 24 showing that SARS-CoV-2 had arrived in the European region (42).

With the widespread of the virus, on January 30, WHO declared COVID-19 a public health emergency of international concern (43). As of 31 January 2020, there were 9,826 cases reported worldwide, including 9782 cases in the Western Pacific, 17 in South-East Asia, 9 in the Region of the Americas, 14 in the European Region, and 4 in the Eastern Mediterranean (44).

FEBRUARY 2020 — COVID-19 continues to spread globally. On February 11, the new coronavirus disease was officially named COVID-19 by the WHO. In the same month, the first death case outside China was reported in the Philippines on February 2 (45). By the end of February, 37 countries reported their first COVID-19 cases, including the first case in Egypt, the African continent, on February 14 (46), bringing the spread of COVID-19 to a total of 62 countries or territories with 85,403 reported cases and 2,924 deaths (45). Among those, China, South Korea, Iran and Italy heavily experienced their first epidemic wave.

MARCH 2020 — The spread of COVID-19 has severely affected the world, especially the European Region. On 11 March 2020, WHO declared that COVID-19 is a pandemic (47). This month, 148 countries were first infected by SARS-CoV-2, with the reported cumulative cases and death as 750,890 and 36,405, respectively (48). Europe accounted for 63.5% of all cases reported this month, showing severe suffering from the COVID-19 pandemic, while the initial epicentre—China—represented a sharp decline in the number of new cases, with zero new cases in the general community on March 30 (49).

The pandemic has posed severe challenges to healthcare systems, resulting in a shortage of healthcare resources. At the end of February, WHO announced that there was a global shortage of personal protective equipment (PPE), which indeed leaves frontline healthcare workers (HCWs) at grave risk (50). Unfortunately, the resource shortage continued within the next half month. On March 3, WHO warned of a shortage of PPE and a lack of testing capacity. Thus strategy recommendations were elevated accordingly (50,51).

APRIL to DECEMBER 2020 — A study showed that pre-symptomatic and asymptomatic cases could transmit SARS-CoV-2 (52). Therefore, implementing social measures and developing a COVID-19 vaccine are crucial to prevent the spread of COVID-19. On April 16, WHO published guidelines for adjustments to public health and social measures, including mass movement restrictions, commonly referred to as "lockdowns" (53). In May, the WHO adjusted the guidelines of contagious measurements by varying degrees of restrictions among different transmission scenarios in heterogeneous settings. For instance, intensive measures have been implemented in dense places or vulnerable populations such as workplaces, schools and nursing homes, and mass gatherings have been restricted (54).

Despite implementing public health and social measures, the transmission of SARS-CoV-2 continued. A total of 15 countries reported their first cases of COVID-19 from April to June (13 countries in April and 2 in May). As of June 30, the reported cases exceeded 10 million worldwide (55). At the end of August, this number reached around 25 million, with 800,000 reported death (56). Moreover, those numbers reached 33 million and 1 million at the end of September (57). Meanwhile, a survey showed that healthcare systems were

disrupted in almost every affected country, with low- and middle-income countries reporting the most significant difficulties (58). As of October, over 42 million cases and 1.1 million deaths have been reported worldwide (59). Furthermore, 3 new countries and territories were affected by SARS-CoV-2. Subsequently, the cumulative number of globally reported cases reached ~50 million in November, and the European Region accounted for the most significant proportion of new cases and deaths (60).

In December, there were three new variants of SARS-CoV-2, designated as Alpha (B.1.1.7 and Q lineages), Beta (B.1.351 and descendent lineages), and Gamma (P.1 and descendent lineages) spread widely (61). They were first detected in the United Kingdom, South Africa, and Brazil, respectively (62–64). With new variants spreading more extremely and existing vaccines offering less protection against infection, the total number of reported COVID-19 cases had surpassed 79.2 million with over 1.7 million deaths (65), and SARS-CoV-2 had not infected only 17 countries and territories worldwide by the end of 2020.

Scientists have pledged to work with the WHO to speed up the development of a vaccine against COVID-19. Brief chronologies of vaccine development can be found in Section 1.3.2. In addition, the treatment of COVID-19 has been developed since April. At the end of May, WHO announced the clinical management of COVID-19 to standardize the care and management of COVID-19 cases (66). In June, there were several improvements in health technologies for COVID-19 patients. For example, dexamethasone has been used to rescue severe cases (67); and hydroxychloroquine was discontinued for the non-reduction of mortality for hospitalised COVID-19 patients (68). Furthermore, the second summit on COVID-19 research and innovation held by WHO on

July 1 agreed that more trials testing antivirals, immunomodulatory drugs, anti-thrombotic agents, and combination therapies at different stages of the disease were necessary. Moreover, in September, antigen-based rapid diagnostic tests (ART tests) were suggested by WHO in the community, which can identify cases rapidly and alleviate the burden of testing laboratories (69,70).

JANUARY to DECEMBER 2021 — The implementation of the COVID-19 vaccine program has started worldwide. In January 2021, the recommendation of using the Moderna COVID-19 vaccine was issued by WHO (71), and 40 million doses of the Pfizer–BioNTech vaccine were ordered by COVAX (72). From February, the COVID-19 vaccine was distributed globally (73). As a result, half of WHO's Member States will have vaccinated 40% of their population by the end of 2021 (74). Another study showed that 9.25 billion COVID-19 vaccines were administered in 2021. However, most doses (76–78%) were distributed in high- and upper-middle-income countries (75). Also, the vaccination data worldwide was tracked and published on the WHO COVID-19 dashboard from March (1).

By January 25, 70 countries and 31 territories across six WHO regions reported SARS-CoV-2 with the three new variants — Alpha, Beta and Gamma. Peaks of those new variants appeared in January; however, there was no indication of the disease severity caused by them (76–78). As of February 23, there have been 110.7 million cumulative cases and more than 2.4 million deaths, with three main variants circulating simultaneously (79). In March, the cumulative of reported cases increased by 14%, reaching over 126 million with more than 2.7 million deaths (80).

In April 2021, a new variant labelled Delta swept India leading to more than 400,000 daily reported cases (61). In addition, it has been shown to have faster viral replication and greater infectiousness (81–84). Subsequently, the Delta variant sparked a new round of COVID-19 waves in over 170 countries in the remaining year (81,85), including the severe waves observed in the UK (86), the US (87,88), and countries in Africa, Europe and Asia (89–92).

A new variant named Omicron was first reported in South Africa in November 2021 and has since rapidly spread globally (61,93). It has been proven more transmissible but less severe than the Delta variant (94,95). Multiple variants brought the cumulative number of cases to 278 million by the end of 2021(96), and the world was panicking again.

JANUARY 2022 onwards — Several studies have shown that unvaccinated or two doses of COVID-19 vaccine are insufficient to prevent new variants from being infected (i.e., Omicron and Delta variant), so booster vaccine doses are highly recommended (97–100). By mid-August 2022, 29.5 per 100 people had received a booster worldwide (101).

Global vaccination rates are on the rise, and Omicron — a variant that is less severe than other variants, has started to become the dominant variant worldwide (102). The US and other countries have gradually lifted COVID-19 social restrictions since February 2022. For example, compulsory mask-wearing when outside was not required in the US and UK from February (103,104). In addition, from June 2022, rescinding the COVID-19 test requirement for international travel (105,106).

Furthermore, more and more countries removed COVID-19 rules, including China. From December 2022, China has lifted strict intervention policies, including lockdown and mandatory facilities isolation of cases. On December 23, the Healthy Code system was closed. And from January 2023, quarantine for inbound travellers has been removed (107).

1.2 Epidemiology, clinical manifestations, the transmission of COVID-19

Epidemiological characteristics of COVID-19 have been shifting over time with the mutation of virus variants. At the beginning of the transmission, older men (with co-morbidities) are more likely to develop severe symptoms or even die (108). However, the age differentiation has weakened since the Delta variant attack (109). Additionally, initial racial and ethnic disparities in COVID-19 infection rate have been reduced from Asian, Black and Hispanic ethnic groups, although individuals of black and Asian ethnicity are at relative higher risk of COVID-19 infection compared to Hispanic ethnic groups (110).

The median incubation period for SARS-CoV-2 is shorter for the Delta and Omicron variants than for the Alpha (83,111,112). Over time, the generation time of SARS-CoV-2 also gets shorter, indicating a faster transmission (83,113). However, the mortality rate is different, rising first with the predominance of the Delta variant in 2021, then decreasing with the Omicron variant ascendancy in 2022 (114–116). Fever, muscle aches, anosmia, dysgeusia, anorexia, nausea, vomiting, diarrhoea, malaise/fatigue, and reported headaches are the main symptoms of all variants of COVID-19, while shortness of breath is not common when the Omicron predominance (117–122).

The transmission of COVID-19 varies in various stages of the pandemic. The basic reproduction number (R_0) in the early outbreak can be up to 6.5 (123,124), where R_0 represents the basic reproduction rate that measures the number of secondary infections generated from a typical case. With strict nonpharmaceutical intervention policies, a lower attack rate and effective reproduction number than those in the early outbreak were observed in previous studies (125,126). However, with the new variants (Delta and Omicron) overwhelming the world, the effective reproduction number increased again (127–130).

Studies also found that the transmission of COVID-19 differs in various settings. The secondary attack rate of household transmission can be up to 32.4% (131,132). A systematic review found that household adults are more likely to be infected by their infected children (133). In schools, a lower risk of COVID-19 among children who attend day-care/preschools and primary schools was observed, while high schoolers showed a higher risk than adults in general communities (134). Semi-closed and enclosed environments, such as migrant worker dormitories, refugee camps, and cruise ships, have higher rates of infections (135–138). Additionally, HCWs are at grave risk of COVID-19, even if vaccinated, when new variants dominate (139,140).

1.3 Public health interventions and vaccination programmes

1.3.1 Public health interventions

COVID-19 is a highly contagious disease, first validated in an enclosed environment– the Diamond Princess cruise ship off Japan, with an initial reproduction number R_0 estimated above 11 (141–143). Lessons learnt from the

previous pandemic–public health measures can effectively mitigate the spread of infectious diseases (144–146). Therefore, individual- and community-level interventions have been implemented to contain COVID-19 transmissions.

First, wearing a face mask shows strong protection for individuals from COVID-19 infection (147,148), and the WHO has recommended limiting the spread of the virus. Research evidence indicated that using a face mask could significantly reduce the infection risk (148). Keeping good personal hygiene, such as sanitisation of hands and protection of eyes in public and health-care settings from containment, effectively decreases the transmission of infectious diseases (148,149). A study has shown that respiratory tract infections can be reduced by 6%–44% in various settings via hand hygiene intervention (149).

In addition, community-level interventions include contact tracing and quarantine of close contacts, school and workplace closures, reduction of mass gatherings, border temperature screening, travel restrictions (including quarantine of importation, health status declaim and Polymerase Chain Reaction [PCR] test) and surveillance. They have been proven effective in limiting the spread of SARS-CoV-2 and controlling the COVID-19 pandemic when COVID-19 vaccines had been developing in the early period of the pandemic (141,143,150,151). Such as, from 4 February to 19 February 2020, 619 of 3700 passengers and crew (17%) confirmed cases were reported; however, the infection rate would reach 79% if there had been no timely implementation of non-pharmaceutical interventions (143). Thus, policymakers implement those in mitigating COVID-19 transmissions (150,152).

1.3.2 Vaccination programmes

Since April 2020, clinical trials of the COVID-19 vaccine have been in progress. On June 24, China approved the CanSino vaccine for emergency use in the military and high-risk groups (153). Moderna and Pfizer, as the leaders in the race of vaccine development, provided promising results regarding vaccine effectiveness in early-phase clinical trials and on non-human primate models in July (154,155). From August, Phase III trials of Moderna and Pfizer immediately began, and both showed encouraging efficacy rates of vaccines (154,155). As of December 2020, 52 candidate vaccines were in human trials. Among the candidates, the vaccines developed by Pfizer-BioNTech (i.e., BNT162b2 mRNA COVID-19 vaccine) (156,157) and the Moderna (i.e., mRNA-1273 COVID-19 vaccine) (158) were given emergency use authorisation by the FDA in December 2020.

Since July 2020, the COVID-19 vaccines global access (COVAX) facility (159) has accepted more than 150 countries' applications to guarantee their rapid, fair and equitable access to COVID-19 vaccines (160). As a result, as of February 2021, more than 250 million people worldwide have been vaccinated against COVID-19 (161). Furthermore, as of September 2022, 67.9% of the world population has received at least one dose of a COVID-19 vaccine (10).

In addition, research evidence suggested that a two-dose regimen of BNT162b2 provided 95% protection in people 16 years of age or older from diagnosis of infection, and mRNA-1273 conferred 94.1% protection against being diagnosed as COVID-19 in the population of 18 years old or above, respectively (154).

While the two-dose vaccine was less effective in protecting people from Delta and Omicron variants (162), a booster dose showed efficacy (98,100).

1.3.3 Planning and selection of interventions

Effective interventions to the pandemic should be practical, responding to the features of the different variants. They should not cause a worse impact in terms of a social and economic burden than the pandemic itself. Thus, selecting suitable interventions for the pandemic is challenging for policymakers and public health professionals. For example, extreme non-pharmaceutical interventions (NPIs), such as lockdowns, were effective until vaccines were rolled out. Continuing strict NPIs when promoting vaccination globally could pose social and economic burdens (163–165).

The transmission dynamics, the severity and the spread of infection are essential when determining the appropriate intervention strategies. For example, extreme and comprehensive NPIs should be implemented for an early outbreak with higher morbidity/mortality and attack rates and when lacking vaccines and essential healthcare resources, such as the pandemic caused by the SARS-CoV-2 virus (excluding Omicron variants). Furthermore, when these factors are unclear, an effective surveillance system combined with appropriate statistical models to rapidly assess and understand critical features is critical in selecting recommended interventions.

Additionally, heterogeneous infection in different settings is crucial to investigate when effectively tailoring intervention strategies. For instance, in a closed or semi-closed environment with close contact, such as migrant worker dormitories, schools, jails, etc., which has been proven to be a high-risk setting for spreading infectious

diseases, such as COVID-19, comprehensive NPIs are necessary to curb the spread of the pandemic (166,167).

1.4 Overview of thesis

1.4.1 Knowledge gaps and motivation.

As presented in chapter 1.1, before the vaccination programme has been implemented globally, the world experienced the COVID-19 pandemic from an unknown novel coronavirus and lack of healthcare resources to panic with high mortality and morbidity rates.

In the early outbreak of COVID-19, the cases were climbing aggressively. In addition, the lack of healthcare resources made society havoc and pushed HCWs at grave risk, causing a considerable challenge to societies, especially healthcare societies. Meanwhile, the WHO announced that PPE was in shortage, and then the WHO recommended re-use or extended-use policies (168). Moreover, studies showed that N95 respirators (N95s) offer good protection against viral respiratory pathogens, especially during aerosol-generating procedures or when a patient's COVID-19 status is positive or unknown. Accordingly, N95s are in high demand for HCWs compared with other PPE categories due to the limited production capacity. To address the shortage of N95s, healthcare planners have suggested the extensive-use or re-use of N95s (171). A study which screened 27 countries or regions revealed that only five countries (19%) allowed extended use, two countries (7%) mentioned reuse, and three countries (11%) recommended both strategies for rationing N95s (172). Therefore, from the beginning stages of the pandemic, optimising N95 requirements based on

different strategies to maximise the protection of HCWs safety has been an urgent issue.

While timely, stringent and effective intervention policies, such as strict lockdowns or strict border control rules, have been demonstrated to have mitigated deteriorating COVID-19 outbreaks in the early outbreak of COVID-19, they are not viable long-term strategies (173,174). Obviously, COVID-19 infection has setting-specific heterogeneity due to differences in living and/or health conditions (175,176), such as the significant differences between general communities and nursing homes (177–179), migrant workers (MWs) (136,180,181), homeless (182,183) and travellers. In addition, outbreaks in the army (184) and prison (185–187) settings have been reported. Hence, it is essential to update responsive interventions in a timely manner according to the type of infection by real-time forecasting of COVID-19. Furthermore, such forecasting can inform the need to escalate or reduce the degree of interventions, which has proven to be one of the approaches taken by the Singapore government to avoid rapid resurgence following the lockdown.

Vaccines have been proven efficient to COVID-19 in preventing people from severe disease, as shown in Section 1.3.2. However, the cases climb in closed/semi-closed, high-risk, density accommodation settings, such as migrant worker dormitories, jails, schools, etc. In addition, in the event of a viral respiratory pandemic due to new viral pathogens, the world may require a prompt response to curb the transmission, especially in high-risk and density environments to control rapid virus transmission. Therefore, investigation of the impact of contact patterns on virus spread in closed/semi-closed, high-risk,

density environmental settings is critical for better preparation for future pandemics.

1.4.2 Aim and objectives

This PhD thesis aims to use statistical and mathematical models to prepare promptly, understand and evaluate the COVID-19 pandemic and provide reliable evidence for public health officials, healthcare providers and individuals to prepare better and respond to the disease. In addition, this proposed thesis fills current knowledge gaps and assists in evaluating and monitoring the pandemic trends to guide effective non-pharmaceutical public health intervention policies.

The specific aims of the following three chapters are:

1. Chapter 2: To assist planning efforts requiring estimations of respirator demand for the healthcare system to continue operating safely in the coming months during the early outbreak of the COVID-19 pandemic, we estimated N95 respirator demand based on simulated epidemics with multiple proposed N95 using rules. This chapter shows Singapore as an example country, published in *Risk Management Healthcare Policy*, 5 November 2020 Volume 2020:13 Pages 2489 —2496.
2. Chapter 3: To create a more tailored output for use by policymakers as part of the broader COVID-19 surveillance system effort before the vaccination programme is implemented widely, we projected epidemic trends within the general community, migrant worker infections and imported cases separately, as well as projection of active ICU cases. We then developed a weekly updated R Shiny platform to present our model

predictions to policymakers to monitor local pandemic trends and update their policies in a timely manner.

3. Chapter 4: To assess the impact of measures to reduce social contact of cases and contacts in MWs' dormitories, we developed an individual-based model using data collected through a Bluetooth wearable device, two dormitories in Singapore, including one apartment-style and the other a barrack-style were taken as an example. This chapter is in submission.

The above chapters are conducted at different stages of the pandemic—before widespread community transmission had occurred (1), while multiple epidemics were concurrently ongoing but before vaccination programmes were implemented widely (2), and in the period once transmission levels had fallen (3)—and thus make use of different degrees of information.

CHAPTER 2: ESTIMATE THE PERSONAL PROTECTIVE EQUIPMENT FOR THE COMING COURSE OF A PANDEMIC: A MULTIPLE SCENARIOS SIMULATION STUDY

This work has been published as: Yinxiaohe Sun, Hirono Otomaru, Sharon Esi Duoduwa Quaye, Jyoti Somani, Natasha Bagdasarian, Darius LL Beh, Dale A Fisher, Alex R Cook, Borame L Dickens. Scenarios to Manage the Demand for N95 Respirators for Healthcare Workers During the COVID-19 Pandemic. Risk Management and Healthcare Policy. Volume 13:2489-2496

* Risk Management and Healthcare Policy 2020 13 2489-2496 was originally published by, adapted and used with permission from Dove Medical Press Ltd. The copyright remains with the authors and Dove Medical Press Limited and no transfer of copyright should be inferred or implied by its reproduction in this thesis.

2.1 Abstract

Background

By estimating N95 respirator demand based on simulated epidemics, we aim to assist planning efforts requiring estimations of respirator demand for the healthcare system to continue operating safely in the coming months.

Methods

We assess respiratory needs over the course of mild, moderate and severe epidemic scenarios within Singapore as a case study using a transmission dynamic model. The number of respirators required within the respiratory isolation wards and intensive care units was estimated over the course of the epidemic. We also considered single-use, extended-use and prolonged-use strategies for N95s for use by HCWs treating suspected but negative (misclassified) or confirmed COVID-19 patients.

Results

Depending on the confirmed to misclassified case ratio, from 1:0 to 1:10, a range of 117.1 million to 1.1 billion masks are required for single-use. This decreases to 71.6–784.4 million for extended-use and 12.8–148.2 million for prolonged-use, representing a 31.8%–38.9% and 86.5%–89.1% reduction respectively.

Conclusions

An extended-use policy should be considered when short-term supply chains are strained but planning measures are in place to ensure long term availability. With severe shortage expectations from a severe epidemic, as some European countries have experienced, prolonged-use is necessary to prolong supply.

Word count = 199

2.2 Introduction

The COVID-19 pandemic is posing severe challenges to healthcare systems, leading to shortages of PPE which leaves frontline HCWs at grave risk. Among the lessons learned during the outbreaks of SARS-CoV in 2003 and the MERS-CoV in 2012 were the need for stringent infection control in healthcare settings, clear criteria for isolation and quarantine measures, and continued evaluation of the effectiveness of PPE in infection prevention (188,189).

The WHO has warned of the risk of the global supply of PPE rapidly depleting (190). Despite these effort, hospitals have faced severe shortages of PPE (191). Shortages of PPE lead to HCWs being dangerously ill-prepared to care for patients with COVID-19 (51). Studies showed that N95s offer good protection

against viral respiratory pathogens, especially during aerosol generating procedures or when a patient's COVID-19 status is positive or unknown (192–194). N95s are of critical importance for confronting the SARS-CoV-2 pandemic among the various PPEs. Strategies to optimise the usage of N95s are therefore vital. These should focus on conserving N95s use while providing adequate protection for HCWs working with COVID-19 patients. HCWs may need to have contact with confirmed or suspected patients and their surroundings within inpatient facilities, often over an entire work shift. Extended-use—or even reuse—of N95s has therefore been suggested, with recommendations in place to use respirators for up to 4 hours if N95s is in short supply (168,195,196). A study which screened 27 countries or regions revealed that only 5 countries (19%) allowed extended use; 2 countries (7%) mentioned reuse; and 3 countries (11%) recommended both strategies for rationing N95s (168). Several ways to decontaminate N95s for reuse have also been well studied (197).

The potential extended-use of N95s requires serious consideration as the demand for respirators will increase in response to rising case counts, especially among those with severe disease who require intensive care and prolonged hospitalization. One in five cases in China had severe symptoms requiring long stays at hospital facilities (198), while a quarter of those required admission to the intensive care unit (ICU) due to complications of acute respiratory distress syndrome, arrhythmia and shock, which significantly increase hospital stay (199). The same study observed a median hospital stay of 10 days (IQR 7–14) among those discharged. Per patient, multiple N95s are consequently required to avoid transmission between HCWs and patients.

In Singapore, a city of ~5.6 million people in the Malay Archipelago, 51,197 cases were diagnosed until 28 July 2020 (200). This number is expected to rise in the coming months, and to ensure adequate provisions it is important to project the number of N95s required. Here we estimate the number of N95s required over the course of the epidemic under three strategies: single-use, extended-use and prolonged-use. By estimating the demand, we aim to aid planning efforts assessing what preparations are required over the coming months for the healthcare system to run with the maximized possible protection for its workers. Although described for the Singapore setting, our results hold for planners in similarly sized cities elsewhere.

2.3 Methods

We developed a Susceptible-Exposed-Infectious-Recovered (SEIR) model to estimate the number of cases with COVID-19 being present in healthcare facilities, and necessary N95s use under three policies: single-use, extended-use and prolonged-use. For planning purposes, we developed three variants of the SEIR model which describe the relative outbreak size, labelled as mild, severe and moderate.

2.3.1 Epidemic Scenarios and model structure

In the Susceptible-Exposed-Infectious-Recovered model (SEIR), we classified the outcomes of COVID-19 infection into six levels, relating to the need for hospitalisation or being admitted to the ICU, and the risk of onward transmission (Figure 1). These were: level 1 for asymptomatic infections; level 2a for mild symptoms that remain undiagnosed at healthcare facilities; level 2b for mild symptomatic ambulatory cases which are identified; level 3 for those

who required hospitalisation but not the ICU for their recovery; level 4 for those who entered ICU on admission; and level 5 for those dying among levels 3 and 4. We combined levels 3–5 as severe and levels 1–2 as mild. We also combined levels 2b-4 as known and levels 1–2a as cryptic, i.e., cases which are not documented by the healthcare system.

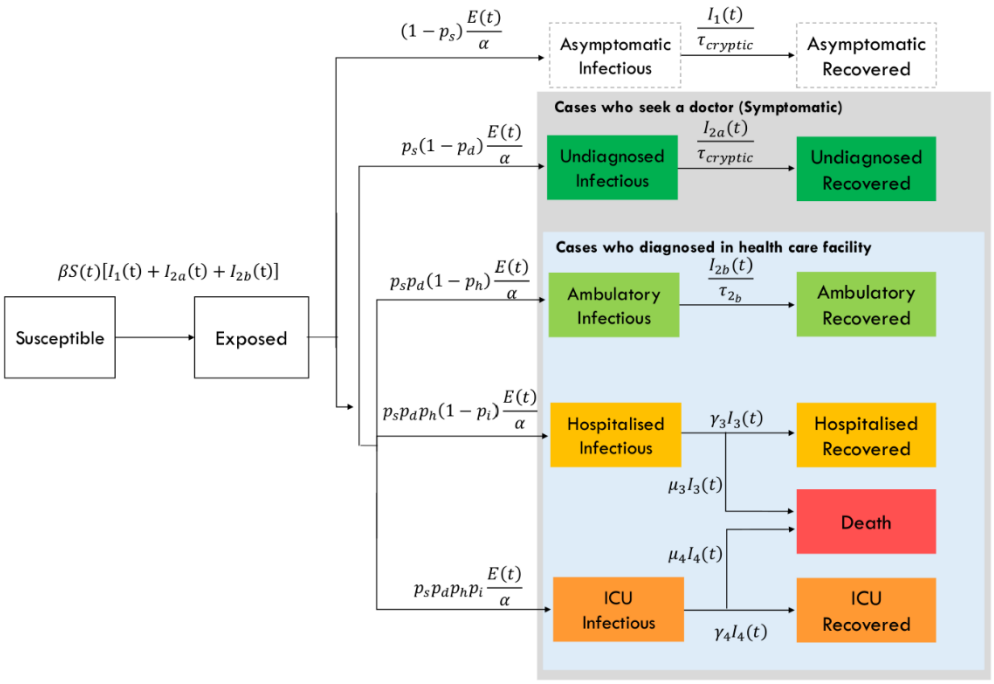


Figure 1. Flow chart of the model structure.

The susceptible population (S) is modelled to become infected at rate $\beta S[I_1(t) + I_{2a}(t) + I_{2b}(t)]$, i.e., we make the assumption that severe cases do not lead to onward transmission as they are isolated. Excluding births and migration over the course of the epidemic due to its relatively short time length, the rate of change for the susceptible population is,

$$\frac{dS(t)}{dt} = -\beta \mathrm{S(t)}[I_1(\mathrm{t}) + I_{2a}(\mathrm{t}) + I_{2b}(\mathrm{t})].$$

The model assumes a latent period, α, so that

$$\frac{dE(t)}{dt} = \beta \mathrm{S(t)}[I_1(\mathrm{t}) + I_{2a}(\mathrm{t}) + I_{2b}(\mathrm{t})] - \frac{E(t)}{\alpha}.$$

A proportion p_s of infections seek medical care after the latent period. Of these, a proportion p_d are not diagnosed—for instance, because they have mild symptoms and are unlinked to a known cluster, or test yielded false negative due to a lower viral load. We assume both groups are infectious for $\tau_{cryptic}$ days before recovery,

$$\frac{dI_1(t)}{dt} = \frac{E(t)}{\alpha}(1 - p_s) - \gamma_{cryptic} I_1(t)$$

$$\frac{dR_1(t)}{dt} = \gamma_{cryptic} I_1(t),$$

$$\frac{dI_{2a}(t)}{dt} = \frac{E(t)}{\alpha} p_s(1 - p_d) - \gamma_{cryptic} I_{2a}(t), \text{ and}$$

$$\frac{dR_{2a}(t)}{dt} = \gamma_{cryptic} I_{2a}(t),$$

where,

$$\gamma_{cryptic} = \frac{1}{\tau_{cryptic}}$$

Among those who are diagnosed, a fraction p_h need to be hospitalised. For those who need not, we have:

$$\frac{dI_{2b}(t)}{dt} = p_s p_d (1 - p_h) \frac{E(t)}{\alpha} - \gamma_{2b} I_{2b}(t), \text{ and}$$

$$\frac{dR_{2b}(t)}{dt} = \gamma_{2b} I_{2b}(t),$$

here,

$$\gamma_{2b} = \frac{1}{\tau_{2_b}}$$

A fraction p_i of hospitalised patients are admitted to the ICU ward directly. This yields:

$$\frac{dI_3(t)}{dt} = p_s p_d p_h (1 - p_i) \frac{E(t)}{\alpha} - \gamma_3 I_3(t) \ - deaths, \text{ and}$$

$$\frac{dR_3(t)}{dt} = \gamma_3 I_3(t)$$

where,

$$\gamma_3 = \frac{1 - CFR_{hosp}}{\tau_3}$$

for those not being admitted to ICU, and

$$\frac{dI_4(t)}{dt} = p_s p_d p_h p_i \frac{E(t)}{\alpha} - \gamma_4 I_4(t) \ - deaths, \text{ and}$$

$$\frac{dR_4(t)}{dt} = \gamma_4 I_4(t)$$

where,

$$\gamma_4 = \frac{1 - CFR_{icu}}{\tau_4}$$

for those admitted to the ICU. The mortality rate for patients hospitalized and in the ICU is assumed to be

$$\frac{dD_3(t)}{dt} = \mu_3 I_3(t),$$

$$\frac{dD_4(t)}{dt} = \mu_4 I_4(t)$$

where, $D_3(t)$ and $D_4(t)$ indicate death from hospitalization and from ICU on t respectively, and

$$\mu_3 = \frac{CFR_{hosp}}{\tau_{3_death}},$$

$$\mu_4 = \frac{CFR_{icu}}{\tau_{4_death}}$$

A fixed incubation period (α) of 5 days was used across all scenarios (201). The median number of days from the first symptom to death (τ_{3_death}, τ_{4_death}) was 14 (Range: 6–41) (202). The time to recovery was set as 10 days (IQR: 7–14) in the general hospital setting (τ_3) and 14 days in ICU (τ_4) (199). The start time of the outbreak was assumed to be on 27 February 2020 with an estimated 100 initial infectious cases in the community. Among all diagnosed cases (p_d), we assumed that 80% of patients had mild symptoms, 14% required hospitalization (p_h)—assuming only patients who need it are admitted—and 6% were admitted to the ICU (p_i) (203).

A review summarised that the mean R_0 ranged from 1.9 to 6.5 in the early outbreak, where 13 publications (out of 17) estimated that the range of R_0 was

between 2.0 and 3.0 (123). According to the previous estimation for Chinese data, the doubling time is 7 days in the early stage of the outbreak, the growth rate of which was calculated as ~10% by *Rule of 70* (204) . In Singapore, there were 33 cases in the following two weeks after the first cases were identified on January 23. We assumed an exponential phase of the early outbreak where $N_t = N_{t0} \times e^{0.1 \times T}$ where T is the number of days between t and the first observed case at t_0. In the initial two weeks, $N_{t0} = 8$ as 8 case importation events were recorded where the infection was not acquired locally. This generated the expected cumulative number of cases with a seven-day doubling time (time where a single case generates another case so that it derives an R_0 of approximately 2) so as to be comparable to the observed number of cumulative cases in the first several weeks in Singapore. Contact tracing measures took place to prevent the further spread of disease, reducing the outbreak's growth (Figure 2). Therefore a R_0 of 2 was set in the *severe* scenario assuming limited control and 1.5 for the *mild* and *moderate* scenario under the assumption that the outbreak could be controlled better than in the early growth period in China.

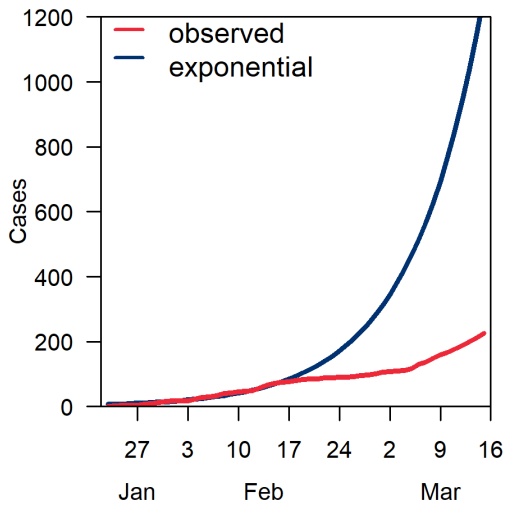


Figure 2. Exponential distribution versus the observed data presentation.

We varied the parameters in the mild to severe scenarios regarding the: (i) proportion of cases seeking medical attention (30%, 50% and 70% separately), (ii) proportion of cases requiring hospitalisation (10%, 20% and 30% respectively), (iii) proportion of cases admitted in ICU among hospitalised cases (20%, 30% and 40% from mild scenarios to severe scenarios), (iv) case fatality rate (0.43%, 1.8% and 4.2%) (172), and (v) the basic reproduction number (1.5, 1.5, and 2.0) (123) (Details can be found in Table 1).

Table 1. Parameters used across different scenarios

Parameters	**Notation**	**Severe**	**Moderate**	**Mild**	**Reference**
Population size (million)	N	5.6	5.6	5.6	
R_0	R_0	2	1.5	1.5	
Cases initially infected	$E(0)$	100	100	100	
Incubation period	α	5	5	5	(201)
Recovery time (cryptic case)	$\tau_{cryptic}$	10	10	10	
Recovery time (ambulatory case)	τ_{2b}	10	10	10	
Recovery time (hospitalized case)	τ_{3a}	10	10	10	(199)
Recovery time (ICU case)	τ_4	14	14	14	(199)
Time to death (hospitalized case)	τ_{3_death}	14	14	14	(202)
Time to death (ICU case)	τ_{4_death}	14	14	14	(202)
% seeking doctor	p_s	70	50	30	
% diagnosis	p_d	89	89	89	(205)
% hospitalization	p_h	30	20	10	(203,206)
% admission ICU among hospitalized cases	p_i	40	30	20	(203,206)
Case fatality ratio (hospitalized case)	CFR_3	4.2	1.8	0.43	(172)
Case fatality ratio (ICU case)	CFR_4	4.2	1.8	0.43	(172)

All analyses were performed by R version 3.6.2 (207).

2.3.2 N95 respirators Policies

Three N95s use policy scenarios were suggested based on the frequency of these inpatient visits: single-use, extended-use and prolonged-use. Under the single-use policy scenario, new N95s are used by all staff for each patient encounter. For the extended-use policy scenario, new N95s are used after every three patient encounters for nurses, while other HCWs don a fresh set for each encounter. For prolonged-use, all staff change N95s after seeing 10 ICU patients concurrently and/or after every four hours.

We itemized the required N95s use as per the initial single use protocol at the National University Hospital (NUH), Singapore, to quantify the number of N95s used per day in the ICU and respiratory isolation wards. We identified four main HCWs types interacting daily with patients: nurses, doctors, ancillary and environmental staff. According to the guidance from consultants in NUH, the nurses encountered the patients the most and operated on two main shifts. Of these, the nurses encountered the patients the most and operated on two main shifts. The day shift in both wards ran from 6am to 12am and the night shift from 12am until 6am the next day. Doctors are assumed to have a total of 10 visits with a patient every 24 hours in both the isolation and ICU wards. Nurses in the ICU are expected to have an encounter every 30 minutes during the day shift and once per hour during the night shift in the ICU, and every 2 hours during the day shift and 4 hours for the night shift in the isolation wards. Only one nurse is required for ~80% of the work but occasionally assistance from a colleague is needed when attending to patient needs. Ancillary staff—a mixture

of radiology technicians, respiratory therapists, and other roles—were assumed to have a total of 10 visits in the ICU and 4 visits in isolation wards every 24 hours. Environmental staff were assumed to have 2 encounters every 24 hours, regardless of the location (Figure 3 and Figure 4).

Each ICU patient is isolated in a room of two patients. One nurse and one doctor attends to these two ICU patients. The nurse encounters patients every 30 minutes during the day shift and every hour during the night shift. Only one nurse is required for ~80% of the work but occasionally assistance from a colleague is needed when attending to patient needs. The doctor and ancillary staff each have a total of ten encounters with a patient every 24 hours and the environmental staff have 2 encounters with a patient every 24 hours (Figure 3).

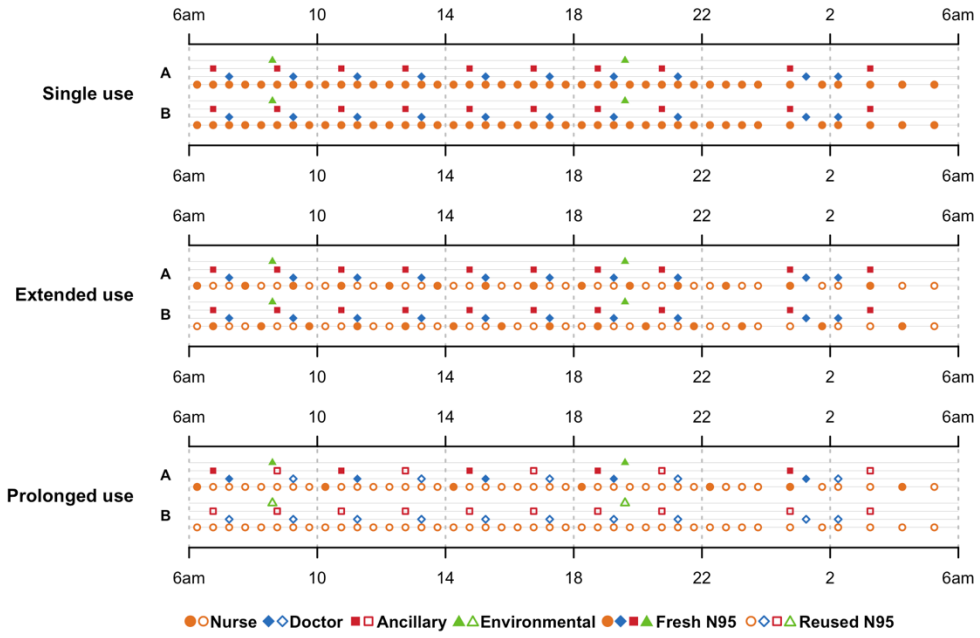


Figure 3 Schematic of N95s utilisation under three protocols in ICU (A and B represent 2 patients, separately). Schematic of N95s utilisation under three protocols in ICU (A and B represent 2 patients, separately). Each symbol represents an interaction with a patient by a healthcare worker. Closed symbols represent new N95s being donned before that interaction with the patient; open symbols represent extensive use of the N95s for that interaction.

Within the isolation wards, ten patients are attended individually with room dividers by three nurses and two doctors. Nurses also require assistance from another nurse for ~20% of their procedures. The nurses visit the wards every 2 hours during the day shift and every 4 hours during the night shift. Every 24 hours, doctors make a total of ten visits, ancillary staff four visits, and environmental staff 2 visits. During these visits, medical personnel will encounter each of the ten patients each time. We assume that two nurses would attend to three patients and thus each one would attend to four on each visit (Figure 4).

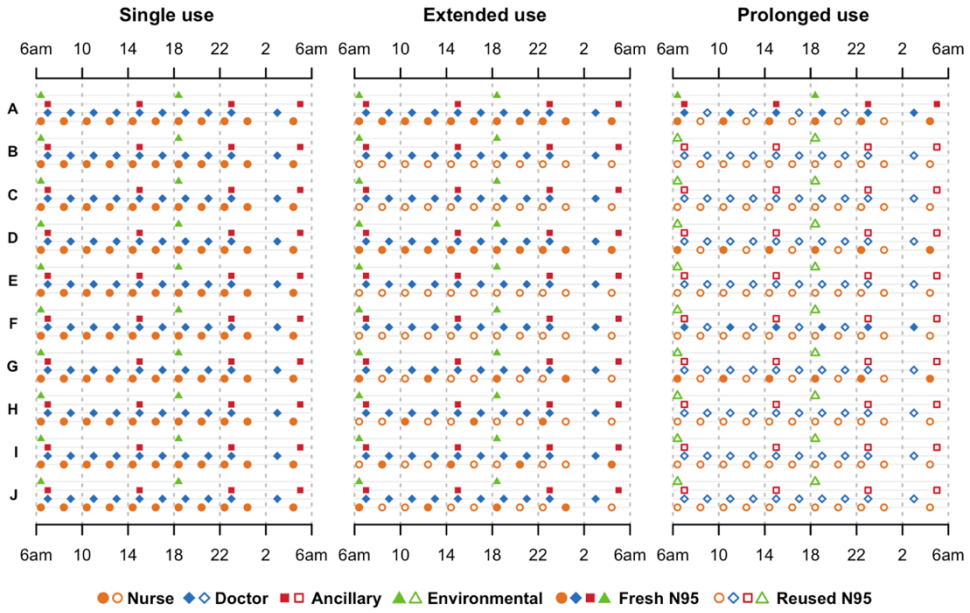


Figure 4. Schematic of N95 utilisation under three protocols in respiratory isolation wards. Each symbol represents an interaction with a patient by a healthcare worker. Closed symbols represent new N95s being donned before that interaction with the patient; open symbols represent extensive use of the N95s for that interaction.

Daily demand of N95s in ICU and isolation wards under different policies were summarised in Table 2 and Table 3.

Table 2. Summary of the average demand of N95s per patient per day in the ICU where 39 sets are required under the extended-use policy and 6 sets under the prolonged-use policy.

Types of staff	Single-use policy	Extended-use policy	Prolonged-use policy
Nurses and assistant nurses	50.4	16.8	4
Doctors	10	10	1
Ancillary staff	10	10	1
Environmental staff	2	2	0
Total	~72	~39	~6

Table 3. Average number of N95s being used in the isolation wards.

Types of staff	Single-use policy	Extended-use policy	Prolonged-use policy
Nurses and assistant nurses	13.2	4	2.16
Doctors	10	10	1.2
Ancillary staff	4	4	0.4
Environmental staff	2	2	0.2
Total	~29	20	~4

2.3.3 Ratio of Confirmed and Suspected but Negative Cases

Suspected cases are divided into negative cases and positive cases based on two consecutive coronavirus RT-PCR tests. Negative cases included those with mild symptoms that remain undiagnosed at healthcare facilities (level 2a) and suspected but negative cases. The positive cases included mild symptomatic ambulatory cases (level 2b), those requiring hospitalisation (level 3) and ICU cases (level 4). All suspected cases, who were quarantined in the isolation wards, were assumed to wait for three days in the isolation ward until laboratory confirmation. After three days, mild symptomatic ambulatory cases (level 2b)

and mild symptoms that remain undiagnosed (level 2a) are discharged from the isolation ward.

To assess the effects of case misclassification, six ratios were used to reflect the proportion of suspected but negative cases with positive cases, ranging from 0–10 (0, 0.5, 1, 2, 5, 10). A zero ratio (1:0) has no case misclassification and conversely a ratio of 10 indicates significant uncertainty in preliminary diagnosis upon admission where a substantial proportion are incorrectly identified as being COVID-19 positive but later are confirmed to be negative. A separate set of analyses were conducted where rapid testing was implemented, or a zero ratio, at 2 to 5 months from the epidemic start point, to estimate its potential effects on usage of N95s.

2.4 Results

2.4.1 Epidemiological features in simulated outbreaks under scenarios

A total of 4.5 million infections (80% of the total population) were projected to occur in the severe scenario and 2.6 million (45.6%) in the moderate or mild scenario over the span of a year. Hospitalized cases reached a peak at approximately 6.5 months (day 194), and 11 months (day 324) after the first case. At the peak, 65,000 hospitalised cases were estimated to occur in the severe epidemic, 15,000 in the moderate and 5,000 in the mild scenario (Figure 5A). The peak in ICU admission occurred 3–4 weeks later than hospital

admission (day 197 and day 328). At this time, the number of patients staying in the ICU amounted to 57,000, 9,000 and 2,000 (Figure 5B).

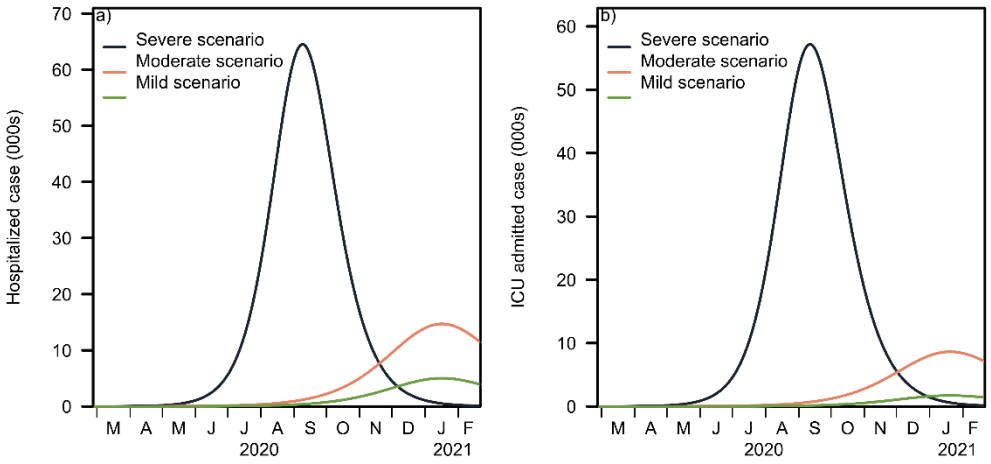


Figure 5. The epidemic curves in different scenarios (A) Hospitalized cases (B) ICU admitted cases. The colours indicate the scenarios. Black: Severe scenario, orange: moderate scenario, and green: mild scenario.

2.4.2 Final N95 respirators usage comparison according to different scenarios

HCWs working in an ICU will use 72, 39, or 6 N95s per case per day in the single-use, extended-use and prolonged-use policies respectively. Workers in the respiratory isolation ward will use 29, 20, or 4 per case per day in these respective policies (Table 2 and Table 3).

In the moderate scenario with the ratio of confirmed cases versus suspected cases at 1:0, the number of required N95s was found to be 117.1 million, 71.6 million and 12.8 million under the three policies. A reduction of 38.9% and 89.1% was observed in the final consumption when an extended-use and prolonged-use policy was implemented (Figure 6D and Table 4). In the severe scenario, the final consumption amounted to 512.8 million, 304 million, and 52.4 million for each policy. The implementation of an extended or prolonged-use policy created further savings of 40.7% and 89.8% (Figure 6A and Table 4). In the

mild scenario, a total of 34.3 million, 21.9 million and 4.1 million N95s were estimated to be required for three policies. The reduction with extended and prolonged-use was 36.2% and 88.1% compared to the single-use policy (Figure 6G and Table 4).

Table 4. Total N95s consumption (millions) for different scenarios

Epidemic Scenario	**Policy**	**Confirmed cases versus Suspected cases**					
		1:0	**2:1**	**1:1**	**1:2**	**1:5**	**1:10**
Severe	single use	512.8	633.8	552.3	996.6	1722.1	2831.4
	extended use	304	387.4	373.4	637.6	1138.0	1971.9
	prolonged use	52.4	69.1	73.5	119.2	219.2	386.0
Moderate	single use	117.1	166.2	180.6	313.4	607.8	1098.4
	extended use	71.6	105.5	122.7	207.0	410.0	748.4
	prolonged use	12.8	19.6	24.3	39.9	80.5	148.2
Mild	single use	34.4	63.9	87.2	152.2	328.8	623.2
	extended use	21.9	42.2	59.6	103.1	225.0	428.0
	prolonged use	4.1	8.2	11.9	20.4	44.7	85.3

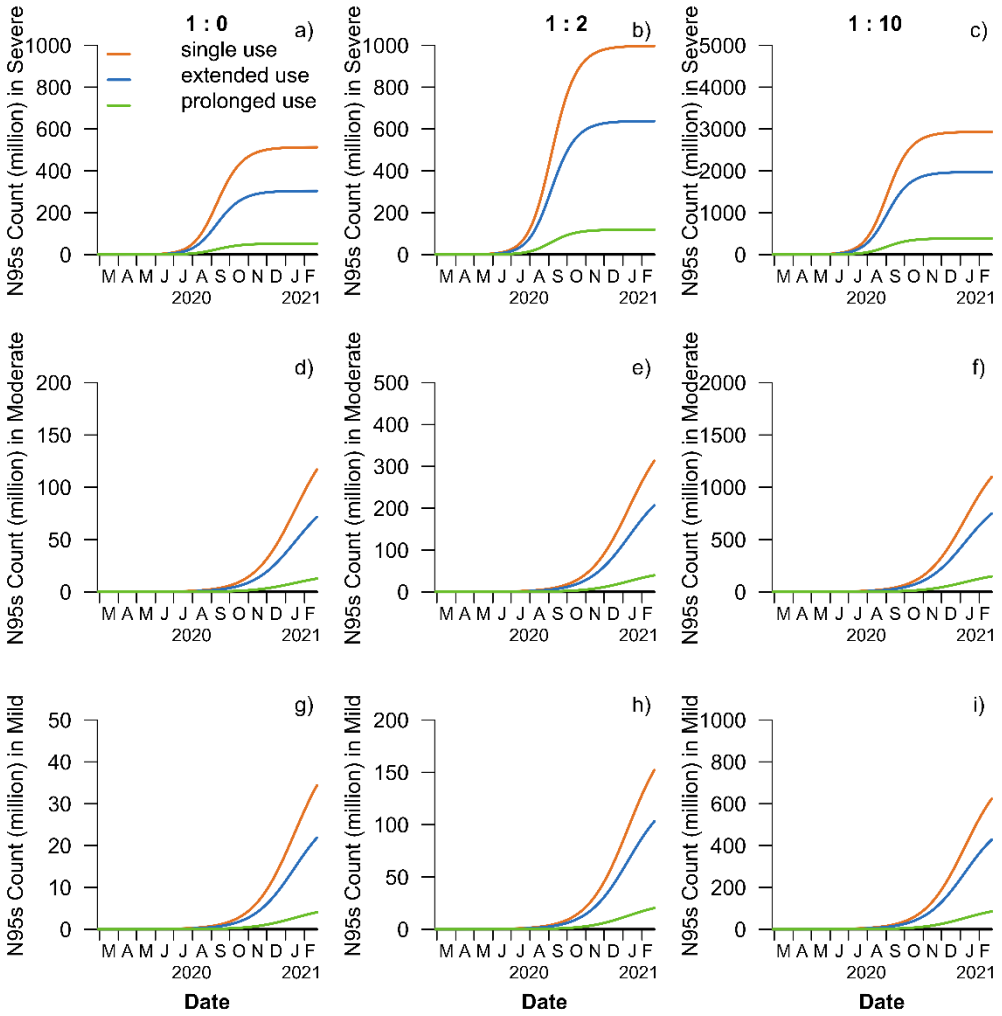


Figure 6. The cumulative N95s consumption for selected ratios of confirmed cases versus suspected cases. The simulated scenarios in plots A-C are severe, D-F moderate, and G-I mild. The ratios of confirmed cases versus suspected cases for: A, D and G is 1:0; B, E and F is 1:2; C, F and I is 1:10. The line colours describe policies of N95s usage where the red line shows single use, blue line shows extended use, and green line shows prolonged use.

Under the mild scenario with the ratio of confirmed cases versus suspected cases at 1:0, the peak month of N95s consumption occurs 11 months later with the consumption of 10.3 million sets during the peak month according to the single-use policy, and the reduction of 35.9% for extended-use and 87.4% in the prolonged-use policy. The peak month occurs six months in the severe scenario (Table 5) where N95s consumption reached 188.1 million. Compared to the single-use policy, the extended-use policy reduces N95s consumption by 40.6% and the prolonged-use policy saves N95s consumption by 89.7% (Table 5).

Table 5. N95s consumption (millions) in the peak month for different scenarios

Epidemic Scenario	Policy	Confirmed cases versus Suspected cases					
		1:0	2:1	1:1	1:2	1:5	1:10
Severe	single use	188.1	234.1	280.8	375.2	660.5	1137.9
	extended use	111.7	143.6	175.9	241.1	438.2	767.6
	prolonged use	19.4	25.7	32.2	45.3	84.7	150.7
Moderate	single use	35.4	49.4	63.4	91.6	176.3	317.5
	extended use	21.7	31.3	41.0	60.4	118.8	216.3
	prolonged use	3.9	5.8	7.8	11.7	23.3	42.8
Mild	single use	10.3	18.8	27.2	44.2	95.0	179.8
	extended use	6.6	12.4	18.2	29.9	65.0	123.5
	prolonged use	1.3	2.4	3.6	5.9	12.9	24.6

We found that the N95s consumption was sensitive to different ratios of confirmed cases versus suspected cases across the epidemic scenarios. With a ratio of 1:2, 313.4 million, 207 million and 39.9 million N95s were estimated for use in each policy within the moderate scenario. The consumption increased ~3 times in the severe epidemic scenario and reduced by ~0.5 times in the mild scenario (Figure 6B, E, H). With a high misclassification rate from a ratio of 1:10, N95s consumption amounted to 1.10 billion, 748.4 million, and 148.2 million in the moderate scenario across the three policies. This was

substantially reduced by 31.8% and 86.5% within the extended and prolonged-use policies (Figure 6F and Table 4).

In the severe scenario with the ratio of confirmed cases versus suspected cases at 1:10, 2.84 billion, 1.98 billion, and 386 million N95s were estimated to be used, representing a 30.4% and 86.4% reduction (Figure 6C and Table 4). In the mild scenario, N95s consumption was estimated to be 623.2 million, 428 million and 85.3 million, reducing final N95s consumption by 31.3% and 86.3% in the extended and prolonged-use policies (Figure 6I and Table 4).

When the ratio of confirmed cases versus suspected cases at 1:10, the peak month of N95s use in the moderate scenario was in 11 months where the N95s needs are 317.5 million according to single-use, 216.3 million for extended-use and 42.8 million for prolonged-use. However, for the severe scenario in the peak month occurring in seven months, N95s consumption increases to a high of 1.14 billion, 767.6 million and 150.7 million with these policies (Table 5).

2.4.3 N95 respirators usage with implementation of rapid testing

We also considered N95 usage in three epidemic scenarios under different policies with the introduction of rapid testing kits. The introduction of rapid diagnostic kits can reduce N95 demand but are still required. In the severe scenario, a 5–month delay in switching to the rapid diagnostic testing results in the usage of 620.6 million N95s with single-use, 413.1 million with extended-use and 80.1 million with prolonged-use policies. In the same epidemic scenario, if the rapid diagnostic is released within 2 months of the outbreak, the amount of N95s usage reduces substantially to 348.1 million, 225.1 million and 42.5 million respectively (Figure 8A). In the mild and

moderate epidemic scenario, the peaks of N95s use are reduced and delayed (Figure 6 and Figure *7*). In the moderate scenario, after a 5-month delay in switching to rapid diagnostic testing, 90.7 million, 60.8 million and 11.9 million N95s are required. A further reduction is observed if introduced at 2 months with 83.0 million, 55.4 million and 10.8 million N95s in demand (Figure 8B). Under the mild epidemic scenario, a 5–month and 2–month delay in the introduction of rapid diagnostic testing results in N95 usage of 33.3 million and 28.6 million for single-use, 22.4 million and 19.2 million for extended-use, and 4.4 million and 3.8 million for prolonged-use (Figure 8C).

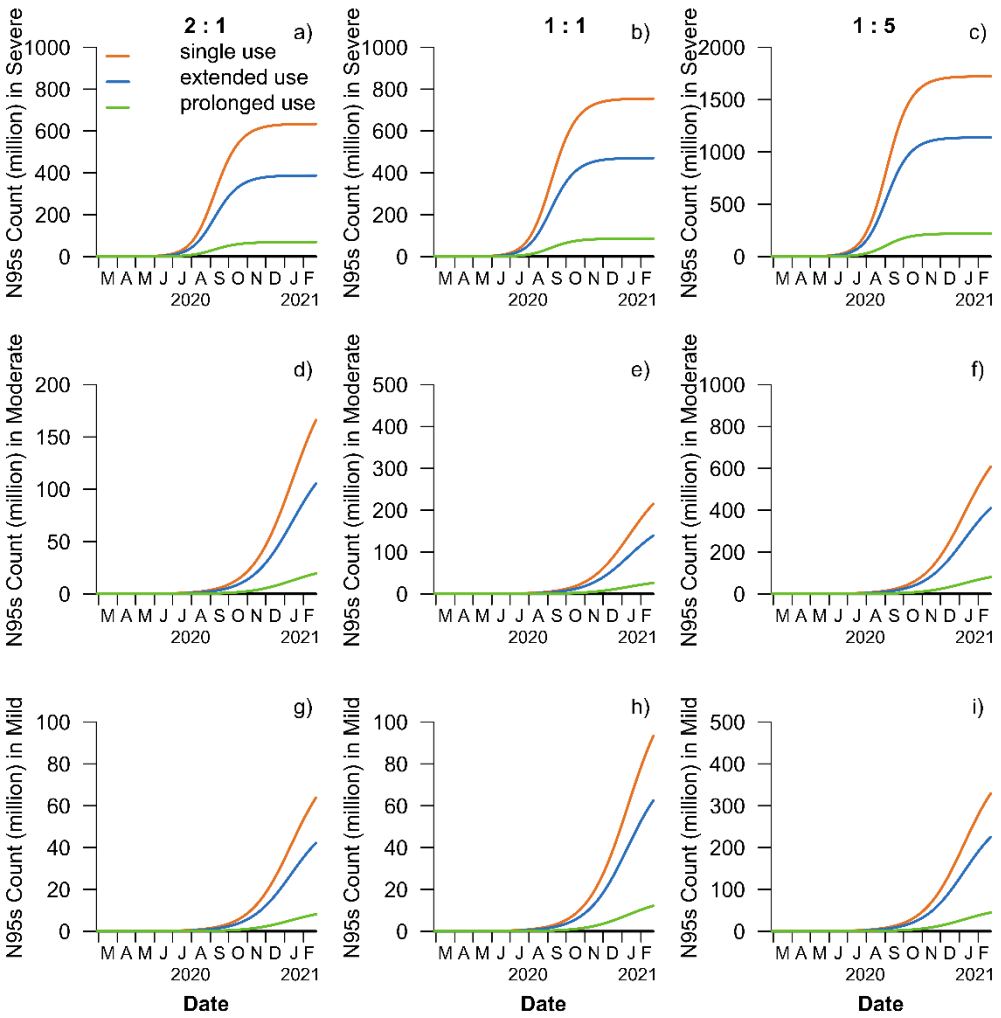


Figure 7. Cumulative N95s consumption when considering confirmed cases versus suspected cases as 2:1, 1:1 and 1:5. Plots A-C reflect the severe epidemic scenario, D-F the moderate scenario, and G-I the mild scenario. The policies for

N95s usage are represented in colour (red for single use, blue for extended use, and green for prolonged use).

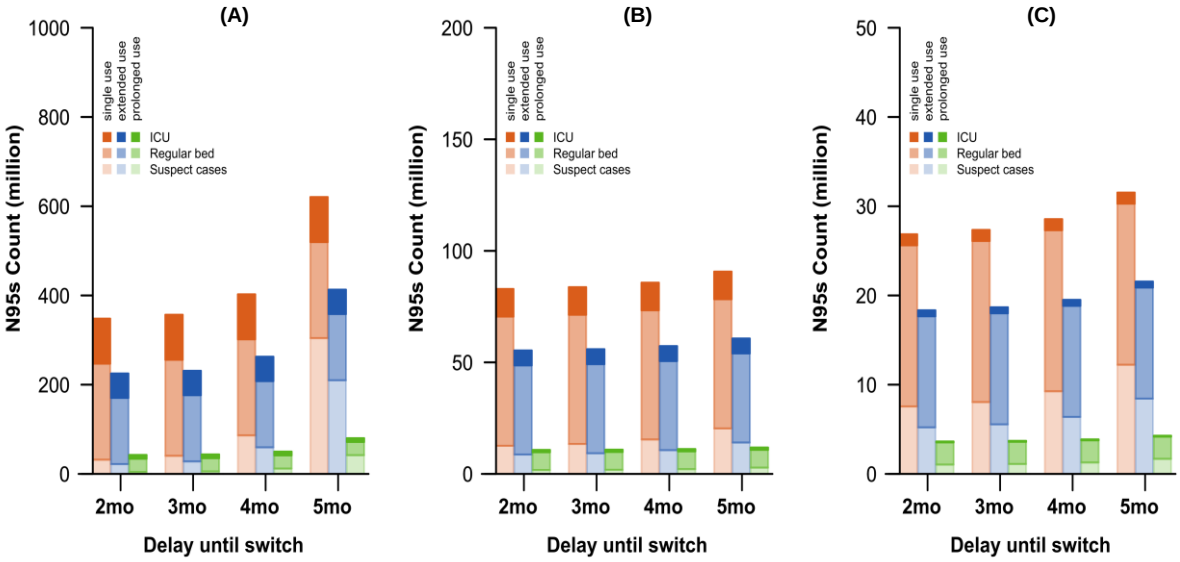


Figure 8. N95s use across time until the switch to rapid testing in different epidemic scenarios where (A) is severe, (B) is moderate and (C) is mild. The orange line series show single use, blue the extended-use and green the prolonged-use policies. Each bar is stratified by the levels of health care setting. Darker colours describe ICU cases, medium shading the cases in regular beds and lighter shades the suspected COVID -19 cases.

2.5 Discussion

Optimal N95 use is required for national and global supplies to be sustained, ensuring HCWs safety and allowing for proper patient care across all healthcare facilities. We projected N95s usage across a wide range of scenarios where for a moderate epidemic scenario, our projections of consumption for a single-use policy with no misclassification of disease at hospital entry (ratio 1:0) was found to be 117.1 million, 71.6 million and 12.8 million. This reduced by 38.9% and 89.1% when extended and prolonged-use policies are implemented. These considerable reductions can extend the lifetime of N95s stocks across time and

allow for a wider distribution between different regions or countries to reduce the stress on the demand. When more case misclassification occurred at a ratio of 1:10, N95s use increased to 1.10 billion, 748.4 million, and 148.2 million on, which would be considerably more challenging to obtain and distribute effectively. The introduction of a rapid diagnostic kit at any point of the epidemic which can immediately ascertain if an individual is positive of SARS-CoV-2 is therefore advantageous and should be immediately implemented when possible provided test accuracy is high.

Our approach can be utilised elsewhere with considerations of the implementation of such extended-use policies. This is especially paramount at this time as the WHO, despite already shipping more than 900,000 surgical masks, 62,000 N95s and 34,000 face shields with other PPEs to 133 countries to supplement supplies for those at contingency or crisis capacities, has stated that overall global supplies are rapidly depleting (208). This is further complicated by the fact that the PPE supply chain landscape is complex. Numerous national PPE stockpiling systems exist, such as the US Strategic National Stockpile, Canada's National Emergency Strategic Stockpile, Australia's National Medical Stockpile and Taiwan's 3-tier stockpile (209). These often require multiple private contractors and vendors to balance procurement and deployment to facilitate demand and require prioritization in allocating resources to different hospitals and healthcare centres with the ongoing supply strain. Manufacturers, particularly in China, are also struggling to ramp up production due to the travel restrictions and quarantining procedures in place which requires other strategies to be considered such as extended-use.

During the last pandemic (influenza A(H1N1-2009)), Hashikura and Kizu recommended a minimum of 8 weeks of PPE supply for stockpiling with 4 sets being utilised per day by high risk workers, 2 sets for medium and low risk groups, 2 surgical masks for each worker and inpatient, and 1 for every suspected case (210). Recommendations at the Ministry of Health, Singapore, are more risk averse by aiming to maintain a 3 to 6-month stockpile, with individual capacities set for each medical institution (211). The overall stockpile capacity is a function of the renewal rate among workers, the protective robustness of the equipment and corresponding costs. Based on our findings, the extended-use policy could be utilised within wards before supply chains are stressed where multiple infected patients are being visited simultaneously, provided infection risk can be mitigated. Should infection become widespread with significant patient load at healthcare facilities, prolonged-use may have to be utilised to prevent supply shortages. Where possible, it is recommended that gowns and N95s use be prioritized for high-risk aerosol generating procedures including endotracheal intubation and bronchoscopy (168). With 42% of 52 ICU patients requiring invasive mechanical ventilation and 13.5% of this group acquiring the infection in hospital, considerable transmission risk exists (212,213). This risk is further exacerbated by the presence of environmental contamination from patients (214).

Singapore is currently at an orange level of its Disease Outbreak Response System Condition, which reflects a high risk of importations of cases and large clusters of cases occurring within foreign worker dormitories. Although PPE, including N95s, only forms a component of the protective procedures, it remains crucial alongside limiting the number of encounters, implementation of

negative pressure isolation rooms, physical barriers and exclusion of non COVID-19 patients by testing. Communication with the public remains the first barrier of defence with recommendations of appropriate mask usage and prioritization of supplies for HCWs and symptomatic individuals highlighted. Public engagement will not be sufficient however as infection events will continue to occur, therefore countries facing even greater short and long term shortages of N95s could potentially consider the policy findings here for implementation (215).

As with all modelling studies, many assumptions were made. Firstly, the SEIR model did not account for any heterogeneities in the infection or symptomatic rates as a result of factors such as the age structure. We also did not consider the effects of ongoing control methods, primarily social distancing, on the number of infections through time. In addition, we did not consider the infection risk to patients from HCWs who have acquired infection from other infected patients. This is due to Singapore's outbreak in the community kept being relatively small, making estimations of the control's effects very challenging. Secondly, parameters such as the ratio of confirmed and suspected cases remain largely unknown. Thirdly, the number of N95s used assumes all HCWs adhere to the guidelines, not counting for accidents or personal preferences, which may inflate the requirements. It is furthermore unknown as to whether the reuse will cause an increased risk of infection among HCWs due to mask degradation. Fourthly, hospital procedures may change where fewer workers stay longer in N95s within the respiratory isolation wards so as to avert N95s use. In addition, the proportions allocated to each level here may change according to specifications set out by policymakers should resources become more limited.

Lastly, the shift schedules of HCWs in NUH in isolation and ICU wards were provided by experts rather than collected from nurses, doctors and other HCWs via survey form directly. This is due to the access restriction in the early outbreak in hospital settings.

Determined by the epidemic size, an extended-use policy should be considered where short-term supply chains are strained but planning measures are in place to ensure long-term availability. With severe shortage expectations from a severe epidemic, prolonged-use can be proposed as a necessary policy to significantly prolong supply. Should infection become widespread with significant patient load at healthcare facilities, extensive reuse, via the extended-use and prolonged-use policy, may have to be utilised to prevent supply shortages.

2.6 Motivation for Next Chapter

This research work implemented SEIR model to provide evidence to decision-makers healthcare settings. Research has shown that the demand for N95s needs under different scenarios and uses policies, which help assist healthcare resource distribution. However, the approach and methodology proposed for estimating the epidemic size in this chapter cannot account for the heterogeneous infection types. Also, the next step for healthcare managers to consider is the hospitalisation and ICU needs. The next chapter, i.e., Chapter 3, showed the active ICU needs from the cases of different infection types.

CHAPTER 3. ESTIMATING THE NUMBER OF COVID-19 CASES BY INFECTION ORIGIN IN SINGAPORE

3.1 Abstract

Background:

Heterogeneity in transmission rates has been observed among different groups in the wider population, which are important to assess when predicting the number of infections and the impact on severe disease among them.

Methods:

We used case data from 19 January 2020 to 31 December 2020 to model past and current infections in a Bayesian framework for three distinct infection groups within Singapore: the general community, foreign workers and those imported. Specifically, mobility data was utilised for the fitting of general community cases across time and an SEIR model for dormitories with the fitting of staged efficacies for the lockdown intervention. We also estimated the number of ICU cases that evolved from each infection group. Model performance was assessed through one-week ahead projections in a rolling window.

Results:

Our models performed well throughout the period of estimation. For foreign worker cases, we project an average of 1 daily case (range: 0–6) in the next 90 days from 31 December 2020. For the general community, 2 are expected (range: 0–16), and for imported cases 8 (range: 2 –14).

Conclusions:

Case projections for the ongoing planning of interventions require frameworks that utilise tailored methods for each subgroup population.

Word counts = 198

3.2 Introduction

The COVID-19 pandemic has been marked by heterogeneous impacts on different population groups, dividing society through factors such as race, age, and social capital (216,217). In western countries, the hardest-hit groups are racial and ethnic minority groups, low-income groups, or among those where working from home is not practical such as migrant communities, HCWs, prisons, and nursing homes (136,139,179,185,218,219). In lower-income countries, high-density urban areas and slums have been especially hard hit as maintaining social distancing is challenging (220,221). Thus attack rates of 48.7% (95% CI: 22.1%–76.8%) to 76% (95% CI: 66.6%–97.9%) have been estimated in cities in India (222) and Brazil (223) in September 2020 and October 2020, respectively. Furthermore, despite sharing similarities in income levels and climatic geographies, countries have had remarkably different outbreak patterns among groups as has been observed for Thailand and Vietnam when compared to Indonesia and the Philippines (224) in 2020. Issues causing such disparities can include misdiagnosis, access to care, low awareness of hygiene and housing density which can make it challenging to estimate cases and apply interventions effectively (225,226).

Singapore is no exception in terms of heterogeneity. Since its first case in January 2020, Singapore has collated details of cases throughout its outbreak,

which are publicised in the national media and government websites (227), and further broken down into general community, foreign worker dormitory and imported cases. Case data highlights the different phases of the pandemic with an early phase marked by low importation levels from China and limited local spread (Jan–Feb 2020), followed by more significant numbers of imported cases from Europe and the Americas with substantial local spread (Mar 2020). With the implementation of lockdown measures (Apr–Jun 2020 (228,229)) which coincided with a massive spread in migrant workers (MWs) (Apr–Aug 2020), a phase of high levels of imported cases with limited local spread followed (Aug 2020 to end 2020). Overall for 2020, the proportion of cases in Singapore has been highly skewed towards MWs (93% of the total) and travellers (3%), with relatively few in the general resident population (4%) (230). The MWs were generally young (median age: 44 years old (231)) and healthy. Hence, very few of the infected population required care in ICU (232,233).

Consequently, policy responses have differed between these three groups. To control spread within the general community, a multi Ministry team set up regional screening centres, contact tracing protocols, enforced crowd management, mandatory mask-wearing and isolation procedures for test positives (234,235). For foreign-worker dormitories, the inter-mixing of workers was limited across different blocks or sites, and movement tracking was performed with regular 14-day swabbing of uninfected workers (236). Imported case identification has been largely addressed through border policy control which began with travel restrictions from Hubei, China to multiple countries globally. Those who were allowed to enter the country initially quarantined at home and then moved to quarantine in approved hotels with

testing. Throughout 2020, many policies have been introduced due to concerns of healthcare capacity, especially ICU needs, and increasing case numbers which are unlinked within disease clusters, with the overall aim of introducing economic and social normalcy to the national population. This has and continues to require informed projections of case counts separated by the origin of infection for effective planning of interventions moving forward.

Here, we therefore describe how we have projected epidemic trends within the general community, foreign worker cases, imported cases, and each of their ICU needs, over the course of the pandemic in 2020 to create a more tailored output for use by policymakers as part of the wider COVID-19 surveillance system effort. We demonstrate the importance of estimating cases across heterogeneous origin when tallying national totals and provide real-time projections across these settings.

3.3 Materials and Methods

3.3.1 Data Sources

Daily reported case data published by Singapore's Ministry of Health were utilised to derive case projections. We extracted data for the period 19 January 2020, the date of Singapore's first case, to 31 December 2020. The dataset distinguishes cases by their origin—general community, imported, and foreign worker—because of notable differences in observed infection rates (236) and different public health measures needed to control infection in these three groups. All cases in Singapore were isolated in healthcare institutions until recovery, some were admitted to ICU. The number of cases in the ICU was also

estimated by accounting for transferring from heterogeneous groups from different origins—general community, imported, and foreign workers.

Mobility data have been found to be significantly correlated with COVID-19 case counts (237,238). We, therefore, obtained this dataset from the COVID-19 Community Mobility Reports (CMR), published by Google (239), for the equivalent time period. Baseline pre-outbreak mobility, defined by the CMR as averaged mobility from 3 January to 15 February 2020, was used as the reference value. From January 19 to February 15, when CMR data were not available, the reference value was assumed. Thereafter, the data describe the amount of time spent, relative to the pre-COVID-19 baseline, in six location types: residential; transit stations; retail and recreation; grocery stores and pharmacies; parks; and workplaces. For the residential category, the change in duration was presented; for other categories, the change in the total number of visits was available.

The CMR data were used to model and project general community cases, for which social distancing measures were a dominant strategy for epidemic mitigation in Singapore. In contrast, mandatory quarantining of affected foreign worker dormitories led to their movement was heavily restricted upon lockdown, which was not reflected in the CMR data. The number of imported cases were assumed not to be affected by time spent in these six location types.

3.3.2 Modelling

Separate models were created for each of the three subgroups and the number of cases in ICU. The total number of new COVID-19 cases in Singapore, N_t, on day t, consisted of general community cases (N_t^C), foreign worker cases

(N_t^f), and imported cases (N_t^i). The number currently in ICU on day t, from any source, was N_t^{ICU} (Figure 9). These were described in turn.

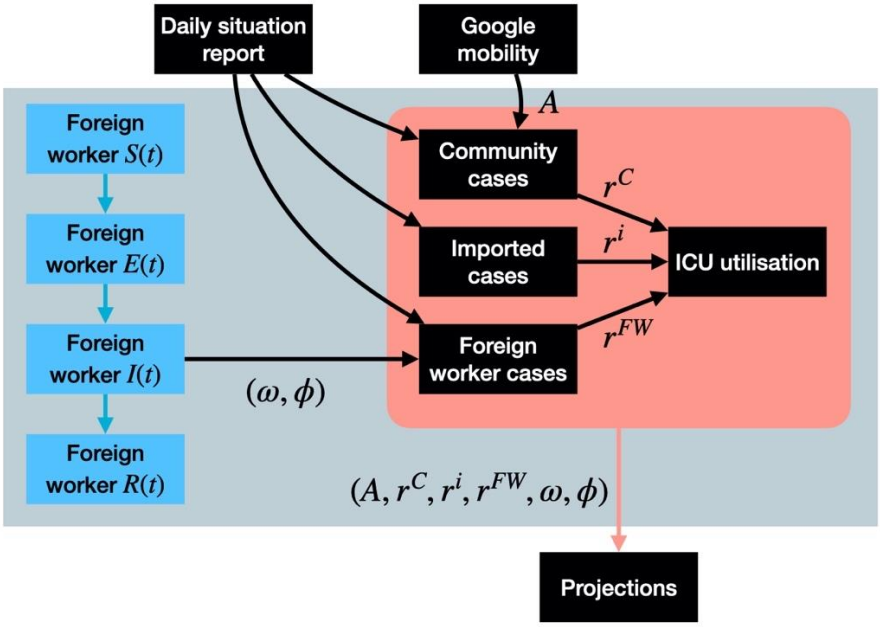


Figure 9. Model structure. Cases originating from the general community, importation and foreign worker dormitories transfer to ICU with risk r^C, r^i, and r^{FW}, respectively. Google mobility data and daily situation reported cases were used to estimate and project the general community cases; daily situation reported cases was used for estimation and projections of imported cases and foreign worker cases. And SEIR model was implemented to assess the epidemic curve in foreign worker subgroup.

General community cases: We assumed a Poisson distribution for the data generation process,

$$N_t^c \sim Pois(\mathrm{E}[N_t^c]).$$

The number of general community cases on day t was assumed to depend on those observed on day $t - d^c$, therefore,

$$E[N_t^c] = a_0 + A_{t-d^c} \times N_{t-d^c}^c$$

where A_{t-d^c} denoted the percentage increase or decrease in general community cases from the previous case count on day $t - d^c$, d^c was an estimated lag

period representing the delayed effect of mobility on cases, and a_0 was a small correction factor allowing for cryptic routes of transmission. When we incorporated mobility,

$$A_t = \exp\{a_1 + a_2 m_t^R + a_3 m_t^T + a_4 m_t^P + a_5 m_t^G + a_6 m_t^W + a_7 m_t^{RR}\},$$

where relative to the baseline, m_t^R indicated changes in residential mobility, m_t^T in transit stations, m_t^P in parks, m_t^G in groceries and pharmacies, m_t^W in workplaces, and m_t^{RR} in retail and recreation sites.

Imported cases: We also assumed a Poisson distribution for the number of imported cases,

$$N_t^i \sim Pois\left(\mathrm{E}\left[N_t^i\right]\right).$$

As countries elsewhere started to close their borders when early outbreak of COVID-19, there was a large rise in the number of imported cases into Singapore as Singaporeans rushed to return before border closures. We described the increase in importations early in 2020 with a Gaussian function, $p(t)$,

$$p_1(t) = \frac{1}{\sigma_1\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{t-\mu_1}{\sigma_1}\right)^2\right\}.$$

Near-complete border closure followed this rush of returnees for residents of all countries except Singaporean citizens and long-term residents from 17 June 2020 (ν_1) when the first green lane agreements came into force (240). Beginning 21 August 2020 (ν_2), a more liberal border regime began with international flights resuming but with travellers subject to strict quarantine procedures (241). During this period, the number of international arrivals was

controlled to impose limits to the number of cases arriving, and as a result, the number of cases was relatively stable after ν_2. The modelled change in imported cases pre and post enforced border control policies among imported cases were denoted as ρ_1 and ρ_2, respectively. We also assumed that imported cases stabilises after ν_2 at a level of ρ_2 above the baseline of ρ_0. The model was thus:

$$\mathrm{E}\left[N_t^i\right] = \rho_0 + \rho_1 p_1(t) \qquad \text{for } t \leq \nu_1,$$

$$\rho_0 + \rho_1 p(t) + \frac{(\rho_2 - \rho_1 p(\nu_1)) \times (t - \nu_1)}{\nu_2 - \nu_1} \qquad \text{for } \nu_1 < t < \nu_2, \text{ and}$$

$$\rho_0 + \rho_1 p(t) + \rho_2(\nu_2 - \nu_1) \qquad \text{for } t \geq \nu_2.$$

Foreign worker cases: During lockdown, dormitories housing foreign workers with outbreaks were designated as isolation areas under Singapore's Infectious Diseases Act to prevent spread within, between and from dormitory residents. Movement between dormitories and worksites was mostly stopped to prevent the mixing of residents though the presence of shared facilities such as toilets prevented complete segregation within blocks. Because of the explosive and widespread nature of the dormitory outbreaks (242), we used a dynamic transmission model to account for herd immunity. After Mid-July 2020, dormitories were "cleared" through mass testing, after which their residents were allowed back to worksites, although any dormitory subsequently reporting a positive case was subject to restrictions. We, therefore, utilised a deterministic SEIR compartment model with four clinical stages of infection —susceptible, exposed (infected but not infectious), infectious and removed (recovered and subsequently immune, or deceased).

At the baseline transmission rate (β_0) with the parameter β_1, which controls transmission rate decline, the susceptible population (S^f) was modelled to become infected at the rate β_t, where

$$\beta(t) = \beta_0 exp\{-\beta_1 t\},$$

which was used to control the rate of infection,

$$\frac{dS^f(t)}{dt} = -\frac{\beta(t)S^f(t)}{n^f} I^f(t).$$

The model assumed a latent period, $1/\kappa$, so the exposed population was determined as,

$$\frac{dE^f(t)}{dt} = \frac{\beta(t)I^f(t)S^f(t)}{n^f} - \kappa E^f(t).$$

The infectious period was denoted as $1/\delta$, therefore the infective population is,

$$\frac{dI^f(t)}{dt} = \kappa E^f(t) - \delta I^f(t),$$

Here, $p_l, l = 0,1,2,3$ reflected the proportion of cases receiving tests, varying across different phases of the intervention strategy. It was separated by time points denoted by shortages in testing kits, resuming where possible to normal testing procedures, and increases in testing capacity to 'clear' cases in dormitories.

The recovered were represented as,

$$\frac{dR^f(t)}{dt} = \delta I^f(t).$$

We assumed a testing capacity per day (ω) with a positive identification testing rate (φ). Therefore, the average number of daily dormitory cases was,

$$E[N_t^f] = \begin{cases} \omega & \text{for } I_t^f > \omega/\varphi, \\ \varphi \times I_t^f & \text{for } I_t^f \leq \omega/\varphi \end{cases}$$

with a Poisson observation model,

$$N_t^f \sim Pois\left(E\left[N_t^f\right]\right).$$

ICU utilisation: To model the number of ICU cases (N_t^{ICU}), we also utilised a Poisson distribution. The risk of imported cases, general community cases and foreign worker cases becoming ICU cases were denoted as r^c, r^i, r^f, respectively. Thus, the expected number of ICU cases on day t was,

$$E[N_t^{ICU}] = \sum_{\tau=t-1-d^{ICU}}^{t-1} r^c N_\tau^c + r^i N_\tau^i + r^f N_\tau^f$$

where the risk of needing ICU treatment changed across time due to increasing mild COVID-19 diagnoses, thus, $r^k = a^k \times \exp(-b^k \times t)$ for $k = \{c, i, f\}$.

3.3.3 Model Fitting

Model parameters were estimated using Metropolis-Hastings (MH) algorithm Markov Chain Monte Carlo (MCMC) method where non-informative priors were assigned to each parameter. A rolling window was utilised with a time index from the January 19 until time t for projections. The parameter t included 365 days in total, from 1 March 2020 to 28 February 2021, where the addition of each day led to an additional value being available for prediction in the likelihood function. The average of all predicted values was taken for each day for projections across time to create incremental daily updates with out-of-sample validation (described later) and a two-week projection rolling window. Each of these simulations was run with 10,000 iterations with a burn-in of 1000.

We explored the parameters for general community cases, imported cases and foreign worker cases as Θ^c, Θ^i, Θ^f, and Θ^{ICU}, where

$\Theta^c = \{a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7, d^c\}$,

$\Theta^i = \{\rho_0, \rho_1, \rho_2, \mu_1, \sigma_1\}$,

$\Theta^f = \{\beta, \kappa, \delta, E_0, \varphi, \omega\}$,

and

$\Theta^{ICU} = \{r^c, r^i, r^f, d^{ICU}\}$.

For general community cases, the posterior distribution was defined as,

$$P(\Theta^c | N^c_{1:t}, M_{1:t}) \propto P(N^c_{1:t} | \Theta^c, M_{1:t}) P(\Theta^c)$$

where $M_{1:t}$ reflected the CMR dataset with six location types from time index 1 to t. For projections of future case counts, we utilised changes in CMR to determine the risk of case increase,

$$m^s_t = l^s_0 + l^s_1 f_t$$

where $f_t \sim N(\mu_t, \sigma)$, $\mu_t = a_0 \frac{b^c}{\Gamma(\mathrm{b})} t^{b-1} e^{-ct}$, and $s = \{R, T, P, G, W, RR\}$ represented the different types of locations in CMR. The projection of mobility changes were shown in Figure 10.

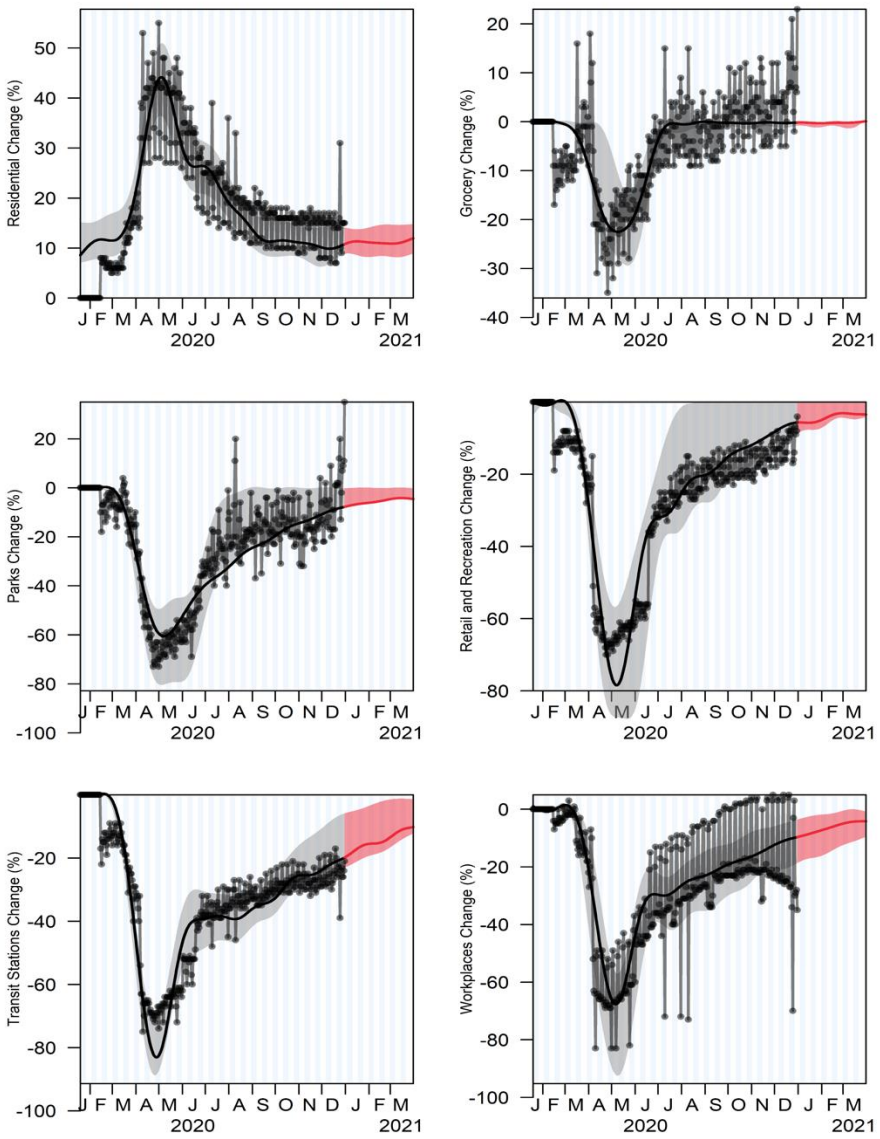


Figure 10. Estimations and projections and projections of 6 types of mobility changes from baseline. The grey polygons represent the 95% CI in the estimation period and the pink polygons are the equivalent for the projection period. The black points-line represent the observed google mobility changes.

The equivalent posterior distributions for imported cases and foreign worker cases were defined as,

$$P\left(\Theta^j \middle| N_{1:t}^j\right) \propto P\left(N_{1:t}^j \middle| \Theta^j\right) P\left(\Theta^j\right), \qquad j = \{i, f\}$$

and active ICU cases as,

$$P\left(\Theta^{ICU}\middle|N_{1:t}^{c}, N_{1:t}^{i}, N_{1:t}^{f}, N_{1:t}^{ICU}\right) \propto$$

$$P\left(N_{t}^{ICU}\middle|\Theta^{ICU}, N_{1:t}^{c}, N_{1:t}^{i}, N_{1:t}^{f}\right)P(\Theta^{ICU}).$$

To obtain posterior draws, we utilised the symmetric proposal distribution, and the acceptance probability was obtained through the MH algorithm.

Projections of cases by origin with the projection window T days were generated from the posterior predictive distribution, derived by sampling from the posterior distribution of the parameters for each submodel and simulating the model beyond the last observation, as follows:

Step 1. **Sample parameter.** A parameter vector Θ^{j} was drawn from the posterior distribution of the parameters given the information to time t, i.e., $P\left(\Theta^{j}\middle|N_{1:t}^{j}\right)$ for $j = \{i, f, c, ICU\}$.

Step 2. **Generate projected data.** Starting with one step ahead ($k = 1$), we calculated the mean for the number of new cases, $E[N_{t+k}^{j}]$, based on the estimated models described in the Modelling section above. For each type of infection j, N_{t+k}^{j} was then drawn from the Poisson distribution given $E\left[N_{t+k}^{j}\right]$. This was repeated, incrementing k from 1 until $k = T$.

Step 3. We repeated steps 1 and 2 until a large enough sample was generated.

After generating the model fits and posterior predictive distribution, we smoothed the quantiles using generalised additive models in the 'mgcv' package in R (243) for visual clarity.

For out of sample validation, projection errors were measured as the difference between the projections and the corresponding reported case counts across the period of 1 March 2020 to 31 December 2020 for all infection types and active cases in ICU, and then averaged over the one-week window following the time the projection is made to get the mean prediction error.

All statistical analysis was performed by R, version 3.6.0 (207). Source codes are available at https://github.com/syxh/HeterogeneityProjection. The COVID-19 case projections with 90 days ahead as of 31 December 2020 was provided as a worked example.

3.4 Results

A total of 58 599 cases were reported in 2020. Peak numbers of imported, general-community and foreign worker cases occurred in March 2020, March 2020, and April 2020, respectively (Figure 11A and B). Over this time horizon, 93% of cases were among the foreign worker community. On 7 April 2020, in response to the surge of cases in all three groups, Singapore started an 8-week lockdown, following which cases in the general community fell, from 38 general community cases on 7 April to four on 2 June 2020. Due to strict border control, there were almost no imported cases during this time. In the initial period of confinement, the daily number of foreign worker cases rose rapidly, reaching 1401 cases on April 20, before dropping to 540 cases on June 2, at which point the rest of Singapore emerged from lockdown, though foreign worker dormitories remained isolated as outbreak areas until July–August, when they were finally declared ‘clear’ of the virus. From June 2, a staged process of relaxation of control measures was introduced, and from then to 31 December 2020, spread in the general community was effectively controlled by

the intervention policies, with an average of 3 cases per day (range: 0–24; Fig Figure 11A). From mid-July, when the last foreign worker dormitory was 'cleared' of SARS-CoV-2, the number of cases has remained low (on average 10 per day, range 0–98, Figure 11B).

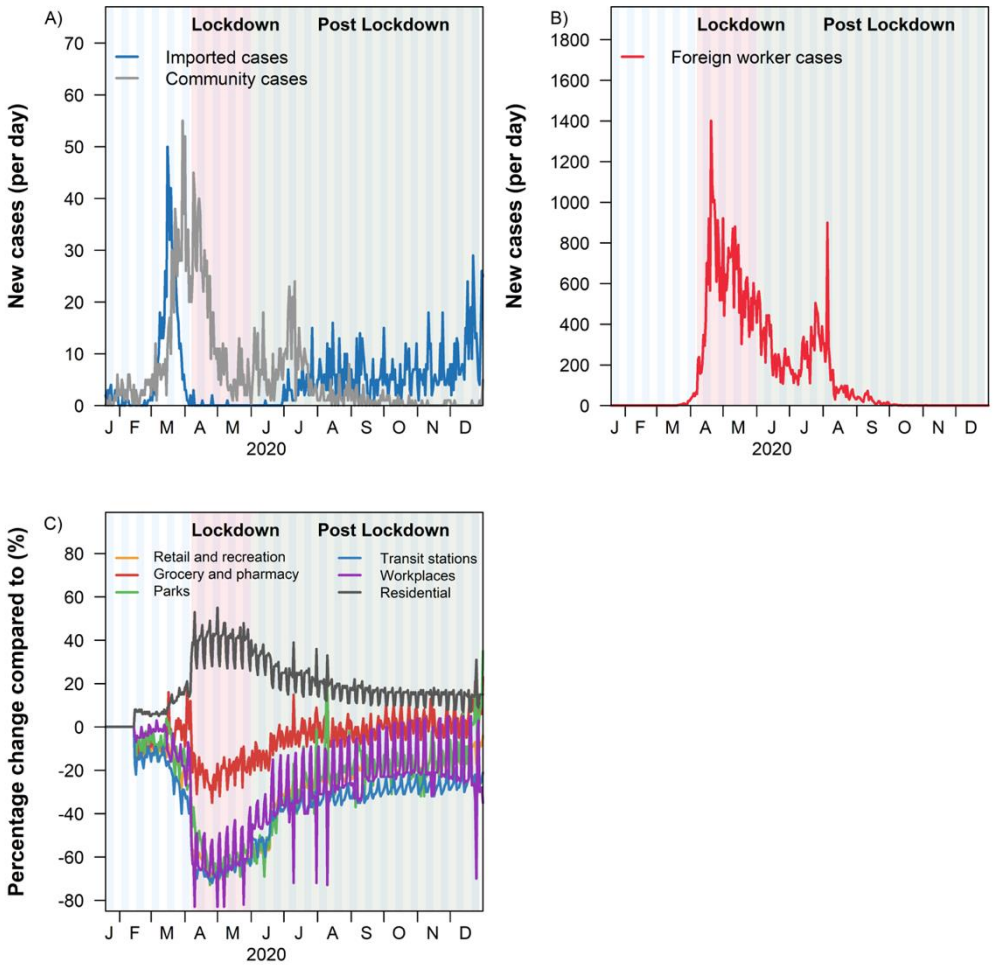


Figure 11. Description of COVID-19 cases in Singapore as of November 13. A) imported and general community cases; B) foreign workers cases. Note the difference in scales; C) the overall trend of google mobility changes from 15 February 2020 to 31 December 2020.

As in many settings, the lockdown policy in Singapore greatly impacted residents' lifestyle, with more time spent at home, and less time elsewhere, which when combined with other policies such as the test, isolate, trace and quarantine programme, was able to contain the infection. We show the overall

trends of Google mobility changes (239) from 15 February 2020 to 31 December 2020 in Figure 11C. Changes to time spent in residential locations showed an n-shaped curve that peaked at +55% on 1 May 2020 in the midst of lockdown. Time spent in public transport, retail and recreation, parks, and workplaces fell from the start of the COVID-19 outbreak, with a u-shaped response that reached a trough in April–May 2020. A rush to stockpile before border closure with Malaysia and general lockdown was evident in visible spikes in grocery and pharmacy store visits on 17 March and 6 April 2020, respectively.

Model performance was assessed through specific forecast windows for all types of infection and active ICU cases (Figure 12). Projections were affected by the initial scarcity of information in data in the early period of the pandemic or updating policies, and the resulting uncertainty in the parameters. For instance, algorithms overestimated the number of foreign worker cases and the number of people in ICUs and underestimated general community cases from the information before lockdown. From mid-April, predictions of general community cases began to forecast with reduced error at an average projection error from –22.0 to 7.0 cases per day (Figure 13A). The projected number of general community cases fell from the end of April, mirroring the fall in the data, followed by a slight rise in mid-July when a general election took place, before declining (Figure 12A). For imported cases, our algorithm correctly characterised the low levels of importations during strict border control from April to June, and the increment in cases with the relaxation of border control from July (Figure 12B). The forecast of infection among foreign workers was accurately declining in general due to intervention policies in place. However,

the projection error of foreign worker cases increased before lockdown and after the clearing of dormitories with mass testing due to rapid and substantial increases in testing capacity (Figure 12C and Figure 13C). Among the projected number of active ICU cases, projections stabilised from July where projection errors ranged from 0 to 2 bed per day (Figure 13D). Projections at other times are presented in Video Files S1, S2, S3 and S4 of general community cases, imported cases, foreign workers cases and ICU utilisation, respectively (https://github.com/syxh/HeterogeneityProjection). Across time, on average 89.0%, 76.1% and 44.8% of reported general community cases, imported cases and foreign worker cases fell within the 95% Credible Interval (CI) of the estimations, respectively, with the equivalent of 79.0% for ICU cases.

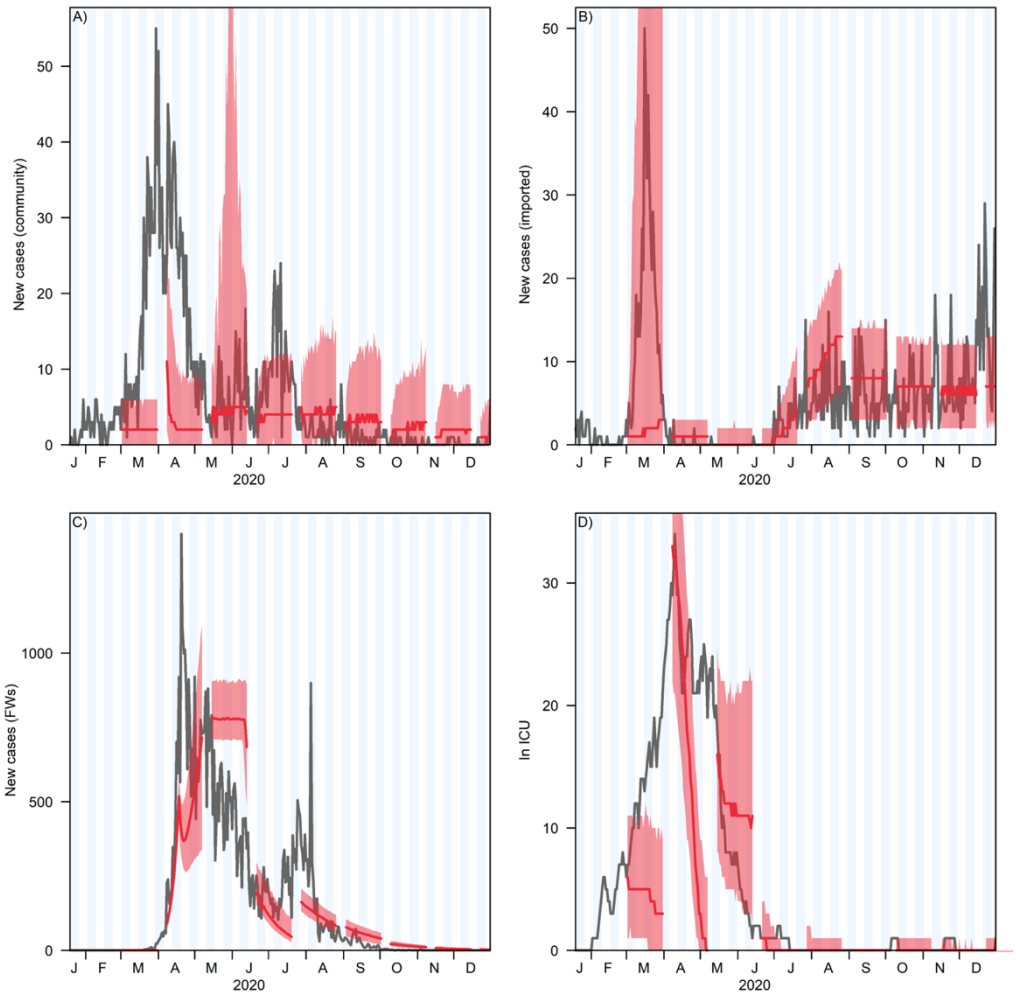


Figure 12. Evaluation of projections with 30-day rolling windows. The panels show A) general community cases, B) imported cases and C) foreign worker cases. Panel D) shows the active number of ICU cases. In each panel, the redlines represent model-based projections, and the red polygons represent corresponding 95% credible intervals (95% CI). Each segment of the projected data in red polygons represents the estimates from one 30-day forecast made at a previous point in time.

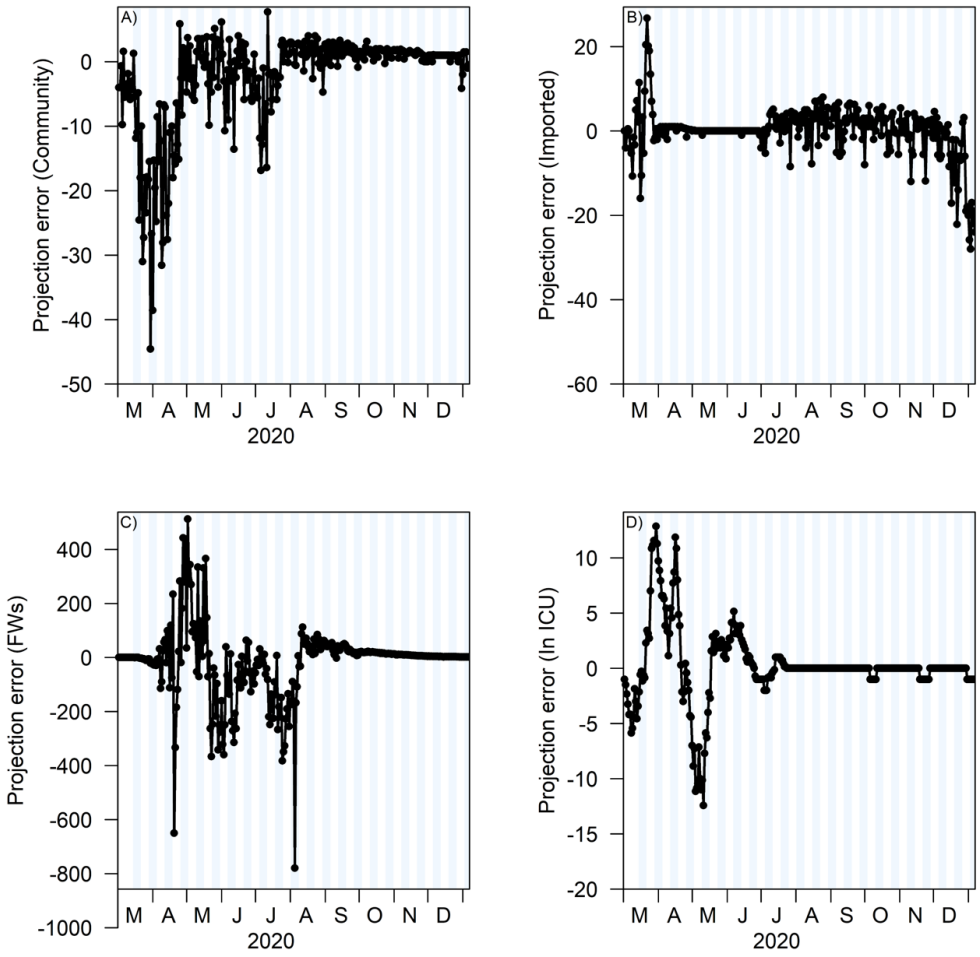


Figure 13. Quantification of projection error. The projection accuracy across time by aggregation of daily projection models.

We presented a 90-day projection from 31 December 2020 by infection type (Figure 14) and the cumulative cases for this time window (Figure 15). The case projections in the general community showed a minimum of 1 case to a maximum of 4 cases per day (range of 95% CI: 0–16, Figure 14A), reaching to 1908 (CI:1526–2967) general community cases 90 days ahead (Figure 15A).

With the relaxation of border control policies, case projections of importation have since stabilised at an average of 8 (range of 95% CI: 2–14) cases per day (Figure 14B), leading to 2493 (95% CI: 2336–2647) imported cases in total (Figure 15B). Among foreign workers, the maximum projection for this time window was on average 1 case per day (range of 95% CI: 0–6, Figure 14C). The total number of cases in the dormitories was expected to reach 54,324 (95%CI: 52,910–61,564) after 90 days from 31 December 2020 (Figure 15C). Projections of ICU utilisation from September were expected to decrease even further and stabilise at ~0 case (range of 95% CI: 0–1, Figure 14D).

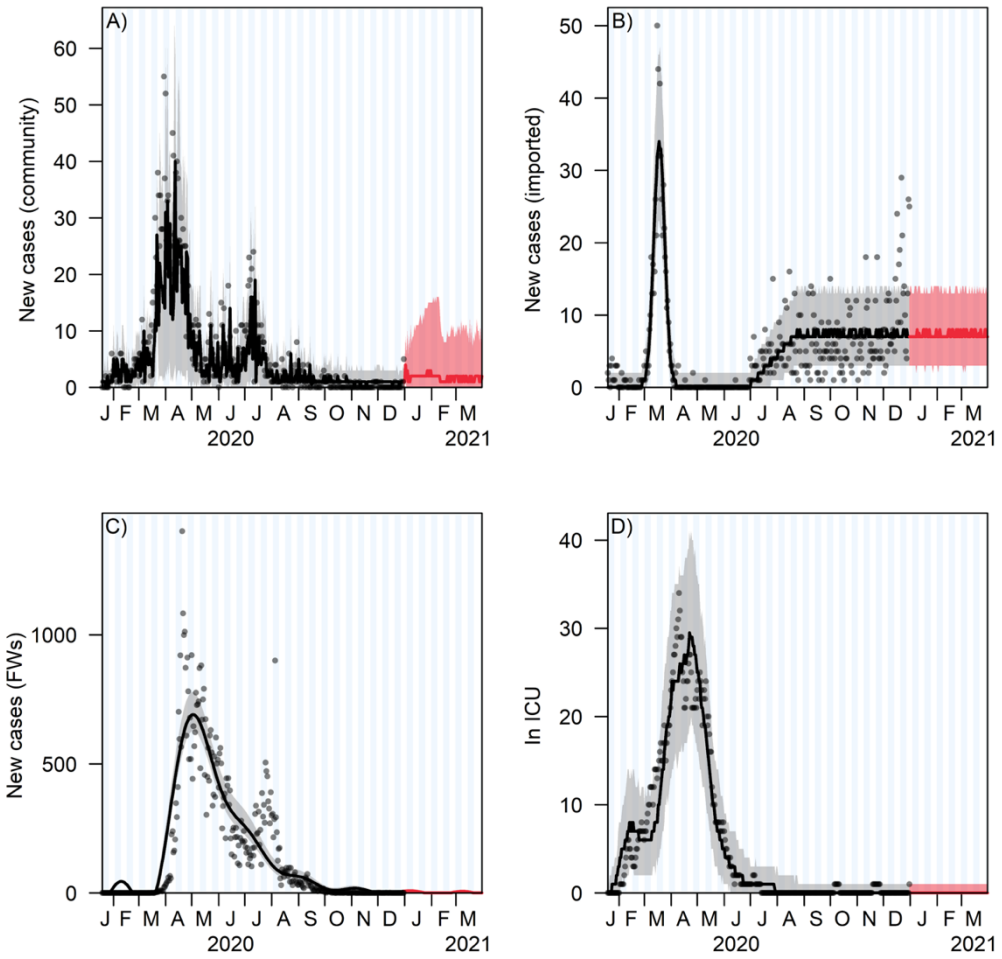


Figure 14. 90-day hindcast from 31 December 2020. Daily number of cases by the origin of infection where the panels show A) general community cases, B)

imported cases, and C) foreign worker cases. Panel D) shows the active number of ICU cases. The grey polygons represent the 95% CI in the estimation period and the red polygons are the equivalent for the projection period. The points represent the observed case data.

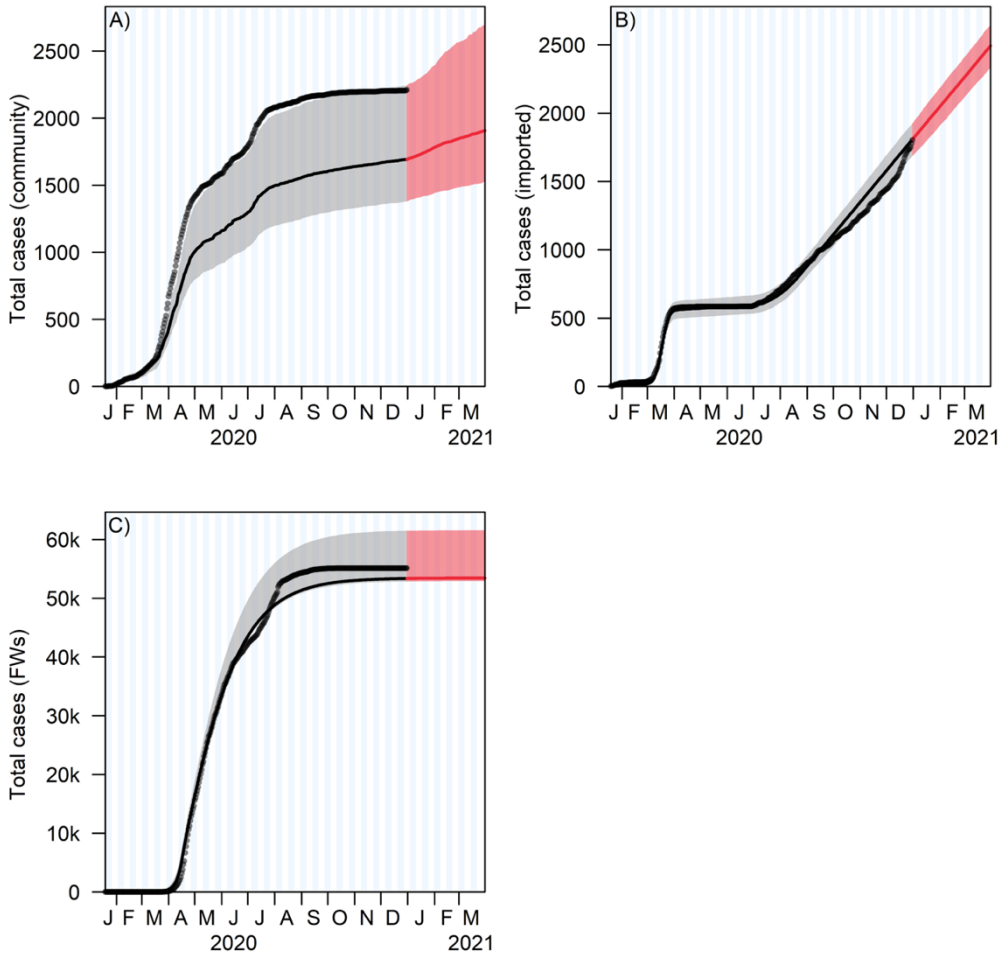


Figure 15. 90-day hindcast of cumulative cases from 31 December 2020. The panels show total number of A) The total number of community cases would be increased but should be under control within first 90-days in 2021, according to current projection of mobility changes. B) the number of imported cases would be increased to 1,850 in total under border control policies of December 2020, and C) the total number of foreign worker cases would be stabilised at ~54,300 cases in the first three months of 2021. The grey polygons represent the 95% CI in the estimation period and the red polygons are the equivalent for the projection period. The points represent the observed case data.

3.5 Discussion

Groups with different social characteristics have shown significant heterogeneity in the spread of SARS-Cov-2 in Singapore (244). The rapid response of intervention policies within the community succeeded in limiting transmission rates to low levels (245), preventing a widespread and national outbreak. With herd immunity unlikely to occur at high level of case suppression, the use of traditional SEIR models may be inappropriate to represent this disease state (246). Our model instead used community mobility data, representing the movement of residents in the general community, as a proxy of the magnitude of disease spread (237). It showed good performance in the fitting and projections for community cases, which has also been observed elsewhere (33).

Foreign workers, in contrast, experienced an epidemic in comparison (136) due to high-density housing conditions, insufficient sanitation and the daily mixing of transient and regular contacts from other dormitories at work sites before COVID-19 pandemic. Starting from April 2020, the transmission rate within dormitories was observed to be relatively high in Singapore with herd immunity developing across multiple dormitory sites. SEIR models have been widely used in high-density environmental settings such as prisons and nursing homes as they are relatively small populations with high contact rates, as observed in a US prison where a COVID-19 outbreak had an estimated basic reproduction number of 8.44 (95% CI: 5.00–13.13) in the early part of the outbreak (247). The dormitory model within this study utilised testing rates (248) when available to improve predictions, substantially improving fitting, which has

wider applicability for other SEIR models operating within semi-closed environments with nonpharmaceutical intervention policies in place.

For border importation estimation, shifting control policies, changing travellers flows and the occurrence of overseas epidemics can substantially complicate projection modelling. Studies (249,250) have estimated imported cases by prevalence in source countries and flight volumes, providing short term predictions. However, such models require large datasets and face complications such as cross-country prevalence differences due to varying national public health practices, data collection issues and political containment decisions. We developed a relatively simple model to project imported cases using historical daily imported case data and border control policies over time, assuming for Singapore that border control measures would remain constant for the foreseeable future.

In estimating ICU utilisation of COVID-19, rates may vary by age group and physical condition (232,233). In Singapore, as foreign worker cases have comprised 93% of all cases, they are generally young adult males in good physical condition (251,252). Thus, the need for ICU has not been regarded as a major issue. One study forecasted ICU utilisation, accounting for characteristics of the subgroup population through patient-specific data, including the length of stay and demographics (253). It however required real-time patient-level data to ensure forecast accuracy. Other studies (254) forecast ICU utilisation by compartmental models. The significant limitation of these models is that they assumed a homogeneous proportion from infected individuals to ICU, which however not applicable in reality. Our model

projected ICU utilisation based on the number of cases reported daily by the heterogeneous severity caused by the type of infection.

Several limitations, however, exist in our estimations. Firstly, our model should not be utilised as an early outbreak detection warning system as the introduction of new variants such as B.1.617 variant (255,256) can cause substantially different outbreak dynamics to occur. Secondly, we assumed that mobility was directly related to the general community cases and the primary driver of the fluctuations where policy factors and adherence to control measures may factor in this relationship. Thirdly, changes in border control policies can also cause projections to differ from observed case numbers. Thus, our projection model requires knowledge of upcoming policy changes to reflect transmission spread. However, it is able to provide evidence for policymakers to observe the long-term effects of ongoing policies. Fourthly, we do not account for any interactions between the different infection groups, which are likely to exist. This may be minimised within dormitories under strict isolation measures, and for border control which has strict testing and quarantining measures in place. Lastly, the asymptomatic status or age structure of individuals was not accounted for as identifying the former is challenging, and age-structured mobility data is required for the latter for community projections.

Despite these limitations, our methodological framework and findings can be used for policymakers in countries with heterogeneous transmission communities, such as the notable outbreaks being observed in large homeless populations (182), nursing homes (178,179,257), MWs (136,181), prisons (185–187,258), slums (220), and the arrivals at borders, to better understand containment policies through projections. We also recommend that countries

can record case registration by source of infection to better understand transmission.

3.6 Motivation for Next Chapter

This chapter presented methods for projecting epidemic trends in the general community, foreign worker dormitory and importation groups, respectively, as well as the projection of active ICU cases, during the period from the onset of the COVID-19 pandemic to the widespread implementation of the vaccination programme. This suggests that recording cases by source type not only benefits academic research, but also that a more comprehensive COVID-19 surveillance system is more effective in helping policymakers create targeted outputs due to subgroup heterogeneity. In Singapore, most cases in the early COVID-19 pandemic were from foreign worker dormitories which are high-risk closed settings. Understanding the transmission within dormitories is critical when reopening during the post-pandemic period. In chapter 4, a Bayesian hierarchical model was implemented to characterise the nature of the contact network of foreign workers in the post-COVID-19 restriction period. In addition, a realistic branching process model was developed using empirical contact networks to explore the impact of policies that used varying degrees of stringency in quarantine to mitigate spread. The results are essential for mitigating emerging infectious disease outbreaks in high-risk closed settings, such as COVID-19.

CHAPTER 4. USE OF BLUETOOTH CONTACT TRACING TECHNOLOGY TO MODEL COVID-19 QUARANTINE POLICIES IN HIGH RISK CLOSED POPULATIONS

4.1 Abstract

Containment measures in high-risk closed settings, like migrant worker dormitories, are critical for mitigating emerging infectious disease outbreaks and protecting potentially vulnerable populations in outbreaks such as COVID-19. The direct impact of social distancing measures can be assessed through wearable contact tracing devices. Here we developed an individual-based model using data collected through a Bluetooth wearable device that collected 33.6 million and 52.8 million contact events in two dormitories in Singapore, one apartment-style and the other a barrack-style, to assess the impact of measures to reduce social contact of cases and their contacts. The simulation of highly detailed contact networks accounts for different infrastructural levels, including room, floor, block and dormitory, and intensity in terms of being regular or transient. Via a branching process model, we then simulated outbreaks that matched the prevalence during the COVID-19 outbreak in the two dormitories and explored alternative scenarios for control. We found that strict isolation of all cases and quarantine of all contacts would lead to extremely low prevalence but that quarantining only regular contacts would lead to only marginally higher prevalence, but substantially fewer total man-hours lost in quarantine. Reducing the density of contacts by 30% through the construction of additional dormitories was modelled to reduce the prevalence by 14% and 9% under smaller and larger outbreaks, respectively. Wearable contact tracing devices

may be used not just for contact tracing efforts but also to inform alternative containment measures in high-risk closed settings.

Word counts = 238

4.2 Introduction

The coronavirus disease 2019 (COVID-19) pandemic has been the first in which digital contact tracing (DCT) tools have been widely deployed as part of several national responses, including in Australia (259,260), China (261), several states of the US (262) and multiple European countries including France (263), Germany (264), Ireland (265), Italy (266), the Netherlands (267), Switzerland (268) and the United Kingdom (269). The ability to rapidly contact trace can assist in timely case identification, allowing for better case management and disease spread control by notifying individuals that they may be infected at an earlier stage (270). The magnitude of outbreak peaks can also be reduced if, following such alerts, infected individuals modify their behaviour to reduce social interactions or to isolate themselves either before they become infectious or at an earlier stage of their infectious period. Early detection is especially important in closed or semi-closed populations—such as residential care homes, boarding schools, military camps, prisons and dormitories—as the dense social network structure in such settings lends itself to explosive outbreaks, as has been witnessed both during COVID-19 (271,272) and before (273,274). Contact networks in such settings are characterised by a high degree of connectivity between individuals within the subpopulation and fewer connections beyond the subpopulation than members of the general population have.

Singapore was the first country to roll out DCT tools with the launch of TraceTogether in March 2020, days after the WHO declared COVID-19 to be a pandemic (275). TraceTogether came in the form of a physical token and smartphone app, evolving through multiple versions to monitor access to key areas, proximity to others and testing status of individuals. At the beginning of the pandemic however, an initial form of TraceTogether under the name of *BluePass* device (BP), comprising of a physical token only, was rapidly deployed to MWs who work in heavy industries and construction as they typically live in high density dormitories. These dormitories observed large outbreaks during Singapore's lockdown in the early days of the pandemic—during which 56% of the residents had documented infections (242,276). These BP were to be carried when inside and outside their dormitories (277), were designed to be sufficiently durable to withstand use at worksites, and have a unique identifier which allowed for interaction events between devices to be tracked for the quarantining of close contacts of cases. After dormitory residents emerged from lockdowns in July and August 2020, they were subject to long-lasting restrictions on their movement, only being allowed to communicate with others that stayed in the same dormitory or worked in the same location to prevent further outbreaks.

In this study we pursue two aims, taking advantage of the high volume of social interaction data recorded in an extraction of around 1000 residents of two dormitories working for the same company. The first was to quantify the nature of the contact network itself, to gain insights on local heterogeneities that would possibly hinder control. The second was to use the empirical contact network to

develop a realistic branching process model to explore the impact of policies that used varying stringency of quarantine to mitigate spread.

4.3 Materials and Methods

Migrant workers and their dormitory. There are two types of MWs in Singapore; 1) those working in construction, marine shipyard, and processing industries (CMP) who are assigned to 43 licensed purpose-build dormitories and 2) those migrant domestic helpers residing predominantly in employer's homes (278). In this study, we focus on the CMP MWs staying in dormitories, thus, we refer to MWs as CMP MWs. Licensed purpose-build dormitories do not have individual bedrooms but instead share. In our study site, each bedroom typically has a capacity of 16 to 24 residents and come in two designs. The first is a traditional apartment style, which provides facilities such as showers, toilets, and kitchens within each room unit. There are 8 room units per floor 5 floor levels in each dormitory block, and 6 blocks in total. The second is a denser barracks-style, where multiple units in the same floor share a communal shower room, toilet, and kitchen, and there are 12 room units located on the same floor. Each dormitory block has 6 floors, and 3 blocks in total.

From the early outbreak of COVID-19 to re-opening orders implemented in 2022, migrant worker dormitories have implemented long-lasting restrictions on MWs movement. These include all MWs being only allowed to stay in their home-dormitories and workplaces, and MWs only having contact with those in the same dormitory and at work sites. To understand the contact pattern among MWs during the peak of the COVID-19 pandemic period, we therefore selected MWs who were working for the same company but stayed in two dormitories

to observe the impacts of dormitory and worksite contact. Although the two dormitories are 7km apart, it is expected that MWs between dormitories will have contact with each other at worksites.

Data collection in selected dormitories. Starting October 2020, all MWs in Singapore were to carry BP for contact tracing purposes (277). Each BP has a unique registration ID and MWs are required to carry their own BP at all times to track interactions (277). In February 2021, BPs were updated with improvements in design. As to the time to collect contact information, MWs were only allowed to communicate with others in the same dormitory or at workplaces, and all MWs assigned to the same dormitory should work for the same company. In collaboration with Keppel Corporation, we conducted fieldwork in two MW dormitories, Acacia Lodge (Dorm A) and Cassia@Penjuru (Dorm C), to obtain records of interactions from BP devices.

In the data collection procedure, I was responsible for poster design to recruit MWs, coordinated the collection of contact trackers, collected the contact records via BP on site, and extracted contact records from the designated server. I also helped to set up the necessary field equipment before we went to the dormitory to collect the contact records.

We recruited 1611 MWs using convenience sampling, which is a non-probability sampling method to obtain sample from a group of people who are easily to contact or to reach, with approximately equal numbers in each dormitory, and after obtaining verbal informed consent, extracted their records of interactions for the period of February to April 2021. Study participants also filled in a brief questionnaire, which we used to collect some demographic

information such as their age, country of origin, and address within the dormitory. No explicitly identifying information was recorded, but BP device ID was included to link with. The questionnaire was translated into 6 languages, including English, Bangladesh, Tamil, Hindi, Burmese and Chinese. The English version of questionnaire was attached in **Appendix 1**. The study was approved by NUS-IRB-2021-92.

Each BP sends a signal via Bluetooth technology every 7 minutes, which detects and records BPs within a ~3m radius. Records of interactions were stored within the device for up to nine weeks with a theoretical maximum of 8 million records.

Classification of contact type. Because of the ongoing restrictions at the time of the study, MWs interacted relatively constrainedly, with workmates and roommates typically having longer and more frequent interactions, with interactions between individuals living on different floors or in different blocks being more random and of shorter duration. We therefore classify contacts into two distinct categories: regular and transient. A regular contact of an individual is defined to be their contact with whom the two individuals have interactions at least four days a week with each interaction lasting more than 15 minutes, following Singapore's Ministry of Health definitions (279). A transient contact is defined when the criteria for the interactions do not meet this threshold (Figure 16).

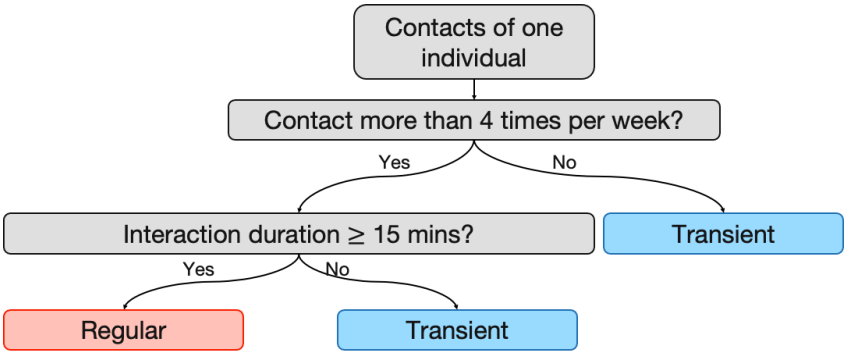


Figure 16. Definition of regular and transient contacts

Description of collected contact network (G). To investigate the occurrence of interactions across time, we presented the number of interactions (in hour scale) each day of week for regular contacts and transient contacts. To understand the exposure between MWs, the mean contact duration per day between two MWs were calculated. We also presented it in violin plots We then summarised the mean number of contacts for regular and transient contacts across the study period, and the mean contact duration per day among all MWs. The exposure time of each individual was formulated as the number of contacts that the MW had multiplied by the duration of interactions (hrs) and is measured as man-hours.

Simulation of the contact network overlaying the two dormitories ($\mathcal{G}$). Pairs of individuals were classified according to their proximity in the structure of the dormitory, defined by sharing the same room, a different room of the same floor, a different floor of the same block, or a different block in the same dormitory. We let $\phi_h = \{\phi_{h_i}\}_{i \in I}$ denote the probability of a pair of proximity h setting (where h represents a distance-type, that is room, floor, block and dormitory; and I corresponds to the total number of rooms, floors, dormitories accordingly) being classified as regular. We listed ϕ_h in Table S2. A Bayesian hierarchical

model for each degree of proximity was constructed as follows: Let y_i denote the observed number of pairs of regular contacts in the i-th group (in a specific degree of proximity) and N_i the total number of pairs of possible contacts within the i-th group. The joint posterior of $\phi_h, \alpha_h, \beta_h$ is therefore

$$p(\phi_h, \alpha_h, \beta_h|y) \propto p(y|\phi_h, \alpha_h, \beta_h)p(\phi_h|\alpha_h, \beta_h)p(\alpha_h, \beta_h)$$

$$= \prod_{i \in I} \phi_{h_i}^{y_i} \left(1 - \phi_{h_i}\right)^{N_i - y_i} \frac{\Gamma(\alpha_h + \beta_h)}{\Gamma(\alpha_h)\Gamma(\beta_h)} \phi_{h_i}^{\alpha_h - 1} \left(1 - \phi_{h_i}\right)^{\beta_h - 1} p(\alpha_h, \beta_h),$$

where α_h and β_h are the hyperparameters of the beta distribution associated with the degree of proximity h. The Metropolis-Hastings Markov Chain Monte Carlo (MCMC) algorithm was used to sample from the joint distribution. We ran a single MCMC chain with run length of 100,000, with the first 10,000 samples discarded as burn-in. Geweke's convergence diagnostic was calculated for each chain. To simulate our regular contact network ($\mathcal{G}_R$), we built an adjacency matrix by drawing from binomial distributions using ϕ_h, obtaining from the collected contact data, and information on the structure of the dorms (Table 6), sequentially filling in the number of contacts from the draws starting from the room level up until the dormitory level.

Table 6. Overview of dataset

	Dorm A	**Dorm C**
Layout distribution		
Number of blocks	6	3
Number of floors in each block	5	6
Number of rooms on each floor	8	12
Number of MWs in each room	20	16

Total number of MWs	4,800	3,456
Overview of collected data		
Number of participants	815	796
Number of participants with completed data entries*	520	491
Total interaction records included in the study	33.6M	52.8M

*: data entries include BP data extraction and questionnaire filling

The same process was repeated for transient contacts, albeit with different estimates of the corresponding ϕ_h s. The network of transient contacts is represented as $\mathcal{G}_T$.

Model structure and inference. The framework of the infection model assumes that physical interaction among MWs led to spread events via a branching process model (Figure 17).

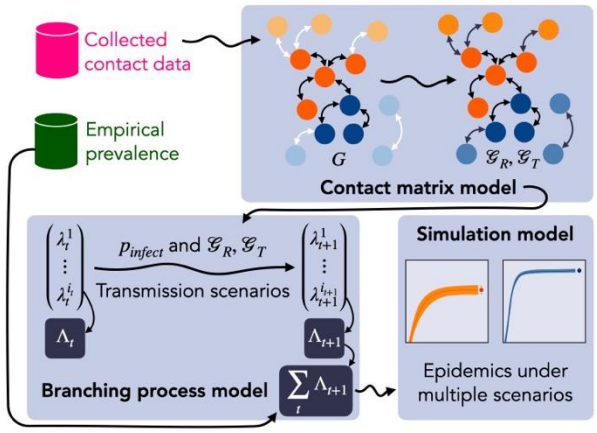


Figure 17. Model Structure. Here, G and $\mathcal{G}$ represent the collected contact network, and the simulated contact network, respectively. With the simulated contact network, we used an SIR-style branching process model to simulate daily cases (Λ_t) from day 1 to day 120 under different infection rates (p_{infect}), since the first wave of the outbreak in Singapore dormitories was about 4 months. We then looked for infection rates by individually matching cumulative cases simulated for different infection rates with reported prevalence. Thereafter, we simulated the epidemic scale of multiple intervention policies under these two scenarios by SIR-style branching process model, including a

smaller outbreak (matching what was observed in Dorm A) and a larger outbreak (Dorm C).

We considered four possible transmission scenarios (Figure 18). With infected i as each infected individual, we assume that the MW's contact data was extracted on the first day of being symptomatic as part of routine surveillance when cases are identified. Let us denote his contagious interval as τ and his duration of delay to be isolated as d ($d \sim Weibull(k, \lambda)$ (with more details in Table 7).

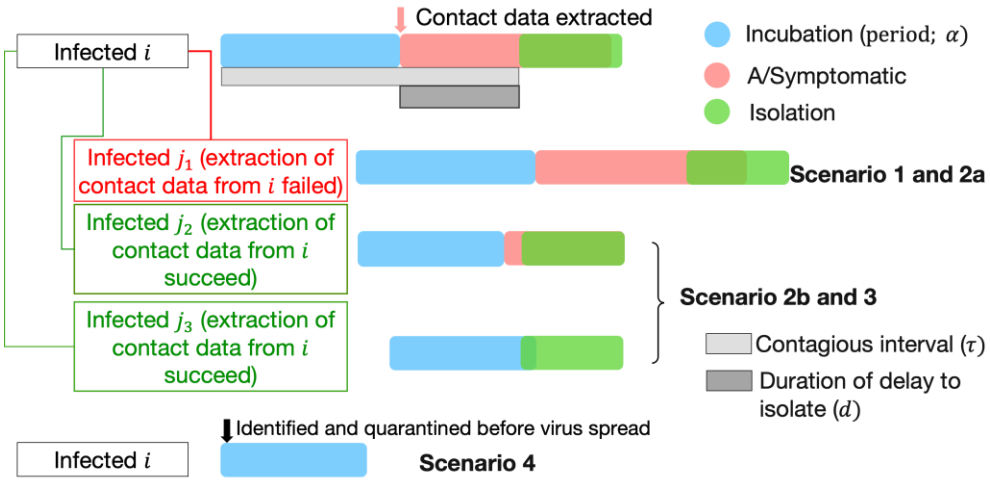


Figure 18. Transmission scenarios from infected i.

Scenario 1. No matter whether infected i was a symptomatic case or not, the extraction of contact data was not successful from infected i:

Although j_1 was infected by i, he was only isolated when he became symptomatic. This is expected to cause largely unknown subsequent infections. Here, $\tau = 14$.

Scenario 2. When infected i was an asymptomatic case, the extraction of contact data was successful from infected i:

> ***Scenario 2a.*** Infected i was hidden. His contact j_1 was only isolated when he became symptomatic. Here, $\tau = 14$.
>
> ***Scenario 2b.*** Although infected i was asymptomatic, he was identified by the mass testing program in dormitory named Rostered Routine Testing (RRT) (280). Thereafter his contacts were identified via the BP. Here, $\tau = min\,(14, \alpha + d)$.

Scenario 3. Infected i was a symptomatic case and his contacts were identified successfully via BP. Here $\tau = min\ (14, \alpha + d)$.

Scenario 4. Infected i was an asymptomatic case where he was identified from RRT, and he is isolated before he infects other MWs. Here, $\tau = 0$. This is expected to occur within the early outbreak where isolation facilities are able to accommodate for the numbers of new infections.

Table 7. Epidemiological parameters

Item	**Description**	**Statistics**
Incubation Period (α)	$D_{incubation} \sim Gamma(\alpha = 10.8, \beta = 0.51)$	$mean = 5.5$ $standard\ deviation = 2.8$
Delay to Isolation (d)	The time period from the (pre-) symptomatic to isolation, $d \sim Weibull(k, \lambda)$ Where $k = 10, \lambda = 1$ (scenario 1,2,3 in table S3); $k = 10, \lambda = 3$ (scenario S1,S2,S3 in table S3)	$mean = \lambda\Gamma(1 + \frac{1}{k})$ $standard\ deviation = \lambda\sqrt{\Gamma\left(1 + \frac{2}{k}\right) - \Gamma\left(1 + \frac{1}{k}\right)^2}$
Contagious Interval (τ)	The time period from the day of infection to the day of isolation or end of infection. Further details can be found in the branching process section	N.A.
Traced Status	There are 3 stages across time when we sample the traced status of probability weight for r_traced: 1) Low traced rate ($r_{traced} = 0.1$) in the early outbreak of COVID-19 in the dormitory due to unawareness and limited resources. Assumed for the first 10 days. 2) Medium traced rate ($r_{traced} = 0.5$) when dormitory managers noticed the importance of contact tracing in the pandemic. Assumed for day 11 to day 20. 3) High traced rate ($r_{traced} = 0.7$) when migrant workers were given the *BluePass* devices (BP). We assumed the average traced rate to be 70% as compliance and technical issues	N.A.

	may occur. Assumed for day 21 onwards.	
Symptomatic Status	We sample the symptomatic status of probability weights for 0.6	N.A.

To simulate the spread of COVID-19, a susceptible, infectious, recovered (SIR)-style model of individual based contact networks was presented, and infected individuals became immune without being able to infect others in the time of the study. Branching process modelling was then implemented to model new infections, corresponding to the matrices $\mathcal{G}_R$ and $\mathcal{G}_T$. Let I_t be the collection of individuals infected on day t. Suppose that individual i was infected on day t_0, i.e, $i \in I_{t_0}$, and that individual j is a neighbour of i (if j is either a regular or transient contact of i). Then, on day $t > t_0$, i infects j with probability $p(t, t_0)$,

$$p(t, t_0) = \begin{cases} \min(\tau, t - t_0)\theta & \text{if } i \text{ and } j \text{ regular contacts} \\ 0.009\min(\tau, t - t_0)\theta & \text{otherwise} \end{cases},$$

where $\min(\tau, t - t_0)\,\theta$ is the infectiousness of i on day t relative to his COVID-19 infection on day t_0, and θ is the infection rate parameter. We also assumed that transmission to regular contacts and transient contacts differs by a multiplicative constant. The constant value of 0.009 was selected as it represents the ratio of interaction duration between regular and transient contacts, which was obtained from the collected contact tracing data. The error of the ratio is small (95% CI of this ratio if 0.0094–0.0096).

The prevalence study among MWs in Singapore showed that the proportion of infected during the course of the first wave the was 33% for Dorm A and 66% for Dorm C (242). To determine the infection rate parameter θ that results in

this proportion becoming infected, we first performed a grid search on the parameter $\theta \in (0, 1)$. We then fit the values of θ and the simulated infection proportion to obtain the relationship between θ and the infected proportion. The runtime of each simulated outbreak was 120 days, i.e., approximating the duration of lockdown in 2020. 1,000 simulations were performed of each θ of the same contact network. The probability of becoming an initial case was assumed to be the same for each MWs, however. To make epidemic models under different θ comparable, we made all initial cases the same.

Simulation under nonpharmaceutical interventions (NPIs). Using the infection rate values corresponding to the infection proportion of 33% and 66% (242), we simulated the interventions with multiple COVID-19 measures with 1,000 simulations of the same contact network. Multiple COVID-19 measures under different intervention policies in the dormitory were listed in Table 8. We adjusted the length of delay of the cases isolation (d) and length of contact quarantine (if any) in the simulation process, including 1 day (Baseline), 3 days and 14 days (case isolation and contact quarantine policies are lifted). The type of contact to be quarantined are also varied, including quarantining all contacts, only quarantining regular contacts and no quarantine of contacts. We also considered building a new dorm to reduce contacts by 30% by splitting the entire population from two dormitories to three.

The baseline scenario was to isolate all positive cases following a 1-day delay. The first alternative scenario was to quarantine all contacts with a 1-day or a 3-day delay whilst the second alternative scenario was to quarantine all regular contacts with a 1-day or a 3-day delay, and the third alternative scenario was to lift case isolation and contacts quarantine orders. Additionally, we simulated the

epidemic size if a new dormitory was built to reduce the density of contact events in the existing dormitories. The length of isolation and quarantine was assumed to be 7 days, and the number of man-days lost to quarantine or isolation was tallied.

Table 8. Scenarios settings in our simulation studies.

Scenario	Isolation Delay Length (days)	Contact Quarantine	Only Regular Contact Quarantine	Isolation & Quarantine Length (days)	Explanation
1	1	No	No	7	Isolate cases with a 1-day delay, which is the baseline scenario.
S1	3	No	No	7	Isolate cases with a 3-day delay.
2	1	Yes	No	7	Isolate cases and quarantine all contacts with 1-day delay
S2	3	Yes	No	7	Isolate cases and quarantine all contacts with a 3-day delay
3	1	Yes	Yes	7	Isolate cases and quarantine regular contacts with 1-day delay
S3	3	Yes	Yes	7	Isolate cases and quarantine regular contacts with a 3-day delay
4	14	No	No	-	No isolation of cases, no quarantine of contacts
5	1	No	No	-	Build a new dormitory to randomly split ~30% of MWs from the two existing dormitories, other police orders are under the baseline scenarios setting.

4.4 Results

A total of 1,611 MWs' data (19.5%) was collected among the estimated 8,256 MWs in the two dorms (815 in Dorm A and 796 in Dorm C). By the time of data extraction, 600 of their BP devices had experienced device failure (mostly from running out of power); these individuals were therefore excluded from the study. The remaining 1,011 MWs with 82.7 million interaction records were included in this study Table 6).

Contact patterns from collected data

Participants were male, mostly aged 25 to 39 (60%), and most originated from South Asia: 47% were from Bangladesh, 37% from India; the remain were mostly from Myanmar, Thailand, Malaysia and China. We found no statistically significant differences in the mean number of contacts across age groups (ANOVA test p-value: 0.09) or their country of origin (ANOVA test p-value: 0.25). However, contact type (regular versus transient; Chi-square test p-value< 0.001) differed significantly in terms of the average number of contacts observed. As to each MW, the mean number of regular contacts was 12 per day and transient contact was 161 with other MWs. The mean number of contacts of each MW was not significantly different among age groups and origin country groups (Table 9).

Table 9. Characteristics of the dormitory residents within Dorm A and Dorm C dormitories.

Characteristics	Group	Number of senders	Mean Number of Contacts	P Value
Contact Type	Regular	982	12	<0.001[#]
	Transient	1011	161	

Characteristics	Group	Number of senders	Mean Number of Contacts	P Value
Age*	<24	53	175	0.09[$]
	25-29	170	187	
	30-34	238	183	
	35-39	230	172	
	40-44	161	164	
	>45	151	159	
Origin Country*	Bangladesh	471	179	0.25[$]
	China	15	137	
	India	375	172	
	Myanmar	93	159	
	Others	49	178	

: Exclude the sources of incomplete questionnaire answers (N=1003).

#: Students' t-test

$: ANOVA test

We found no evidence of age-based assortative mixing. However, there was evidence of assortative mixing by country of origin, with those originating from China, Myanmar, Bangladesh and other countries (not India) spending more time with those from their same country of origin for both regular contacts and transient contacts; Indians had assortative mixing with other Indians and Myanmar (Figure 19). We also found that MWs who had regular contacts with the same group of MWs interacted with them almost daily (81%) whereas with transient contacts, most interacted across different group of MWs, mainly on a weekly or monthly basis (

Figure 20).

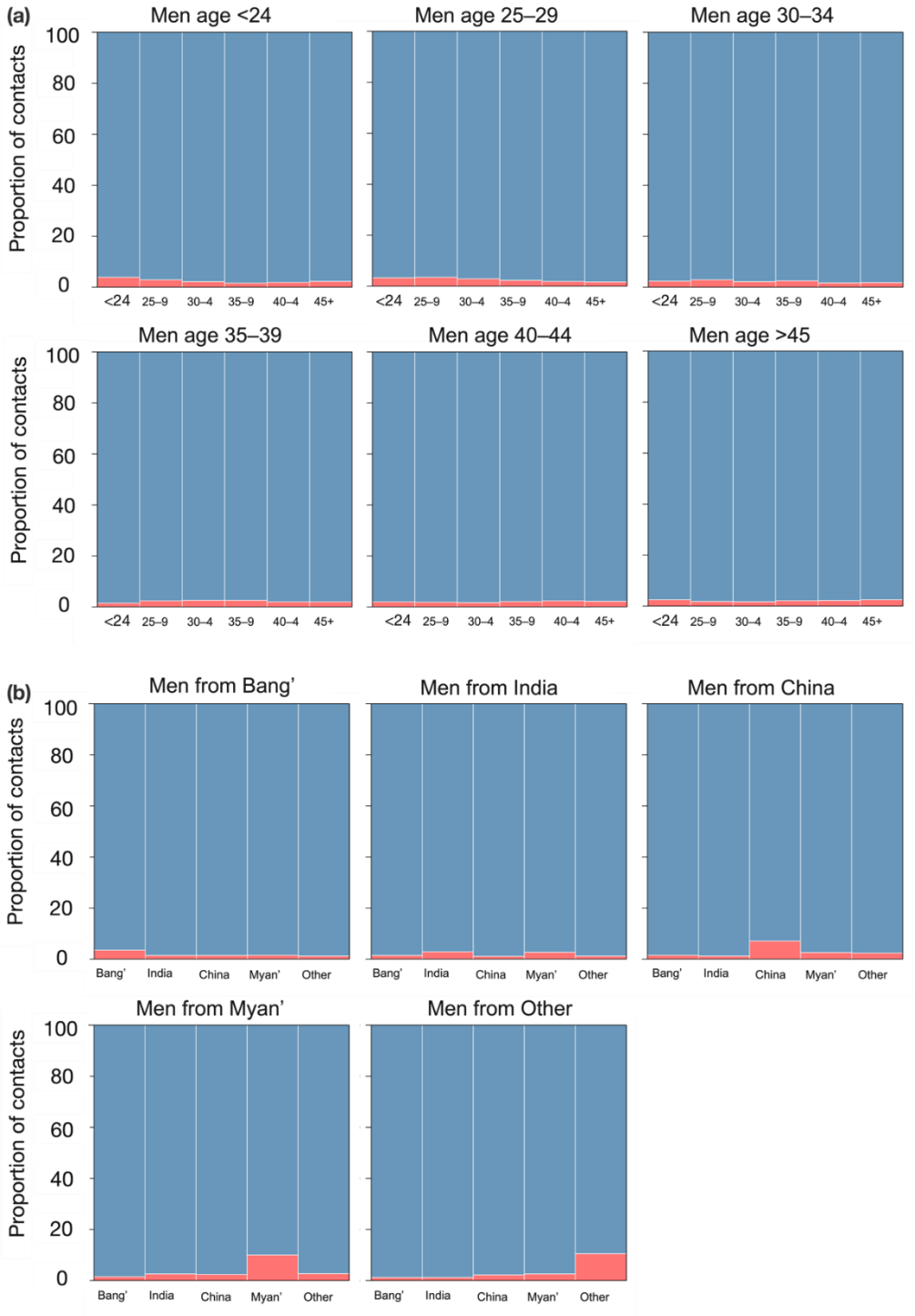


Figure 19. Demographic characterises comparison by different contact types. (a) Contact types in age groups. (b) Contact types in the original countries. The red colour indicated regular contact, and the blue colour indicated transient contact.

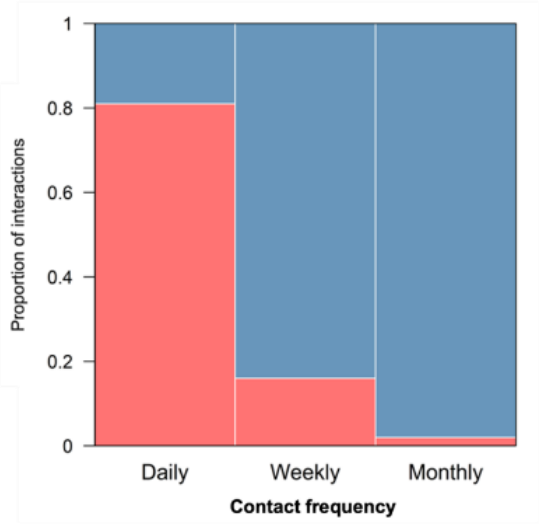


Figure 20. Overview of interaction frequency of regular contacts and transient contacts. The red colour indicated regular contact, and the blue colour indicated transient contact. Monthly regular contacts indicated that the pair of MWs contacted at least 4 times (each time the contact duration was greater than 15 mins) within one week in each month during study period. Weekly regular contacts indicated that the pair MWs contacted at least 4 times (each time the contact duration was greater than 15 mins) per week. Daily regular contacts indicated that the pair MWs contacted daily, and they contacted with each other at least 15 mins per day.

In addition, there were no notable trends in interaction event behaviour on different days of the week (WAVK test (281) p-value of regular contacts: 0.18; WAVK test p-value of transient contacts: 0.55). However, there were strong hourly trends (WAVK test p-value of regular contacts: 0.01; WAVK test p-value of transient contacts: 0.01). The peak for interactions of regular contacts was between 18:00 and 06:00 daily, which is the typical resting time for MWs. In contrast, the peak time of interactions of transient contacts was at the time of mass transportation to and from work and on their lunch breaks (Figure 21).

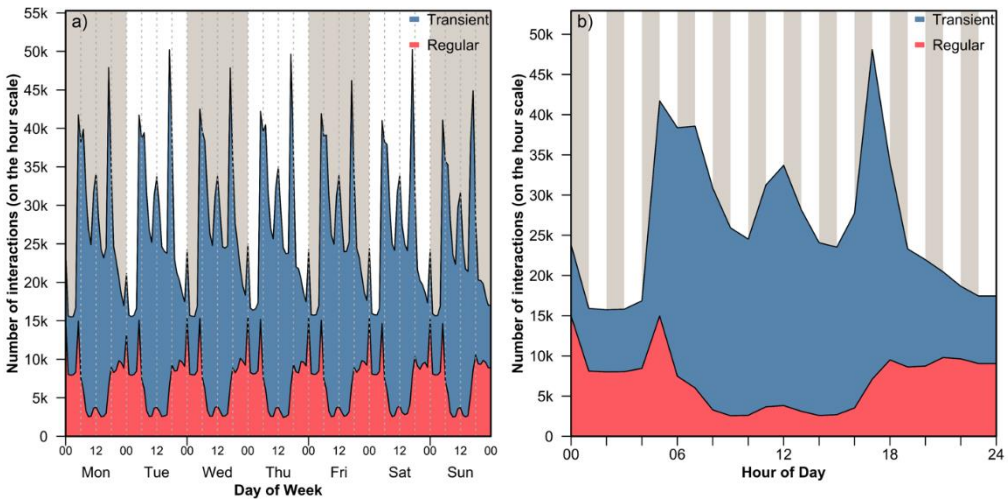


Figure 21. a) The average number of interactions that occurred per hour across the day of the week; b) the average number of interactions that occurred per hour in a day across regular and transient events.

For interaction duration in a day, when compared to transient contacts (mean: 5 minutes), regular contacts (mean: 7 hrs 3 mins) spent significantly longer periods together (p-value < 0.001). The details of interaction duration within a day can be found in the supplementary information (Figure 22). Furthermore, we found that regular contacts usually occurred with people sharing the same room, and transient occurred between individuals of different dormitories, blocks or floors. The mean interaction duration in a day within the same room of regular contacts was 8 hours 52 minutes; shorter durations of interaction events happened for contacts within the same floor and same block. There were few regular contacts with relatively short interaction duration in the same dormitory but not in the same blocks, and between the different dormitories. The duration of transient contacts interactions was on average 8 mins for each interaction; multiple interactions between transient contacts were observed, particularly for those in the same unit, totalling 53 mins over the typical day in multiple short spells (Table 10).

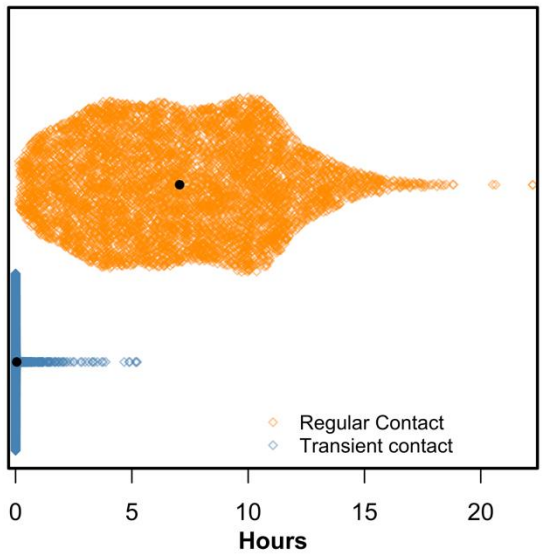


Figure 22 Interaction duration between each pair of MWs within a day. We found that the mean interaction duration of regular contacts was longer than transient contacts. We did however observe that there were some long interaction events between transient contacts, which occurred when MWs had multiple non-consecutive transient contacts across certain days.

Table 10. Summary of regular and transient contacts in a day

Items	**Group**	**Regular**		**Transient**	
		Mean Number of Contacts/days	**Mean Contact Duration/day***	**Mean Number of Contacts/days**	**Mean Contact Duration/day***
Overall		5	7hr 3min	26	5min
Dormitory Setting	Same Unit	2	8hr 52min	1	53min
	Same Floor Only	1	6hr 10min	4	12min
	Same Block Only	1	5hr 42min	7	5min
	Same Dormitory Only	0.5	4hr 22min	7	1min
	Different Dormitory	0.5	3hr 02min	7	2min

*: Interaction duration per day is calculated by summing up all interactions within a day with the same individual over possibly multiple interactions. The mean interaction duration of regular encounter and transient encounter was 53 mins and 2 min in a day, respectively.

Simulation of the contact network of the whole population in study dormitory

According to the likelihood of contact within the different degrees of proximity (room, floor, block, dormitory) obtaining from the collected contact data, we simulated the contact network among 8,256 MWs in two dormitories. There was a mean number of regular and transient contacts per individual of 13 and 169 (Figure 23), respectively, with most of the former being individuals sharing the same room and floor of a block, similar to the empirical data (Table 9).

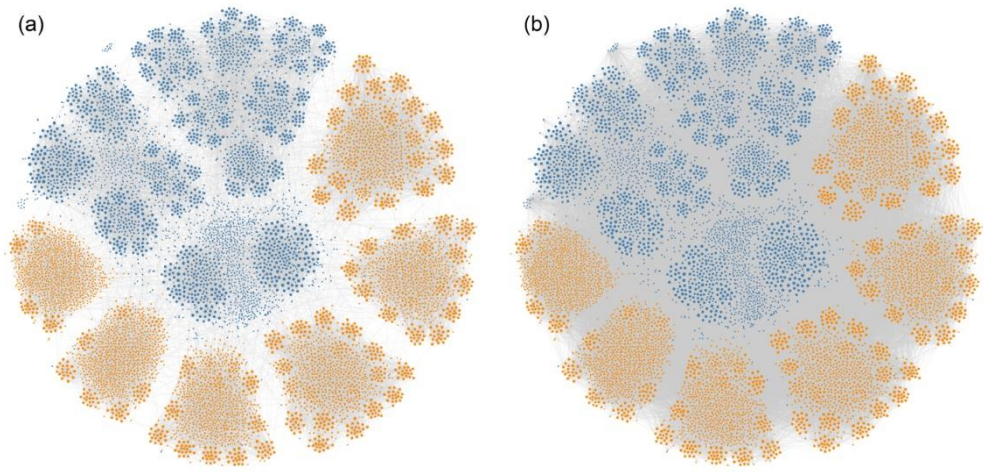


Figure 23. Simulated contact network. Regular (a) and Transient contacts (b) for Dorm A (dark orange) and Dorm C (steel blue).

Simulation of epidemic size under multiple NPIs

The simulated dormitory outbreak ran for four months (the final attack rates were similar when extended to 6 months, Figure 24b). To obtain infection rates that match the empirical final attack rates in the two dormitories (33% and 66%) required an infection risk per person throughout the course of infection of 0.4 (for Dorm A) and 0.87 (for Dorm C) (Figure 24a).

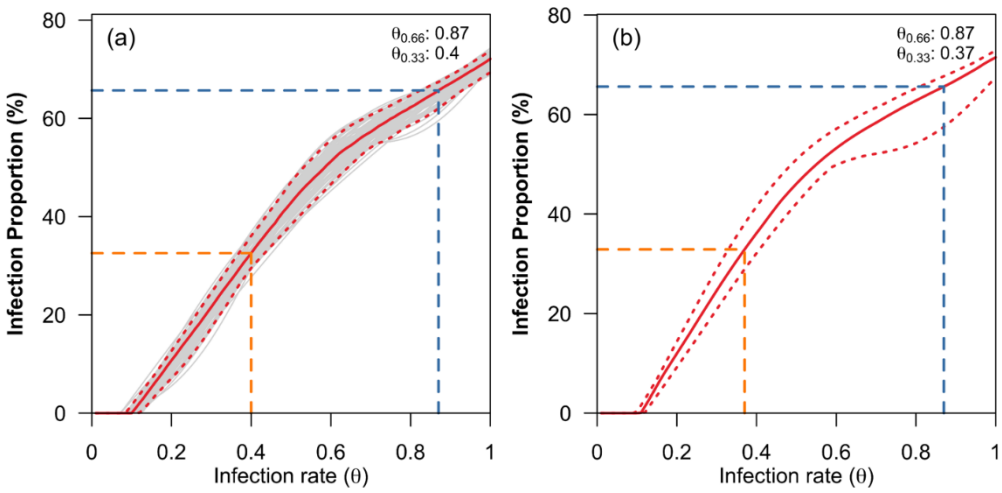


Figure 24. (a) The overall proportions of infected MWs at different infection probabilities within four months. The red solid line represents the median infection rate in Dorms A and C based on 33% and 66% being infected, and the red dashed line is the 95CIs; (b) The overall proportions of infected at different infection probabilities within six months. The red solid line represents the median infection rate in Dorm A and Dorm C based on 33% and 66% being infected, and the red dashed line is the 95CIs. Orange and blue lines indicate Dorm A and Dorm C respectively.

Using an infection rate of 0.40 to represent a smaller outbreak, and 0.87 for a larger outbreak, we then investigated the effects of intervention strategies on epidemic size at 120 days under multiple NPIs: baseline where only positive cases are isolated, quarantining of regular contacts, quarantining of all contacts, and no isolation and quarantining at all. All were run with and without delay in case identification. With the case identification delay we found that if the infection rate of 0.40, the proportion of infected was 39% (95% CI: 30%–45%) when only isolating cases with a moderate delay (i.e., 3 days) (Figure 25a). When all contacts were quarantined with a moderate delay, the proportion of infected was 2.1% (1.7%–2.5%) (Figure 25b). When only regular contacts were quarantined with a moderate delay, a significant decrement in the infected proportion was also observed at 11% (8%–13%) (Figure 25c). If the infection

rate of 0.87, the proportion of infected was 70% (95% CI: 66%–71%) when only isolating cases with a moderate delay (Figure 25d). When all contacts were quarantined with moderate delay, the proportion of infected was 9% (8%–11%) (Figure 25e). When only regular contacts were quarantined with a moderate delay, a significant decrement in the infected proportion was also observed at 46% (42%–48%) (Figure 25f).

For the baseline scenario with a less infectious variant, 31% (95% CI: 28%–41%) of the total resident population across both dormitories would be infected by day 120 (Figure 26a), while losing 22,064 man-days accordingly. When all contacts were immediately quarantined, the infected proportion decreased substantially to 1.1% (1%–1.3%) (Figure 26b) as 66,667 man-days would be lost accordingly. When only regular contacts were immediately quarantined, the proportion was also low at 6% (5%–8%) with 35,504 man-days lost (Figure 26c). Without any isolation or quarantining, the proportion is at its highest at 41% (31%–47%) (Figure 26d). In a lager outbreak, 67% (62%–68%) of the whole resident population were infected under the baseline scenario with 38,892 man-days lost (Figure 26f), decreasing to 33% (28%–35%) when only regular contacts were quarantined while 87,500 man-days lost accordingly (Figure 26h), and to 3.8% (3.3%–4.4%) when all contacts were quarantined with 112,500 man-days lost (Figure 26g). In contrast, the proportion getting infected was high at 72% (71%–73%) when no quarantining or isolation was implemented at all (Figure 26i).

To further investigate the impact of resident density, we reduced the number of social contacts of each resident by moving 30% of residents to a hypothetical new dormitory. The proportion of infected was substantially lower at 19%

(15%–24%) for the baseline scenario with a θ of 0.4 (Figure 26e) and 57% (51%–61%) when θ was set at 0.87 (Figure 26j).

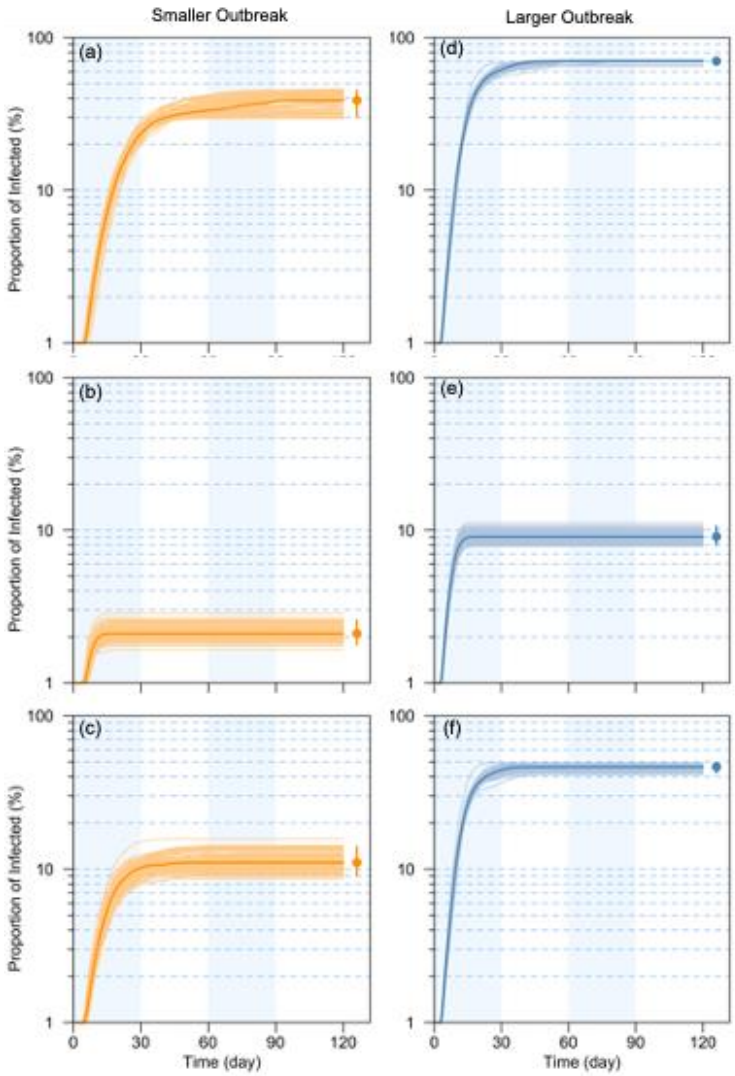


Figure 25. Simulation of epidemic size under multiple intervention scenarios. (a) and (d) Scenario S1: Isolate cases with moderate delay (i.e., 3 days); (b) and (e) Scenario S2: Isolate cases and quarantine all contacts with reasonable delay; (c) and (f) Scenario S3: Isolate cases and quarantine regular contacts with a moderate delay.

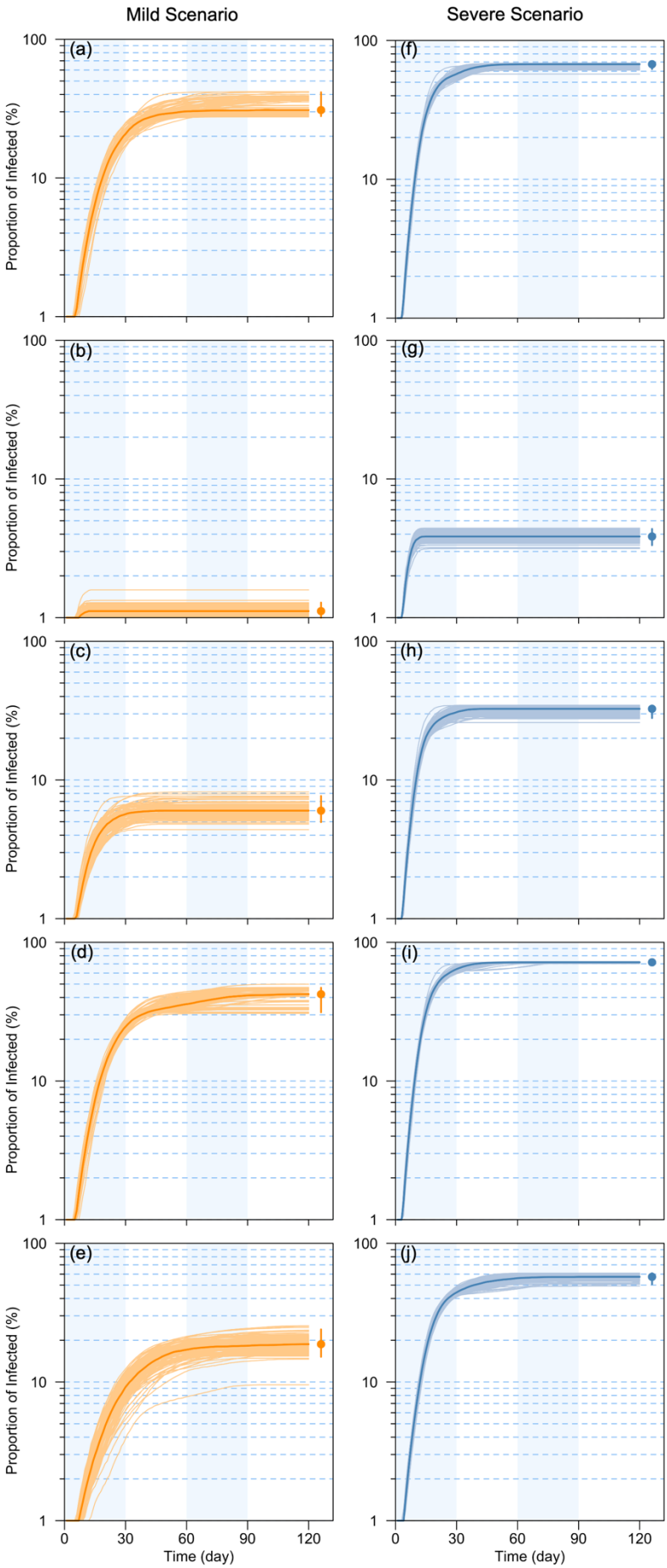

Mild Scenario
Severe Scenario
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
(j)
Proportion of Infected (%)
100
10
1
0
30
60
90
120
Time (day)

Figure 26. Simulation of epidemic size under multiple NPIs under smaller and larger outbreaks (i.e., with θ of 0.4 and 0.87). (a) and (f) Scenario 1: Immediate isolation of cases in the baseline scenario; (b) and (g) Scenario 2: Immediate isolation of cases and quarantining for all contacts; (c) and (h) Scenario 3: Immediate isolation and quarantining of regular contacts; (d) and (i) Scenario 4: No isolation of cases or quarantining of contacts; (e) and (j) Scenario 5: baseline scenario under a 30% reduction in contacts. Daily cases of the epidemic can be found in Figure 27.

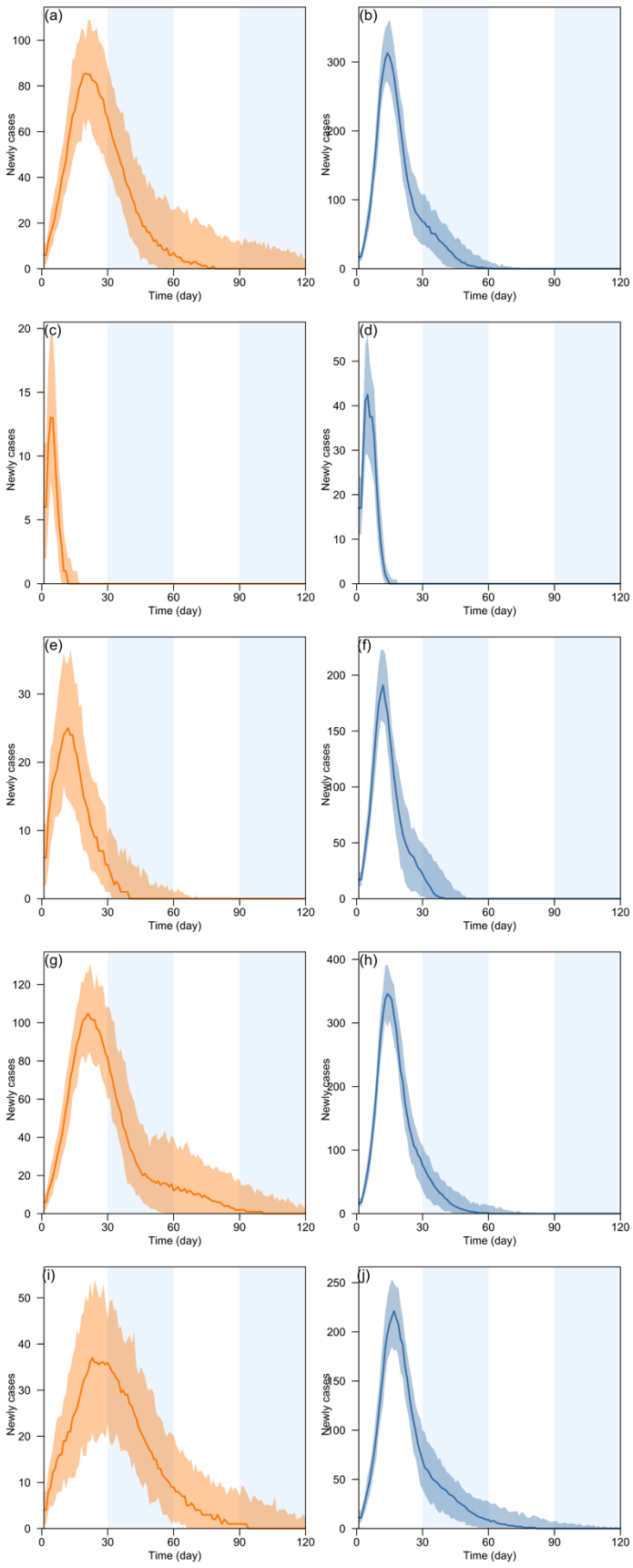


Figure 27. Simulation of epidemic under multiple NPIs under smaller and larger outbreaks (i.e., with θ of 0.4 and 0.87). (a) and (f) Scenario 1: Immediate

isolation of cases in the baseline scenario; (b) and (g) Scenario 2: Immediate isolation of cases and quarantining for all contacts; (c) and (h) Scenario 3: Immediate isolation and quarantining of regular contacts; (d) and (i) Scenario 4: No isolation of cases or quarantining of contacts; (e) and (j) Scenario 5: baseline scenario under a 30% reduction in contacts

4.5 Discussion

DCT data can be coupled with infectious disease models to simulate the spread of disease. In this study, we utilised a branching process model and simulated contact network derived from empirical social interaction data to model an epidemic outbreak in a closed population. The heterogeneous mixing at different infrastructural levels was captured based on the data collected from portable Bluetooth devices, revealing group-based contact patterns and the relations within the dormitory hierarchy in which transmission risk may be elevated. The effectiveness of various NPIs designed to reduce the risk of transmission through blocking of certain interactions was investigated using our model. Through understanding viral spread in closed high-density settings, more informed planning for future outbreaks can be carried out.

We found that the severity of the outbreak between the two study dormitories is significantly different. The infection rate was lower in Dorm A than in the Dorm C (135). It could be plausible that Dorm A, which is laid out more in the style of sparser populated apartment blocks, had less environmental transmission when compared to the barrack style of Dorm C. To show this, we assumed that contacts would be reduced by building new dormitories for MWs, and we simulated the epidemic size with a reduced overall number of contacts by 30%, decreasing infection rates from 31% and 67% to 19% and 57% for a more and less infectious pathogen, respectively. This indicates that reducing

resident density, potentially through the construction of new dormitories could substantially lower the risk of rapid infectious disease spread when new outbreaks emerge.

In the face of rising cases, lockdowns are critical tools which can reduce the spread of a respiratory virus in closed settings. However, costs can be exceedingly high when the virus is highly transmissible (282) due to the containment measures required to ensure compliance, often bringing additional mental health burdens to the residents (283–285) and, despite ongoing efforts at containment in China, these measures do not seem sustainable in other settings. Our simulation results indicate that the quarantining of all or regular contacts can substantially reduce outbreak size, but such methods should be used at the beginning of new outbreaks or when the risk of the illness developing severe complications warrants such constrained measures for the residents, such as the implementation of lockdowns in Wuhan city (27,286) and other cities and territories/countries early in the COVID-19 pandemic (287). Notably, there is a difference between the baseline of only isolating cases and no control at all at. 10%, and 5 % reduction in the infection proportion if only isolating cases compared to no control actions when the infection rate of 0.4 and 0.87, respectively. At a latter part of an outbreak, the isolation of cases may therefore bring relatively few benefits to the population as a whole as the majority have already been infected. Nevertheless, mass testing is strongly recommended and individuals should be given a choice to self-isolate if they are positive or wish to as they have been suspected of being exposed (288).

The simulations revealed that a policy of quarantining of regular contacts only would be relatively efficient, in reducing infections with relatively little impact

on the number of people being quarantined. In contrast, quarantining all contacts would be necessary to prevent epidemic spread, but due to the large number of casual contacts in a high density living environment such as those considered in this paper, such a policy would lead to large numbers of people in quarantine, and without long term means of preventing spread, such quarantine would need to be implemented over too long a time period to be feasible.

Overall, the use of Bluetooth technology via wearable devices such as the BP is useful when modelling epidemic sizes under different quarantine orders, which is critical information for policymakers in their planning efforts for closed settings, especially ones with low vaccination rates. Having stand-alone Bluetooth devices can overcome the disadvantages of common digital communication technologies (289) as such devices do not require any modification to an individual's mobile cellular device to incorporate contact tracing functioning, which reduces the perceived level of intrusion and may encourage participation. Moreover, all BPs are secure, with the devices de-identified, and social interaction records encrypted before being stored locally on each device, so that data retrieval is only possible by accessing the memory storage of the device. In our study, this was obtained by physical extraction, but this was a labour-intensive endeavour that could not be sustained in routine use.

The study has its limitations. We had the assumption that the likelihood of transmission of SARS-CoV-2 was proportional between the two contact categories with transmission to transient contacts occurring at 0.009 times that to regular contacts. This was determined through the proportion of regular and transient mean interaction duration from the DCT data obtained from the BPs.

While we know that the risk of transmission decreases with shorter verbal conversations (290), this could be an underestimation of transmission likelihood if individuals are being infected through surfaces and fomites. In addition, the contact networks we observed were for a period in which residents were able to leave their dormitory for work and in some other limited circumstances for other purposes, and the structure of the network may have been different at earlier and later periods in the pandemic. Also, we assumed that MWs had the same probability of being the initial case in our simulation model. Factors such as heterogeneous mixing and contact behaviour for MWs could not be incorporated for initial cases as this information is largely unknown.

Despite these limitations, using DCT devices was able to investigate heterogeneities in contact within the networks of a closed environment, which has implications on how epidemics spread. The impact of various policies on SARS-CoV-2 transmission can be explored, feeding into policymaking decisions on how to prevent future outbreaks of respiratory illnesses in high risk closed environments where epidemics can rapidly take hold and spread.

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 Summary of major findings

Rapidly climbing COVID-19 cases in the early outbreak posed a severe challenge to healthcare systems, resulting in a shortage of healthcare resources, including PPE (291,292), ICU beds, and inpatient beds. That not only leaves frontline HCWs at grave risk but also challenges policymakers and society as a whole. Hence, a comprehensive approach is necessary for timely estimating PPE demands. Chapter 2 of this thesis estimated the demand for N95s, the critical type of PPEs, to aid planning efforts assessing what preparations are required over the coming months for the healthcare system to run with the maximised possible protection for its workers. This study estimates the epidemic size throughout mild, moderate, and severe epidemic scenarios within Singapore as a case study, using a transmission dynamic model. In addition, we assess the number of N95s required within the respiratory isolation wards and ICUs throughout the epidemic, considering multiple strategies, including single-use, extended-use, and prolonged-use, for N95 respirators for use by HCWs treating suspected but negative (misclassified) or confirmed COVID-19 patients. Also, the study calculates the demands of N95s under different usage strategies over the course of the epidemic if there is rapid testing is implemented. Our approach can be utilised elsewhere with considerations of implementing N95s extended-use policies. This was especially paramount in the early outbreak of COVID-19 as the WHO had stated that global supplies were rapidly depleting overall. This is further complicated by the fact that the PPE supply chain landscape is complex due to the quarantine and lockdown policies in developing countries, such as China, which are the primary producers of PPE.

While timely, stringent and effective intervention policies, as implemented in many countries in early 2020, have been demonstrated to have mitigated deteriorating COVID-19 outbreaks, they are not viable long-term strategies (173) as it is impractical to implement strict lockdowns or border control policies for the months or years needed for mass vaccination to occur (174). COVID-19 infection has setting-specific heterogeneity due to differences in living and/or health conditions (175,176), such as the significant differences between normal communities and nursing homes (179,271,272), MWs (136,180,181), homeless (182,183) and travellers. In addition, outbreaks in army (184) and prison (185) settings have been reported. Hence, it is essential to update responsive interventions in a timely manner according to the type of infection by real-time forecasting of COVID-19. Chapter 3 of this thesis used a Bayesian hierarchical statistical model to estimate epidemic trends within the community, foreign worker infections and imported cases separately, thus providing real-time projections for SARS-CoV-2 cases in Singapore in heterogeneous transmission settings. Such forecasting can inform the need to escalate or reduce the degree of interventions, which has proven to be one of the approaches taken by the Singapore government to avoid rapid resurgence following the lockdowns.

Before the vaccination programme was implemented (i.e., by the end of 2020), Singapore's remarkably heterogeneous disease states existed: a low endemic community state and a high dormitory outbreak state (over 93% of cases belonged to the latter). Consequently, policy responses differ between sites. In dormitory settings, operators are expected to limit the inter-mixing of workers from different blocks or sites, perform movement tracking, and perform regular

14-day swabbing of uninfected workers (236). Hence, understanding the spread between dormitories to improve the intervention strategies to minimise the risk of new disease clusters emerging is essential. Chapter 4 of this thesis took advantage of the high volume of social interaction data recorded in an extraction of around 1,000 residents of two dormitories working for the same company, to quantify the nature of the contact network to gain insights into local heterogeneities that would possibly hinder control. In addition, the study used the empirical contact network to develop a realistic branching process model to explore the impact of policies that used varying stringency of quarantine to mitigate spread. Wearable contact tracing devices may be used not just for contact tracing efforts but also to inform alternative containment measures in high-risk closed settings. The simulation results under multiple policies indicated that strict isolation of all cases and quarantine of all contacts would lead to very low prevalence but that quarantining only regular contacts would lead to only marginally higher prevalence, but substantially fewer total man-hours lost in quarantine. Moreover, the construction of additional dormitories (to reduce the density of contacts by 30%) can reduce the prevalence by 14% and 9% under smaller and larger outbreaks, respectively.

5.2 Public health contributions and implications

There are several public health contributions and implications of this thesis. Firstly, this thesis provides researchers and healthcare policymakers with different degrees of information to adjust intervention policies across different stages of the pandemic—before widespread community transmission had occurred, while multiple epidemics were concurrently ongoing, and in the

period once transmission levels had fallen—and thus make well preparation and knowledge for next infectious disease outbreak.

Secondly, the thesis contributes to an estimated demand for N95s in the early outbreak of infectious disease. Considering the different severe degrees of the epidemic with multiple use policy, should infection become widespread with significant patient load at healthcare facilities, extensive reuse, via the extended-use and prolonged-use policy, may have to be utilised to prevent supply shortages. By doing so, the findings provide information for use in planning purposes which can be utilised not only in the COVID-19 pandemic in Singapore but other outbreaks globally also.

Thirdly, this thesis proposes a methodological framework to project the number of infections by the origin of infection and the impact on server cases among them. This finding can be used by policymakers in countries with heterogeneous transmission communities, such as the notable outbreaks observed in countries with large homeless populations, nursing homes, MWs, prisons, slums, and arrivals at borders, to better understand containment policies through projections.

Finally, the findings of this thesis support that wearable contact tracing devices may be used not just for contact tracing efforts but also to investigate heterogeneities in contact within the networks of a closed environment, which has profound implications on how epidemics spread. Meanwhile, this provides new insights into the contact network in a specific community and can help policymakers choose appropriate containment measures under various restrictive policies by building simulation models on the network.

5.3 Limitations

Limitations were discussed in each project. However, the following limitations are worth emphasising again.

Reinfection was not modelled, instead, lasting immunity was considered. Accounting for reinfection adds excellent value, however, a level of complexity requires a different methodological framework, i.e., an additional comparator to reflect the reinfection group at the population level and factors to become reinfection cases at the individual level will cause computationally burden. However, reinfection was rarely observed in the early outbreak of the COVID-19 pandemic, it also depends on the mutations in SARS-CoV-2 variants. Additionally, this thesis provides a representative view of different COVID-19 waves, each with a mono-dominated variant in each phase. Again, the reinfection situation may be far more complex with a longer-term phase to evaluate.

Chapters 2 and 3 of this thesis are based on data before the massive vaccine implementation programme. As a result, it only provides informative sights to policymakers from 2021, when vaccine implementation programming has started, rather than clear findings to them. Moreover, the models developed may need to characterise importation cases' impact on local transmission fully. But the concern is probably minor for this thesis because strict travelling restrictions and quarantine policies in each country were implemented before the vaccination programme.

This thesis also assumes that infections likely are the same in a similar setting, especially in closed environments. However, the distance may result in different

infection probabilities. Such as, in chapter 4, based on our observations of dormitory layouts during fieldwork, Dorm A is laid out more in the style of apartment blocks, sparser and with more spacing between blocks than Dorm C, which is more like a traditional dormitory design, with blocks placed in close proximity to each other. It could plausibly be that infected workers in one block could expose those in a neighbouring block to infection at night through open windows for instance. There also seemed to be fewer social spaces in Dorm C, which may have led to greater crowding in those available.

5.4 Future studies

As per the limitations of this thesis, several future studies can be implemented to fill in the knowledge gaps.

Future studies are needed to investigate the with a longer duration of follow up would anticipate more reinfections, thus raising the importance to account for such reinfection events within the model. Moreover, the data would also provide a way to estimate the risk of such reinfection events.

Furthermore, infectivity between/within hospitalized and ICU groups by patients to HCWs to patients' contacts can be considered in the model when estimating N95 needs to prepare for future infectious diseases.

In addition, a comprehensive model accounting for multiple pandemic phases and mutation of variants is necessary in future work.

Due to the ventilation properties, the distance between individual is worth considering in the context of the spread of airborne infectious diseases. The effects of such anecdotal evidence would be an interesting future research topic.

Finally, we used the simulated contact network within the whole dormitories (i.e., Dorm A and Dorm C) to calculate the infection risk under the mild and severe situations, separately. However, the heterogeneity in dormitory design was observed of these two dormitories and this might cause the infection risk bias. However, calculating the infection risk with contact records collecting from more dormitories can mitigate this bias.

5.5 Conclusion

This thesis provides a set of statistical modelling studies for discussing and better understanding the preparation, suitability, sustainability, and optimal implementation of control measures. The results obtained are essential for public health decision-makers in deploying limited resources and adjusting intervention policies with different phases of the campaigns.

Reference

1. WHO Coronavirus Disease (COVID-19) Dashboard [Internet]. [cited 2020 Oct 27]. Available from: https://covid19.who.int

2. Coronaviridae Study Group of the International Committee on Taxonomy of Viruses. The species Severe acute respiratory syndrome-related coronavirus: classifying 2019-nCoV and naming it SARS-CoV-2. Nat Microbiol. 2020 Apr;5(4):536–44.

3. Naming the coronavirus disease (COVID-19) and the virus that causes it [Internet]. [cited 2020 Jul 21]. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/technical-guidance/naming-the-coronavirus-disease-(covid-2019)-and-the-virus-that-causes-it

4. Guarner J. Three Emerging Coronaviruses in Two Decades. Am J Clin Pathol. 2020 Mar 9;153(4):420–1.

5. Chan-Yeung M, Xu RH. SARS: epidemiology. Respirology. 2003;8(s1):S9–14.

6. Centers for Disease Control and Prevention (CDC). Severe acute respiratory syndrome--Singapore, 2003. MMWR Morb Mortal Wkly Rep. 2003 May 9;52(18):405–11.

7. de Groot RJ, Baker SC, Baric RS, Brown CS, Drosten C, Enjuanes L, et al. Middle East respiratory syndrome coronavirus (MERS-CoV): announcement of the Coronavirus Study Group. J Virol. 2013 Jul;87(14):7790–2.

8. Drosten C, Günther S, Preiser W, van der Werf S, Brodt HR, Becker S, et al. Identification of a novel coronavirus in patients with severe acute respiratory syndrome. N Engl J Med. 2003 May 15;348(20):1967–76.

9. Kuiken T, Fouchier RA, Schutten M, Rimmelzwaan GF, van Amerongen G, van Riel D, et al. Newly discovered coronavirus as the primary cause of severe acute respiratory syndrome. Lancet. 2003 Jul 26;362(9380):263–70.

10. Ritchie H, Mathieu E, Rodés-Guirao L, Appel C, Giattino C, Ortiz-Ospina E, et al. Coronavirus Pandemic (COVID-19). Our World Data [Internet]. 2020 Mar 5 [cited 2022 Sep 18]; Available from: https://ourworldindata.org/covid-vaccinations

11. Huang C, Wang Y, Li X, Ren L, Zhao J, Hu Y, et al. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. Lancet Lond Engl. 2020;395(10223):497–506.

12. van Dorp L, Acman M, Richard D, Shaw LP, Ford CE, Ormond L, et al. Emergence of genomic diversity and recurrent mutations in SARS-CoV-2. Infect Genet Evol. 2020 Sep;83:104351.

13. Li X, Zai J, Zhao Q, Nie Q, Li Y, Foley BT, et al. Evolutionary history, potential intermediate animal host, and cross-species analyses of SARS-CoV-2. J Med Virol. 2020 Mar 11;10.1002/jmv.25731.

14. Andersen KG, Rambaut A, Lipkin WI, Holmes EC, Garry RF. The proximal origin of SARS-CoV-2. Nat Med. 2020 Mar 17;1–3.

15. Allam Z. The First 50 days of COVID-19: A Detailed Chronological Timeline and Extensive Review of Literature Documenting the Pandemic. Surv Covid-19 Pandemic Its Implic. 2020;1–7.

16. The Fight Against COVID-19 | english.scio.gov.cn [Internet]. [cited 2022 Dec 30]. Available from: http://english.scio.gov.cn/featured/chinakeywords/2020-09/09/content_76686089.htm

17. About the virus [Internet]. [cited 2022 Feb 16]. Available from: https://www.euro.who.int/en/health-topics/health-emergencies/coronavirus-covid-19/novel-coronavirus-2019-ncov

18. Wacharapluesadee S, Buathong R, Iamsirithawon S, Chaifoo W, Ponpinit T, Ruchisrisarod C, et al. Identification of a Novel Pathogen Using Family-Wide PCR: Initial Confirmation of COVID-19 in Thailand. Front Public Health. 2020;8:555013.

19. Update: Cluster of pneumonia cases associated with novel coronavirus – Wuhan, China – 2019 [Internet]. European Centre for Disease Prevention and Control. 2020 [cited 2022 Feb 14]. Available from: https://www.ecdc.europa.eu/en/news-events/update-cluster-pneumonia-cases-associated-novel-coronavirus-wuhan-china-2019

20. Novel Coronavirus – China [Internet]. [cited 2022 Dec 30]. Available from: https://www.who.int/emergencies/disease-outbreak-news/item/2020-DON233

21. Chan JFW, Yuan S, Kok KH, To KKW, Chu H, Yang J, et al. A familial cluster of pneumonia associated with the 2019 novel coronavirus indicating person-to-person transmission: a study of a family cluster. The Lancet. 2020 Jan;S0140673620301549.

22. Archived: WHO Timeline - COVID-19 [Internet]. [cited 2022 Feb 15]. Available from: https://www.who.int/news/item/27-04-2020-who-timeline---covid-19

23. WHO. Novel Coronavirus (2019-nCoV) situation report - 1 [Internet]. 2020 Jan. Available from: chrome-extension://oemmndcbldboiebfnladdacbdfmadadm/https://www.who.in

t/docs/default-source/coronaviruse/situation-reports/20200121-sitrep-1-2019-ncov.pdf

24. WHO. Novel Coronavirus (2019-nCoV) situation report - 3 [Internet]. 2020 Jan. Available from: chrome-extension://oemmndcbldboiebfnladdacbdfmadadm/https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200123-sitrep-3-2019-ncov.pdf

25. Shangguan Z, Wang MY, Sun W. What Caused the Outbreak of COVID-19 in China: From the Perspective of Crisis Management. Int J Environ Res Public Health. 2020 May;17(9):3279.

26. Cyranoski D. What China's coronavirus response can teach the rest of the world. Nature. 2020 Mar 17;579(7800):479–80.

27. Lau H, Khosrawipour V, Kocbach P, Mikolajczyk A, Schubert J, Bania J, et al. The positive impact of lockdown in Wuhan on containing the COVID-19 outbreak in China. J Travel Med. 2020 May 18;27(3):taaa037.

28. Timeline of ECDC's response to COVID-19 [Internet]. [cited 2022 Feb 6]. Available from: https://www.ecdc.europa.eu/en/covid-19/timeline-ecdc-response

29. Wu YC, Chen CS, Chan YJ. The outbreak of COVID-19: An overview. J Chin Med Assoc. 2020 Mar;83(3):217.

30. Marome W, Shaw R. COVID-19 Response in Thailand and Its Implications on Future Preparedness. Int J Environ Res Public Health. 2021 Feb;18(3):1089.

31. Watanabe M. The COVID-19 Pandemic in Japan. Surg Today. 2020 Aug;50(8):787–93.

32. Early Trend of Imported COVID-19 Cases in South Korea. Osong Public Health Res Perspect. 2020 Jun;11(3):140–5.

33. Pung R, Cook AR, Chiew CJ, Clapham HE, Sun Y, Li Z, et al. Effectiveness of Containment Measures Against COVID-19 in Singapore: Implications for Other National Containment Efforts. Epidemiol Camb Mass. 2021 Jan;32(1):79–86.

34. Cuong LV, Giang HTN, Linh LK, Shah J, Sy LV, Hung TH, et al. The first Vietnamese case of COVID-19 acquired from China. Lancet Infect Dis. 2020 Apr 1;20(4):408–9.

35. Bastola A, Sah R, Rodriguez-Morales AJ, Lal BK, Jha R, Ojha HC, et al. The first 2019 novel coronavirus case in Nepal. Lancet Infect Dis. 2020 Mar 1;20(3):279–80.

36. Elengoe A. COVID-19 Outbreak in Malaysia. Osong Public Health Res Perspect. 2020 Jun;11(3):93–100.

37. WHO. Ministry of Health responds to first positive case of new coronavirus [Internet]. [cited 2022 Feb 17]. Available from: https://www.who.int/cambodia/news/detail/28-01-2020-ministry-of-health-responds-to-first-positive-case-of-new-coronavirus

38. Wickramaarachchi WPTM, Perera SSN, Jayasinghe S. COVID-19 Epidemic in Sri Lanka: A Mathematical and Computational Modelling Approach to Control. Comput Math Methods Med. 2020 Oct 16;2020:4045064.

39. Khanna RC, Cicinelli MV, Gilbert SS, Honavar SG, Murthy GVS. COVID-19 pandemic: Lessons learned and future directions. Indian J Ophthalmol. 2020 May;68(5):703–10.

40. Edrada EM, Lopez EB, Villarama JB, Salva Villarama EP, Dagoc BF, Smith C, et al. First COVID-19 infections in the Philippines: a case report. Trop Med Health. 2020 Apr 14;48(1):21.

41. Holshue ML, DeBolt C, Lindquist S, Lofy KH, Wiesman J, Bruce H, et al. First Case of 2019 Novel Coronavirus in the United States. N Engl J Med. 2020 Mar 5;382(10):929–36.

42. Bernard Stoecklin S, Rolland P, Silue Y, Mailles A, Campese C, Simondon A, et al. First cases of coronavirus disease 2019 (COVID-19) in France: surveillance, investigations and control measures, January 2020. Eurosurveillance. 2020 Feb 13;25(6):2000094.

43. COVID-19 Public Health Emergency of International Concern (PHEIC) Global research and innovation forum [Internet]. [cited 2022 Feb 17]. Available from: https://www.who.int/publications/m/item/covid-19-public-health-emergency-of-international-concern-(pheic)-global-research-and-innovation-forum

44. WHO. Novel Coronavirus (2019-nCoV) situation report - 11 [Internet]. 2020 Jan. Available from: chrome-extension://oemmndcbldboiebfnladdacbdfmadadm/https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200131-sitrep-11-ncov.pdf

45. WHO. Novel Coronavirus (2019-nCoV) situation report - 13 [Internet]. 2020 Feb. Available from: chrome-extension://oemmndcbldboiebfnladdacbdfmadadm/https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200202-sitrep-13-ncov-v3.pdf

46. Akande OW, Akande TM. COVID-19 pandemic: A global health burden. Niger Postgrad Med J. 2020;27(3):147–55.

47. WHO Director-General's opening remarks at the media briefing on COVID-19 - 11 March 2020 [Internet]. Available from: https://www.who.int/director-general/speeches/detail/who-director-

general-s-opening-remarks-at-the-media-briefing-on-covid-19---11-march-2020

48. Coronavirus disease 2019 (COVID-19)Situation Report –71 [Internet]. Available from: https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200331-sitrep-71-covid-19.pdf?sfvrsn=4360e92b_8

49. WHO. Coronavirus disease 2019 (COVID-19) situation report - 70 [Internet]. 2020 Mar. Available from: https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200330-sitrep-70-covid-19.pdf

50. Rational use of personal protective equipment for coronavirus disease (COVID-19): interim guidance, 27 February 2020 [Internet]. Available from: https://apps.who.int/iris/handle/10665/331215

51. WHO. Shortage of personal protective equipment endangering health workers worldwide [Internet]. 2020 Mar. Available from: https://www.who.int/news/item/03-03-2020-shortage-of-personal-protective-equipment-endangering-health-workers-worldwide

52. World Health Organization. Coronavirus disease 2019 (COVID-19): situation report - 73 [Internet]. World Health Organization; 2020 Apr. Available from: https://apps.who.int/iris/handle/10665/331686

53. WHO. Considerations in adjustingpublic health and social measures in the context of COVID-19 [Internet]. 2020 Apr. Available from: https://apps.who.int/iris/handle/10665/331773

54. WHO. Public health criteria to adjust public health and social measures in the context of COVID-19: annex to considerations in adjusting public health and social measures in the context of COVID-19 [Internet]. 2020 May. Available from: https://apps.who.int/iris/handle/10665/332073

55. Coronavirus disease (COVID-19) Situation Report–162 [Internet]. Available from: https://www.who.int/docs/default-source/coronaviruse/20200630-covid-19-sitrep-162.pdf?sfvrsn=e00a5466_2

56. Weekly epidemiological update - 31 August 2020 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---31-august-2020

57. Weekly epidemiological update - 28 September 2020 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---28-september-2020

58. World Health Organization. Pulse survey on continuity of essential health services during the COVID-19 pandemic: interim report, 27 August 2020 [Internet]. World Health Organization; 2020. Report No.: WHO/2019-

nCoV/EHS_continuity/survey/2020.1. Available from: https://apps.who.int/iris/handle/10665/334048

59. Weekly epidemiological update - 27 October 2020 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---27-october-2020

60. Weekly epidemiological update - 10 November 2020 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---10-november-2020

61. CDC. SARS-CoV-2 Variant Classifications and Definitions [Internet]. Centers for Disease Control and Prevention. 2020. Available from: https://www.cdc.gov/coronavirus/2019-ncov/variants/variant-classifications.html

62. Tegally H, Wilkinson E, Giovanetti M, Iranzadeh A, Fonseca V, Giandhari J, et al. Detection of a SARS-CoV-2 variant of concern in South Africa. Nature. 2021 Apr;592(7854):438–43.

63. Weekly epidemiological update - 5 January 2021 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---5-january-2021

64. Weekly epidemiological update - 19 January 2021 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---19-january-2021

65. Weekly epidemiological update - 29 December 2020 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---29-december-2020

66. Organization WH. Clinical management of COVID-19: interim guidance, 27 May 2020. World Health Organization; 2020 p. 62 p.

67. WHO welcomes preliminary results about dexamethasone use in treating critically ill COVID-19 patients [Internet]. Available from: https://www.who.int/news/item/16-06-2020-who-welcomes-preliminary-results-about-dexamethasone-use-in-treating-critically-ill-covid-19-patients

68. Coronavirus disease (COVID-19): Solidarity Trial and hydroxychloroquine [Internet]. Available from: https://www.who.int/news-room/questions-and-answers/item/coronavirus-disease-covid-19-hydroxychloroquine

69. World Health Organization. Antigen-detection in the diagnosis of SARS-CoV-2 infection using rapid immunoassays: interim guidance, 11 September 2020 [Internet]. World Health Organization; 2020. Report No.: WHO/2019-nCoV/Antigen_Detection/2020.1. Available from: https://apps.who.int/iris/handle/10665/334253

70. WHO Emergency Use Listing for In vitro diagnostics (IVDs) Detecting SARS-CoV-2 [Internet]. Available from: https://www.who.int/publications/m/item/200922-eul-sars-cov2-product-list

71. Interim recommendations for use of the Moderna mRNA-1273 vaccine against COVID-19 [Internet]. Available from: https://www.who.int/publications-detail-redirect/interim-recommendations-for-use-of-the-moderna-mrna-1273-vaccine-against-covid-19

72. COVAX Announces new agreement, plans for first deliveries [Internet]. Available from: https://www.who.int/news/item/22-01-2021-covax-announces-new-agreement-plans-for-first-deliveries

73. COVID-19 vaccine doses shipped by the COVAX Facility head to Ghana, marking beginning of global rollout [Internet]. Available from: https://www.who.int/news/item/24-02-2021-covid-19-vaccine-doses-shipped-by-the-covax-facility-head-to-ghana-marking-beginning-of-global-rollout

74. WHO Director-General's opening remarks at the media briefing on COVID-19 - 22 December 2021 [Internet]. Available from: https://www.who.int/director-general/speeches/detail/who-director-general-s-opening-remarks-at-the-media-briefing-on-covid-19---22-december-2021

75. de Bengy Puyvallée A, Storeng KT. COVAX, vaccine donations and the politics of global vaccine inequity. Glob Health. 2022 Mar 5;18(1):26.

76. Weekly epidemiological update - 27 January 2021 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---27-january-2021

77. Cele S, Gazy I, Jackson L, Hwa SH, Tegally H, Lustig G, et al. Escape of SARS-CoV-2 501Y.V2 from neutralization by convalescent plasma. Nature. 2021 May;593(7857):142–6.

78. NERVTAG paper on COVID-19 variant of concern B.1.1.7 [Internet]. GOV.UK. Available from: https://www.gov.uk/government/publications/nervtag-paper-on-covid-19-variant-of-concern-b117

79. Weekly epidemiological update - 23 February 2021 [Internet]. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update---23-february-2021

80. WHO. Weekly epidemiological update on COVID-19 - 30 March 2021 [Internet]. 2021 Mar. Available from: https://www.who.int/publications/m/item/weekly-epidemiological-update-on-covid-19---31-march-2021

81. Li B, Deng A, Li K, Hu Y, Li Z, Shi Y, et al. Viral infection and transmission in a large, well-traced outbreak caused by the SARS-CoV-2 Delta variant. Nat Commun. 2022 Jan 24;13(1):460.

82. Reardon S. How the Delta variant achieves its ultrafast spread. Nature [Internet]. 2021 Jul 21; Available from: https://www.nature.com/articles/d41586-021-01986-w

83. Hart WS, Miller E, Andrews NJ, Waight P, Maini PK, Funk S, et al. Generation time of the alpha and delta SARS-CoV-2 variants: an epidemiological analysis. Lancet Infect Dis. 2022 May;22(5):603–10.

84. Ong SWX, Chiew CJ, Ang LW, Mak TM, Cui L, Toh MPHS, et al. Clinical and Virological Features of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) Variants of Concern: A Retrospective Cohort Study Comparing B.1.1.7 (Alpha), B.1.351 (Beta), and B.1.617.2 (Delta). Clin Infect Dis Off Publ Infect Dis Soc Am. 2021 Aug 23;75(1):e1128–36.

85. Delta variant triggers dangerous new phase in the pandemic [Internet]. Available from: https://www.science.org/content/article/delta-variant-triggers-dangerous-new-phase-pandemic

86. Thakur V, Bhola S, Thakur P, Patel SKS, Kulshrestha S, Ratho RK, et al. Waves and variants of SARS-CoV-2: understanding the causes and effect of the COVID-19 catastrophe. Infection. 2022 Apr 1;50(2):309–25.

87. Paul P, France AM, Aoki Y, Batra D, Biggerstaff M, Dugan V, et al. Genomic Surveillance for SARS-CoV-2 Variants Circulating in the United States, December 2020–May 2021. Morb Mortal Wkly Rep. 2021 Jun 11;70(23):846–50.

88. Dutta A. COVID-19 waves: variant dynamics and control. Sci Rep. 2022 Jun 4;12(1):9332.

89. Lee YN. 3 charts show how far Covid delta variant has spread around the world [Internet]. CNBC. [cited 2022 Aug 2]. Available from: https://www.cnbc.com/2021/08/06/covid-charts-show-how-far-delta-variant-has-spread-around-the-world.html

90. Africa: COVID-19 Delta variant cases 2022 [Internet]. Statista. [cited 2022 Aug 2]. Available from: https://www.statista.com/statistics/1249798/number-of-sars-cov-2-delta-variant-cases-in-africa-by-country/

91. Tareq AM, Emran TB, Dhama K, Dhawan M, Tallei TE. Impact of SARS-CoV-2 delta variant (B.1.617.2) in surging second wave of COVID-19 and efficacy of vaccines in tackling the ongoing pandemic. Hum Vaccines Immunother. 17(11):4126–7.

92. Aleem A, Akbar Samad AB, Slenker AK. Emerging Variants of SARS-CoV-2 And Novel Therapeutics Against Coronavirus (COVID-19). In: StatPearls

[Internet]. Treasure Island (FL): StatPearls Publishing; 2022. Available from: http://www.ncbi.nlm.nih.gov/books/NBK570580/

93. Classification of Omicron (B.1.1.529): SARS-CoV-2 Variant of Concern [Internet]. Available from: https://www.who.int/news/item/26-11-2021-classification-of-omicron-(b.1.1.529)-sars-cov-2-variant-of-concern

94. Mohsin M, Mahmud S. Omicron SARS-CoV-2 variant of concern: A review on its transmissibility, immune evasion, reinfection, and severity. Medicine (Baltimore). 2022 May 13;101(19):e29165.

95. Kremer C, Braeye T, Proesmans K, André E, Torneri A, Hens N. Serial Intervals for SARS-CoV-2 Omicron and Delta Variants, Belgium, November 19-December 31, 2021. Emerg Infect Dis. 2022 Aug;28(8):1699–702.

96. WHO. Weekly epidemiological update - 28 December 2021 [Internet]. 2021 Dec. Available from: https://www.who.int/docs/default-source/coronaviruse/situation-reports/20211228_weekly_epi_update_72.pdf?sfvrsn=7a3567d4_3&download=true

97. Accorsi EK, Britton A, Fleming-Dutra KE, Smith ZR, Shang N, Derado G, et al. Association Between 3 Doses of mRNA COVID-19 Vaccine and Symptomatic Infection Caused by the SARS-CoV-2 Omicron and Delta Variants. JAMA. 2022 Feb 15;327(7):639–51.

98. Andrews N, Stowe J, Kirsebom F, Toffa S, Rickeard T, Gallagher E, et al. Covid-19 Vaccine Effectiveness against the Omicron (B.1.1.529) Variant. N Engl J Med. 2022 Apr 21;386(16):1532–46.

99. Abbasi J. Vaccine Booster Dose Appears to Reduce Omicron Hospitalizations. JAMA. 2022 Apr 12;327(14):1323.

100. Barda N, Dagan N, Cohen C, Hernán MA, Lipsitch M, Kohane IS, et al. Effectiveness of a third dose of the BNT162b2 mRNA COVID-19 vaccine for preventing severe outcomes in Israel: an observational study. Lancet Lond Engl. 2021 Dec 4;398(10316):2093–100.

101. COVID-19 vaccine boosters administered per 100 people [Internet]. Our World in Data. [cited 2022 Aug 14]. Available from: https://ourworldindata.org/grapher/covid-vaccine-booster-doses-per-capita

102. Wolter N, Jassat W, Walaza S, Welch R, Moultrie H, Groome M, et al. Early assessment of the clinical severity of the SARS-CoV-2 omicron variant in South Africa: a data linkage study. Lancet Lond Engl. 2022 Jan 29;399(10323):437–46.

103. Requirement for Face Masks on Public Transportation Conveyances and at Transportation Hubs | Quarantine | CDC [Internet]. 2022.

Available from: https://www.cdc.gov/quarantine/masks/face-masks-public-transportation.html

104. Face coverings: when to wear one, exemptions and what makes a good one [Internet]. GOV.UK. Available from: https://www.gov.uk/government/publications/face-coverings-when-to-wear-one-and-how-to-make-your-own/face-coverings-when-to-wear-one-and-how-to-make-your-own

105. Coronavirus Disease 2019 [Internet]. Centers for Disease Control and Prevention. 2022. Available from: https://www.cdc.gov/media/releases/2022/s0610-COVID-19-test.html

106. Travel to England from another country during coronavirus (COVID-19) [Internet]. GOV.UK. Available from: https://www.gov.uk/guidance/travel-to-england-from-another-country-during-coronavirus-covid-19

107. Notice for China-bound Travelers (Updated December 27, 2022) [Internet]. Available from: http://my.china-embassy.gov.cn/eng/fwzc/zytz/202212/t20221228_10997401.htm

108. Hu B, Guo H, Zhou P, Shi ZL. Characteristics of SARS-CoV-2 and COVID-19. Nat Rev Microbiol. 2021 Mar;19(3):141–54.

109. Florensa D, Mateo J, Spaimoc R, Miret C, Godoy S, Solsona F, et al. Severity of COVID-19 cases in the months of predominance of the Alpha and Delta variants. Sci Rep. 2022 Sep 14;12(1):15456.

110. Sze S, Pan D, Nevill CR, Gray LJ, Martin CA, Nazareth J, et al. Ethnicity and clinical outcomes in COVID-19: A systematic review and meta-analysis. EClinicalMedicine. 2020 Dec;29:100630.

111. Lauer SA, Grantz KH, Bi Q, Jones FK, Zheng Q, Meredith HR, et al. The Incubation Period of Coronavirus Disease 2019 (COVID-19) From Publicly Reported Confirmed Cases: Estimation and Application. Ann Intern Med. 2020 Mar 10;M20-0504.

112. Wang Y, Chen R, Hu F, Lan Y, Yang Z, Zhan C, et al. Transmission, viral kinetics and clinical characteristics of the emergent SARS-CoV-2 Delta VOC in Guangzhou, China. EClinicalMedicine. 2021 Oct;40:101129.

113. Kremer C, Braeye T, Proesmans K, André E, Torneri A, Hens N. Observed serial intervals of SARS-CoV-2 for the Omicron and Delta variants in Belgium based on contact tracing data, 19 November to 31 December 2021 [Internet]. medRxiv; 2022. p. 2022.01.28.22269756. Available from: https://www.medrxiv.org/content/10.1101/2022.01.28.22269756v1

114. Xue L, Jing S, Zhang K, Milne R, Wang H. Infectivity versus fatality of SARS-CoV-2 mutations and influenza. Int J Infect Dis. 2022 Aug 1;121:195–202.

115. Li J, Huang DQ, Zou B, Yang H, Hui WZ, Rui F, et al. Epidemiology of COVID-19: A systematic review and meta-analysis of clinical characteristics, risk factors, and outcomes. J Med Virol. 2021 Mar;93(3):1449–58.

116. Nyberg T, Ferguson NM, Nash SG, Webster HH, Flaxman S, Andrews N, et al. Comparative analysis of the risks of hospitalisation and death associated with SARS-CoV-2 omicron (B.1.1.529) and delta (B.1.617.2) variants in England: a cohort study. The Lancet. 2022 Apr 2;399(10332):1303–12.

117. Borges do Nascimento IJ, O'Mathúna DP, von Groote TC, Abdulazeem HM, Weerasekara I, Marusic A, et al. Coronavirus disease (COVID-19) pandemic: an overview of systematic reviews. BMC Infect Dis. 2021 Jun 4;21(1):525.

118. Stokes EK, Zambrano LD, Anderson KN, Marder EP, Raz KM, El Burai Felix S, et al. Coronavirus Disease 2019 Case Surveillance — United States, January 22–May 30, 2020. Morb Mortal Wkly Rep. 2020 Jun 19;69(24):759–65.

119. Chen N, Zhou M, Dong X, Qu J, Gong F, Han Y, et al. Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study. The Lancet. 2020 Jan;S0140673620302117.

120. Modes ME, Directo MP, Melgar M, Johnson LR, Yang H, Chaudhary P, et al. Clinical Characteristics and Outcomes Among Adults Hospitalized with Laboratory-Confirmed SARS-CoV-2 Infection During Periods of B.1.617.2 (Delta) and B.1.1.529 (Omicron) Variant Predominance — One Hospital, California, July 15–September 23, 2021, and December 21, 2021–January 27, 2022. Morb Mortal Wkly Rep. 2022 Feb 11;71(6):217–23.

121. Rashedi R, Samieefar N, Akhlaghdoust M, Mashhadi M, Darzi P, Rezaei N. Delta Variant: The New Challenge of COVID-19 Pandemic, an Overview of Epidemiological, Clinical, and Immune Characteristics. Acta Bio Medica Atenei Parm. 2022;93(1):e2022179.

122. Cascella M, Rajnik M, Aleem A, Dulebohn SC, Napoli RD. Features, Evaluation, and Treatment of Coronavirus (COVID-19) [Internet]. StatPearls [Internet]. StatPearls Publishing; 2022. Available from: https://www.ncbi.nlm.nih.gov/books/NBK554776/

123. Park M, Cook AR, Lim JT, Sun Y, Dickens BL. A Systematic Review of COVID-19 Epidemiology Based on Current Evidence. J Clin Med. 2020 Mar 31;9(4):E967.

124. Alimohamadi Y, Taghdir M, Sepandi M. Estimate of the Basic Reproduction Number for COVID-19: A Systematic Review and Meta-analysis. J Prev Med Public Health Yebang Uihakhoe Chi. 2020 May;53(3):151–7.

125. Maier BF, Brockmann D. Effective containment explains subexponential growth in recent confirmed COVID-19 cases in China. Science. 2020 May 15;368(6492):742–6.

126. Linka K, Peirlinck M, Kuhl E. The reproduction number of COVID-19 and its correlation with public health interventions. medRxiv. 2020 Jul 7;2020.05.01.20088047.

127. Liu Y, Rocklöv J. The reproductive number of the Delta variant of SARS-CoV-2 is far higher compared to the ancestral SARS-CoV-2 virus. J Travel Med. 2021 Aug 9;28(7):taab124.

128. Ito K, Piantham C, Nishiura H. Relative instantaneous reproduction number of Omicron SARS-CoV-2 variant with respect to the Delta variant in Denmark. J Med Virol. 2022 May;94(5):2265–8.

129. Nishiura H, Ito K, Anzai A, Kobayashi T, Piantham C, Rodríguez-Morales AJ. Relative Reproduction Number of SARS-CoV-2 Omicron (B.1.1.529) Compared with Delta Variant in South Africa. J Clin Med. 2022 Jan;11(1):30.

130. Liu Y, Rocklöv J. The effective reproductive number of the Omicron variant of SARS-CoV-2 is several times relative to Delta. J Travel Med. 2022 May 31;29(3):taac037.

131. Wu J, Huang Y, Tu C, Bi C, Chen Z, Luo L, et al. Household Transmission of SARS-CoV-2, Zhuhai, China, 2020. Clin Infect Dis Off Publ Infect Dis Soc Am. 2020 May 11;ciaa557.

132. Fung HF, Martinez L, Alarid-Escudero F, Salomon JA, Studdert DM, Andrews JR, et al. The Household Secondary Attack Rate of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2): A Rapid Review. Clin Infect Dis Off Publ Infect Dis Soc Am. 2020 Oct 12;73(Suppl 2):S138–45.

133. Silverberg SL, Zhang BY, Li SNJ, Burgert C, Shulha HP, Kitchin V, et al. Child transmission of SARS-CoV-2: a systematic review and meta-analysis. BMC Pediatr. 2022 Apr 2;22(1):172.

134. Irfan O, Li J, Tang K, Wang Z, Bhutta ZA. Risk of infection and transmission of SARS-CoV-2 among children and adolescents in households, communities and educational settings: A systematic review and meta-analysis. J Glob Health. 2021;11:05013.

135. Gorny AW, Bagdasarian N, Koh AHK, Lim YC, Ong JSM, Ng BSW, et al. SARS-CoV-2 in migrant worker dormitories: Geospatial epidemiology

supporting outbreak management. Int J Infect Dis. 2021 Feb 1;103:389–94.

136. Koh D. Migrant workers and COVID-19. Occup Environ Med. 2020;77(9):634–6.

137. Plucinski MM, Wallace M, Uehara A, Kurbatova EV, Tobolowsky FA, Schneider ZD, et al. Coronavirus Disease 2019 (COVID-19) in Americans Aboard the Diamond Princess Cruise Ship. Clin Infect Dis Off Publ Infect Dis Soc Am. 2021 May 18;72(10):e448–57.

138. Kondilis E, Papamichail D, McCann S, Carruthers E, Veizis A, Orcutt M, et al. The impact of the COVID-19 pandemic on refugees and asylum seekers in Greece: A retrospective analysis of national surveillance data from 2020. EClinicalMedicine. 2021 Jul;37:100958.

139. Sabetian G, Moghadami M, Hashemizadeh Fard Haghighi L, Shahriarirad R, Fallahi MJ, Asmarian N, et al. COVID-19 infection among healthcare workers: a cross-sectional study in southwest Iran. Virol J. 2021 Mar 17;18(1):58.

140. Gholami M, Fawad I, Shadan S, Rowaiee R, Ghanem H, Hassan Khamis A, et al. COVID-19 and healthcare workers: A systematic review and meta-analysis. Int J Infect Dis IJID Off Publ Int Soc Infect Dis. 2021 Mar;104:335–46.

141. Mizumoto K, Chowell G. Transmission potential of the novel coronavirus (COVID-19) onboard the diamond Princess Cruises Ship, 2020. Infect Dis Model. 2020 Jan 1;5:264–70.

142. Azimi P, Keshavarz Z, Cedeno Laurent JG, Stephens B, Allen JG. Mechanistic transmission modeling of COVID-19 on the Diamond Princess cruise ship demonstrates the importance of aerosol transmission. Proc Natl Acad Sci. 2021 Feb 23;118(8):e2015482118.

143. Rocklöv J, Sjödin H, Wilder-Smith A. COVID-19 outbreak on the Diamond Princess cruise ship: estimating the epidemic potential and effectiveness of public health countermeasures. J Travel Med. 2020 May 18;27(3):taaa030.

144. Bootsma MCJ, Ferguson NM. The effect of public health measures on the 1918 influenza pandemic in U.S. cities. Proc Natl Acad Sci. 2007 May;104(18):7588–93.

145. Fong MW, Gao H, Wong JY, Xiao J, Shiu EYC, Ryu S, et al. Nonpharmaceutical Measures for Pandemic Influenza in Nonhealthcare Settings—Social Distancing Measures. Emerg Infect Dis. 2020 May;26(5):976–84.

146. Lee VJ, Yap J, Cook AR, Chen MI, Tay JK, Barr I, et al. Effectiveness of public health measures in mitigating pandemic influenza spread: a

prospective sero-epidemiological cohort study. J Infect Dis. 2010 Nov 1;202(9):1319–26.

147. Howard J, Huang A, Li Z, Tufekci Z, Zdimal V, van der Westhuizen HM, et al. An evidence review of face masks against COVID-19. Proc Natl Acad Sci. 2021 Jan 26;118(4):e2014564118.

148. Chu DK, Akl EA, Duda S, Solo K, Yaacoub S, Schünemann HJ, et al. Physical distancing, face masks, and eye protection to prevent person-to-person transmission of SARS-CoV-2 and COVID-19: a systematic review and meta-analysis. Lancet. 2020 Jun 27;395(10242):1973–87.

149. Rabie T, Curtis V. Handwashing and risk of respiratory infections: a quantitative systematic review. Trop Med Int Health. 2006;11(3):258–67.

150. Iezadi S, Gholipour K, Azami-Aghdash S, Ghiasi A, Rezapour A, Pourasghari H, et al. Effectiveness of non-pharmaceutical public health interventions against COVID-19: A systematic review and meta-analysis. PloS One. 2021;16(11):e0260371.

151. Lai S, Ruktanonchai NW, Zhou L, Prosper O, Luo W, Floyd JR, et al. Effect of non-pharmaceutical interventions to contain COVID-19 in China. Nature. 2020 Sep;585(7825):410–3.

152. Filonets T, Solovchuk M, Gao W, Sheu TWH. Investigation of the Efficiency of Mask Wearing, Contact Tracing, and Case Isolation during the COVID-19 Outbreak. J Clin Med. 2021 Jun 23;10(13):2761.

153. CanSino's COVID-19 vaccine candidate approved for military use in China. Reuters [Internet]. 2020 Jun 29; Available from: https://www.reuters.com/article/us-health-coronavirus-china-vaccine-idUSKBN2400DZ

154. Polack FP, Thomas SJ, Kitchin N, Absalon J, Gurtman A, Lockhart S, et al. Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine. N Engl J Med. 2020 Dec 31;383(27):2603–15.

155. Baden LR, El Sahly HM, Essink B, Kotloff K, Frey S, Novak R, et al. Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine. N Engl J Med. 2021 Feb 4;384(5):403–16.

156. Commissioner O of the. FDA Takes Key Action in Fight Against COVID-19 By Issuing Emergency Use Authorization for First COVID-19 Vaccine [Internet]. FDA. FDA; 2020. Available from: https://www.fda.gov/news-events/press-announcements/fda-takes-key-action-fight-against-covid-19-issuing-emergency-use-authorization-first-covid-19

157. Information for Healthcare Professionals on COVID-19 Vaccine Pfizer/BioNTech (Regulation 174) [Internet]. GOV.UK. Available from: https://www.gov.uk/government/publications/regulatory-approval-of-

pfizer-biontech-vaccine-for-covid-19/information-for-healthcare-professionals-on-pfizerbiontech-covid-19-vaccine

158. Commissioner O of the. FDA Takes Additional Action in Fight Against COVID-19 By Issuing Emergency Use Authorization for Second COVID-19 Vaccine [Internet]. FDA. FDA; 2020. Available from: https://www.fda.gov/news-events/press-announcements/fda-takes-additional-action-fight-against-covid-19-issuing-emergency-use-authorization-second-covid

159. WHO. COVAX -- Working for global equitable access to COVID-19 vaccines [Internet]. Available from: https://www.who.int/initiatives/act-accelerator/covax

160. More than 150 countries engaged in COVID-19 vaccine global access facility [Internet]. Available from: https://www.who.int/news/item/15-07-2020-more-than-150-countries-engaged-in-covid-19-vaccine-global-access-facility

161. Tatar M, Wilson FA. The largest vaccination campaign in history: A golden opportunity for bundling public health interventions. J Glob Health. 11:03076.

162. Shiehzadegan S, Alaghemand N, Fox M, Venketaraman V. Analysis of the Delta Variant B.1.617.2 COVID-19. Clin Pract. 2021 Oct 21;11(4):778–84.

163. Thompson J, Wattam S. Estimating the impact of interventions against COVID-19: From lockdown to vaccination. PLOS ONE. 2021 Dec 17;16(12):e0261330.

164. Rossman H, Shilo S, Meir T, Gorfine M, Shalit U, Segal E. COVID-19 dynamics after a national immunization program in Israel. Nat Med. 2021 Jun;27(6):1055–61.

165. Onyeaka H, Anumudu CK, Al-Sharify ZT, Egele-Godswill E, Mbaegbu P. COVID-19 pandemic: A review of the global lockdown and its far-reaching effects. Sci Prog. 2021 Jun;104(2):368504211019854.

166. Brauner JM, Mindermann S, Sharma M, Johnston D, Salvatier J, Gavenčiak T, et al. Inferring the effectiveness of government interventions against COVID-19. Science. 2021 Feb 19;371(6531):eabd9338.

167. Bagdasarian N, Fisher D. Heterogenous COVID-19 transmission dynamics within Singapore: a clearer picture of future national responses. BMC Med. 2020 Jun 2;18(1):164.

168. Recommended Guidance for Extended Use and Limited Reuse of N95 Filtering Facepiece Respirators in Healthcare Settings | NIOSH | CDC [Internet]. 2020. Available from:

https://www.cdc.gov/niosh/topics/hcwcontrols/recommendedguidanceextuse.html

169. Howard BE. High-Risk Aerosol-Generating Procedures in COVID-19: Respiratory Protective Equipment Considerations. Otolaryngol--Head Neck Surg Off J Am Acad Otolaryngol-Head Neck Surg. 2020 Jul;163(1):98–103.

170. M Z, Ar E, Rj T, H Z, J W. Masks or N95 Respirators During COVID-19 Pandemic-Which One Should I Wear? J Oral Maxillofac Surg Off J Am Assoc Oral Maxillofac Surg [Internet]. 2020 Dec [cited 2022 Oct 12];78(12). Available from: https://pubmed.ncbi.nlm.nih.gov/32926868/

171. CDC. Masks and Respirators [Internet]. Centers for Disease Control and Prevention. 2020. Available from: https://www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/about-face-coverings.html

172. Mizumoto K, Chowell G. Estimating Risk for Death from Coronavirus Disease, China, January–February 2020 - Volume 26, Number 6—June 2020 - Emerging Infectious Diseases journal - CDC. Available from: https://wwwnc.cdc.gov/eid/article/26/6/20-0233_article

173. Furuse Y, Oshitani H. Viruses That Can and Cannot Coexist With Humans and the Future of SARS-CoV-2. Front Microbiol [Internet]. 2020;11. Available from: https://www.frontiersin.org/articles/10.3389/fmicb.2020.583252/full

174. Erkhembayar R, Dickinson E, Badarch D, Narula I, Warburton D, Thomas GN, et al. Early policy actions and emergency response to the COVID-19 pandemic in Mongolia: experiences and challenges. Lancet Glob Health. 2020 Sep 1;8(9):e1234–41.

175. Wang Y, Teunis P. Strongly Heterogeneous Transmission of COVID-19 in Mainland China: Local and Regional Variation. Front Med [Internet]. 2020;7. Available from: https://www.frontiersin.org/articles/10.3389/fmed.2020.00329/full

176. Tu H, Tu S, Gao S, Shao A, Sheng J. Current epidemiological and clinical features of COVID-19; a global perspective from China. J Infect. 2020;81(1):1–9.

177. Wang Y, Wang Y, Chen Y, Qin Q. Unique epidemiological and clinical features of the emerging 2019 novel coronavirus pneumonia (COVID-19) implicate special control measures. J Med Virol. 2020;92(6):568–76.

178. McMichael TM, Currie DW, Clark S, Pogosjans S, Kay M, Schwartz NG, et al. Epidemiology of Covid-19 in a Long-Term Care Facility in King County, Washington. N Engl J Med. 2020 21;382(21):2005–11.

179. Fallon A, Dukelow T, Kennelly SP, O'Neill D. COVID-19 in Nursing Homes. QJM Int J Med. 2020 Apr 20;hcaa136.

180. Chew M, Koh FH, Wu J, Ngaserin S, Ng A, Ong B, et al. Clinical assessment of COVID-19 outbreak among migrant workers residing in a large dormitory in Singapore. J Hosp Infect. 2020 Sep;106(1):202–3.

181. Alahmad B, Kurdi H, Colonna K, Gasana J, Agnew J, Fox MA. COVID-19 stressors on migrant workers in Kuwait: cumulative risk considerations. BMJ Glob Health. 2020;5(7).

182. Tsai J, Wilson M. COVID-19: a potential public health problem for homeless populations. Lancet Public Health. 2020 Apr 1;5(4):e186–7.

183. Baggett TP, Keyes H, Sporn N, Gaeta JM. Prevalence of SARS-CoV-2 Infection in Residents of a Large Homeless Shelter in Boston. JAMA. 2020 Jun 2;323(21):2191–2.

184. Servies T. Characteristics of U.S. Army Beneficiary Cases of COVID-19 in Europe, 12 March 2020-17 April 2020. MSMR. 2020 Dec;27(12):9–13.

185. Hawks L, Woolhandler S, McCormick D. COVID-19 in Prisons and Jails in the United States. JAMA Intern Med. 2020 Aug 1;180(8):1041.

186. Saloner B, Parish K, Ward JA, DiLaura G, Dolovich S. COVID-19 Cases and Deaths in Federal and State Prisons. JAMA. 2020 Aug 11;324(6):602.

187. Jiménez MC, Cowger TL, Simon LE, Behn M, Cassarino N, Bassett MT. Epidemiology of COVID-19 Among Incarcerated Individuals and Staff in Massachusetts Jails and Prisons. JAMA Netw Open. 2020 Aug 21;3(8):e2018851.

188. Koplan JP, Butler-Jones D, Tsang T, Yu W. Public Health Lessons from Severe Acute Respiratory Syndrome a Decade Later. Emerg Infect Dis. 2013 Jun;19(6):861–3.

189. Lee S il. Costly Lessons From the 2015 Middle East Respiratory Syndrome Coronavirus Outbreak in Korea. J Prev Med Pub Health. 2015 Nov;48(6):274.

190. Coronavirus (COVID-19) events as they happen [Internet]. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

191. Ranney ML, Griffeth V, Jha AK. Critical Supply Shortages — The Need for Ventilators and Personal Protective Equipment during the Covid-19 Pandemic. N Engl J Med. 2020 Apr 30;382(18):e41.

192. Zhang M, Emery AR, Tannyhill RJ, Zheng H, Wang J. Masks or N95 Respirators During COVID-19 Pandemic–Which One Should I Wear? J Oral Maxillofac Surg. 2020 Dec;78(12):2114–27.

193. COVID-19: How much protection do face masks offer? - Mayo Clinic [Internet]. Available from: https://www.mayoclinic.org/diseases-conditions/coronavirus/in-depth/coronavirus-mask/art-20485449

194. Prevention USC for DC and. Interim Infection Prevention and Control Recommendations for Patients with Suspected or Confirmed Coronavirus Disease 2019 (COVID-19) in Healthcare Settings. 2020 Mar 22 [cited 2022 Oct 17]; Available from: https://paho-covid-prod.atmire.com/handle/20.500.12663/675

195. Beckman S, Materna B, Goldmacher S, Zipprich J, D'Alessandro M, Novak D, et al. Evaluation of respiratory protection programs and practices in California hospitals during the 2009-2010 H1N1 influenza pandemic. Am J Infect Control. 2013 Nov;41(11):1024–31.

196. WHO. Rational use of personal protective equipment for coronavirus disease 2019 (COVID-19) [Internet]. 2020 Feb. Available from: https://apps.who.int/iris/handle/10665/331215

197. Rodriguez-Martinez CE, Sossa-Briceño MP, Cortés JA. Decontamination and reuse of N95 filtering facemask respirators: A systematic review of the literature. Am J Infect Control. 2020 Dec;48(12):1520–32.

198. Guan W jie, Ni Z yi, Hu Y, Liang W hua, Ou C quan, He J xing, et al. Clinical Characteristics of Coronavirus Disease 2019 in China. N Engl J Med. 2020 Apr 30;382(18):1708–20.

199. Wang D, Hu B, Hu C, Zhu F, Liu X, Zhang J, et al. Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus–Infected Pneumonia in Wuhan, China. JAMA. 2020 Mar 17;323(11):1061–9.

200. WHO. Coronavirus disease (COVID-19) situation Report - 191 [Internet]. Available from: https://www.who.int/publications/m/item/situation-report---191

201. Linton NM, Kobayashi T, Yang Y, Hayashi K, Akhmetzhanov AR, Jung SM, et al. Incubation Period and Other Epidemiological Characteristics of 2019 Novel Coronavirus Infections with Right Truncation: A Statistical Analysis of Publicly Available Case Data. J Clin Med. 2020 Feb 17;9(2):538.

202. Wang W, Tang J, Wei F. Updated understanding of the outbreak of 2019 novel coronavirus (2019-nCoV) in Wuhan, China. J Med Virol. 2020 Apr;92(4):441–7.

203. 新型コロナウイルス感染症対策専門家会議 (New Coronavirus Infectious Disease Control Expert Meeting) [Internet]. Available from: https://www.mhlw.go.jp/stf/seisakunitsuite/newpage_00011.html

204. Du Z, Wang L, Cauchemez S, Xu X, Wang X, Cowling BJ, et al. Risk for Transportation of Coronavirus Disease from Wuhan to Other Cities in China. Emerg Infect Dis. 2020 May;26(5):1049–52.

205. Li Z, Yi Y, Luo X, Xiong N, Liu Y, Li S, et al. Development and clinical application of a rapid IgM-IgG combined antibody test for SARS-CoV-2 infection diagnosis. J Med Virol. 2020 Sep;92(9):1518–24.

206. Statement on the second meeting of the International Health Regulations (2005) Emergency Committee regarding the outbreak of novel coronavirus (2019-nCoV) [Internet]. Available from: https://www.who.int/news/item/30-01-2020-statement-on-the-second-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-(2019-ncov)

207. R Core Team (2019). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. [Internet]. Available from: https://www.r-project.org/

208. WHO. Rolling updates on coronavirus disease (COVID-19). [Internet]. Available from: https://www.who.int/emergencies/diseases/novel-coronavirus-2019/events-as-they-happen

209. Chen YJ, Chiang PJ, Cheng YH, Huang CW, Kao HY, Chang CK, et al. Stockpile Model of Personal Protective Equipment in Taiwan. Health Secur. 2017 Apr;15(2):170–4.

210. Hashikura M, Kizu J. Stockpile of personal protective equipment in hospital settings: preparedness for influenza pandemics. Am J Infect Control. 2009 Nov;37(9):703–7.

211. MOH | Being Prepared for a Pandemic [Internet]. Available from: https://www.moh.gov.sg/diseases-updates/being-prepared-for-a-pandemic

212. Yang X, Yu Y, Xu J, Shu H, Xia J, Liu H, et al. Clinical course and outcomes of critically ill patients with SARS-CoV-2 pneumonia in Wuhan, China: a single-centered, retrospective, observational study. Lancet Respir Med. 2020 May 1;8(5):475–81.

213. Cheung JCH, Ho LT, Cheng JV, Cham EYK, Lam KN. Staff safety during emergency airway management for COVID-19 in Hong Kong. Lancet Respir Med. 2020 Apr 1;8(4):e19.

214. Ong SWX, Tan YK, Chia PY, Lee TH, Ng OT, Wong MSY, et al. Air, Surface Environmental, and Personal Protective Equipment Contamination by Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) From a Symptomatic Patient. JAMA. 2020 Apr 28;323(16):1610–2.

215. Dong E, Du H, Gardner L. An interactive web-based dashboard to track COVID-19 in real time. Lancet Infect Dis. 2020 May 1;20(5):533–4.

216. Friesen J, Pelz PF. COVID-19 and Slums: A Pandemic Highlights Gaps in Knowledge About Urban Poverty. JMIR Public Health Surveill. 2020 Sep 4;6(3):e19578.

217. CDC. Community, Work, and School [Internet]. Centers for Disease Control and Prevention. 2020. Available from: https://www.cdc.gov/coronavirus/2019-ncov/community/health-equity/race-ethnicity.html

218. Yaya S, Yeboah H, Charles CH, Otu A, Labonte R. Ethnic and racial disparities in COVID-19-related deaths: counting the trees, hiding the forest. BMJ Glob Health. 2020 Jun 1;5(6):e002913.

219. Katikireddi SV, Lal S, Carrol ED, Niedzwiedz CL, Khunti K, Dundas R, et al. Unequal impact of the COVID-19 crisis on minority ethnic groups: a framework for understanding and addressing inequalities. J Epidemiol Community Health. 2021 Oct 1;75(10):970–4.

220. Sahasranaman A, Jensen HJ. Spread of COVID-19 in urban neighbourhoods and slums of the developing world. J R Soc Interface. 2021 Jan 27;18(174):20200599.

221. Bhadra A, Mukherjee A, Sarkar K. Impact of population density on Covid-19 infected and mortality rate in India. Model Earth Syst Environ. 2021 Mar 1;7(1):623–9.

222. Pons-Salort M, John J, Watson OJ, Brazeau NF, Verity R, Kang G, et al. Reconstructing the COVID-19 epidemic in Delhi, India: infection attack rate and reporting of deaths. medRxiv. 2021 Mar 26;2021.03.23.21254092.

223. Buss LF, Prete CA, Abrahim CMM, Mendrone A, Salomon T, de Almeida-Neto C, et al. Three-quarters attack rate of SARS-CoV-2 in the Brazilian Amazon during a largely unmitigated epidemic. Science. 2021 Jan 15;371(6526):288–92.

224. Fauzi MA, Paiman N. COVID-19 pandemic in Southeast Asia: intervention and mitigation efforts. Asian Educ Dev Stud. 2020 Jan 1;10(2):176–84.

225. Hamid H, Abid Z, Amir A, Rehman TU, Akram W, Mehboob T. Current burden on healthcare systems in low- and middle-income countries: recommendations for emergency care of COVID-19. Drugs Ther Perspect. 2020 Oct 1;36(10):466–8.

226. Shadmi E, Chen Y, Dourado I, Faran-Perach I, Furler J, Hangoma P, et al. Health equity and COVID-19: global perspectives. Int J Equity Health

[Internet]. 2020 Jun 26 [cited 2021 May 25];19. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7316580/

227. MOH | Situation Report [Internet]. Available from: https://www.moh.gov.sg/covid-19/situation-report

228. Lai_Hanwei. PMO | PM Lee Hsien Loong on the COVID-19 situation in Singapore on 3 April 2020 [Internet]. Prime Minister's Office Singapore. Lai_Hanwei; 2020. Available from: http://www.pmo.gov.sg/Newsroom/PM-Lee-Hsien-Loong-on-the-COVID19-situation-in-Singapore-on-3-April-2020

229. Dickens BL, Koo JR, Lim JT, Park M, Quaye S, Sun H, et al. Modelling lockdown and exit strategies for COVID-19 in Singapore. Lancet Reg Health West Pac. 2020 Aug 1;1:100004.

230. MOH. 31 December 2020 daily report on COVID-19 [Internet]. Available from: https://www.moh.gov.sg/docs/librariesprovider5/2019-ncov/ceg_20201231_daily_report_on_covid-19_cabinet.pdf

231. Summary Table: Labour Force [Internet]. Available from: https://stats.mom.gov.sg/Pages/Labour-Force-Summary-Table.aspx

232. Pijls BG, Jolani S, Atherley A, Derckx RT, Dijkstra JIR, Franssen GHL, et al. Demographic risk factors for COVID-19 infection, severity, ICU admission and death: a meta-analysis of 59 studies. BMJ Open. 2021 Jan 1;11(1):e044640.

233. Kim L, Garg S, O'Halloran A, Whitaker M, Pham H, Anderson EJ, et al. Risk Factors for Intensive Care Unit Admission and In-hospital Mortality Among Hospitalized Adults Identified through the US Coronavirus Disease 2019 (COVID-19)-Associated Hospitalization Surveillance Network (COVID-NET). Clin Infect Dis. 2021 May 1;72(9):e206–14.

234. COVID-19 (Temporary Measures) (Control Order) Regulations 2020 - Singapore Statutes Online [Internet]. Available from: https://sso.agc.gov.sg/SL-Supp/S254-2020/Published/20200407?DocDate=20200407

235. Visits to popular markets to be restricted based on the last digit of your IC number [Internet]. Available from: http://www.gov.sg/article/visits-to-popular-markets-to-be-restricted-based-on-the-last-digit-of-your-ic-number

236. Containing COVID-19 spread at foreign worker dormitories [Internet]. Available from: http://www.gov.sg/article/containing-covid-19-spread-at-foreign-worker-dormitories

237. Badr HS, Du H, Marshall M, Dong E, Squire MM, Gardner LM. Association between mobility patterns and COVID-19 transmission in the

USA: a mathematical modelling study. Lancet Infect Dis. 2020 Jul;S1473309920305533.

238. Lim JT, Dickens BSL, Choo ELW, Chew LZX, Koo JRH, Tam C, et al. Revealing regional disparities in the transmission potential of SARS-CoV-2 from interventions in Southeast Asia. Proc R Soc B Biol Sci. 2020 Aug 26;287(1933):20201173.

239. COVID-19 Community Mobility Report [Internet]. COVID-19 Community Mobility Report. Available from: https://www.google.com/covid19/mobility?hl=en

240. MOH | News Highlights [Internet]. Available from: https://www.moh.gov.sg/news-highlights/details/gradual-re-opening-of-travel-and-changes-to-border-measures

241. MOH | News Highlights [Internet]. Available from: https://www.moh.gov.sg/news-highlights/details/updates-on-border-and-community-measures

242. Tan IB, Tan C, Hsu LY, Dan YY, Aw A, Cook AR, et al. Prevalence and Outcomes of SARS-CoV-2 Infection Among Migrant Workers in Singapore. JAMA. 2021 Feb 9;325(6):584.

243. Wood SN. Generalized additive models: an introduction with R. Second edition. Boca Raton: CRC Press/Taylor & Francis Group; 2017. 476 p. (Chapman & Hall/CRC texts in statistical science).

244. Bagdasarian N, Fisher D. Heterogenous COVID-19 transmission dynamics within Singapore: a clearer picture of future national responses. BMC Med [Internet]. 2020 Jun 2;18. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7268580/

245. Rocklöv J, Sjödin H. High population densities catalyse the spread of COVID-19. J Travel Med. 2020 18;27(3).

246. Moein S, Nickaeen N, Roointan A, Borhani N, Heidary Z, Javanmard SH, et al. Inefficiency of SIR models in forecasting COVID-19 epidemic: a case study of Isfahan. Sci Rep. 2021 Feb 25;11(1):4725.

247. Puglisi LB, Malloy GSP, Harvey TD, Brandeau ML, Wang EA. Estimation of COVID-19 basic reproduction ratio in a large urban jail in the United States. Ann Epidemiol. 2021 Jan;53:103–5.

248. MOH | News Highlights [Internet]. Available from: https://www.moh.gov.sg/news-highlights/details/measures-to-contain-the-covid-19-outbreak-in-migrant-worker-dormitories

249. Menkir TF, Chin T, Hay JA, Surface ED, De Salazar PM, Buckee CO, et al. Estimating internationally imported cases during the early COVID-19 pandemic. Nat Commun. 2021 Jan 12;12(1):311.

250. Russell TW, Wu JT, Clifford S, Edmunds WJ, Kucharski AJ, Jit M. Effect of internationally imported cases on internal spread of COVID-19: a mathematical modelling study. Lancet Public Health. 2021 Jan 1;6(1):e12–20.

251. Chew MH, Koh FH, Wu JT, Ngaserin S, Ng A, Ong BC, et al. Clinical assessment of COVID-19 outbreak among migrant workers residing in a large dormitory in Singapore. J Hosp Infect. 2020 Sep 1;106(1):202–3.

252. Construction sector: Work Permit requirements [Internet]. Ministry of Manpower Singapore. Available from: https://www.mom.gov.sg/passes-and-permits/work-permit-for-foreign-worker/sector-specific-rules/construction-sector-requirements

253. Baas S, Dijkstra S, Braaksma A, van Rooij P, Snijders FJ, Tiemessen L, et al. Real-time forecasting of COVID-19 bed occupancy in wards and Intensive Care Units. Health Care Manag Sci. 2021;24(2):402–19.

254. Yamana T, Pei S, Kandula S, Shaman J. Projection of COVID-19 Cases and Deaths in the US as Individual States Re-open May 4, 2020. medRxiv. 2020 May 13;2020.05.04.20090670.

255. MOH | News Highlights [Internet]. Available from: https://www.moh.gov.sg/news-highlights/details/reply-to-media-queries-on-the-number-of-imported-cases-who-were-short-term-visitors

256. MOH | News Highlights [Internet]. Available from: https://www.moh.gov.sg/news-highlights/details/moh-reply-in-response-to-media-queries-on-reports-of-new-variant-in-singapore

257. Comas-Herrera A, Zalakaín J, Litwin C, Hsu AT, Lemmon E, Henderson D, et al. Mortality associated with COVID-19 outbreaks in care homes: early international evidence.

258. Franco-Paredes C, Jankousky K, Schultz J, Bernfeld J, Cullen K, Quan NG, et al. COVID-19 in jails and prisons: A neglected infection in a marginalized population. PLoS Negl Trop Dis. 2020 Jun 22;14(6):e0008409.

259. Goggin G. COVID-19 apps in Singapore and Australia: reimagining healthy nations with digital technology. Media Int Aust. 2020 Nov;177(1):61–75.

260. Health AGD of. COVIDSafe app [Internet]. Australian Government Department of Health. Australian Government Department of Health; 2020. Available from: https://www.health.gov.au/resources/apps-and-tools/covidsafe-app

261. Liang F. COVID-19 and Health Code: How Digital Platforms Tackle the Pandemic in China. Soc Media Soc. 2020 Jul 1;6(3):2056305120947657.

262. Surio P, McLean JE, Jain A, Chughtai Z, Ruebush E, Lane JT, et al. Digital Tools Adopted by Public Health Agencies to Support COVID-19 Case Investigation and Contact Tracing, United States, 2020-2021. Public Health Rep. 2022 Aug 30;137(2 Suppl):18S-22S.

263. Touzani R, Schultz E, Holmes SM, Vandentorren S, Arwidson P, Guillemin F, et al. Early Acceptability of a Mobile App for Contact Tracing During the COVID-19 Pandemic in France: National Web-Based Survey. JMIR MHealth UHealth. 2021 Jul 19;9(7):e27768.

264. Hannemann N, Götz NA, Schmidt L, Hübner U, Babitsch B. Perception of a Tracing App During the Covid-19 Pandemic in Germany. Stud Health Technol Inform. 2021 May 27;281:816–7.

265. O'Callaghan ME, Buckley J, Fitzgerald B, Johnson K, Laffey J, McNicholas B, et al. A national survey of attitudes to COVID-19 digital contact tracing in the Republic of Ireland. Ir J Med Sci. 2021 Aug;190(3):863–87.

266. Mazza* C, Girardi* D, Gentile L, Gaeta M, Signorelli C, Odone A. Public health effectiveness of digital contact tracing in the COVID-19 pandemic: A systematic review of available data. Acta Bio Medica Atenei Parm. 2021;92(Suppl 6):e2021439.

267. Bente BE, van 't Klooster JWJR, Schreijer MA, Berkemeier L, van Gend JE, Slijkhuis PJH, et al. The Dutch COVID-19 Contact Tracing App (the CoronaMelder): Usability Study. JMIR Form Res. 2021 Mar 26;5(3):e27882.

268. Salathé M, Althaus CL, Anderegg N, Antonioli D, Ballouz T, Bugnion E, et al. Early evidence of effectiveness of digital contact tracing for SARS-CoV-2 in Switzerland. Swiss Med Wkly [Internet]. 2020 Dec 16 [cited 2022 Nov 3];(51). Available from: https://smw.ch/article/doi/smw.2020.20457

269. Wymant C, Ferretti L, Tsallis D, Charalambides M, Abeler-Dörner L, Bonsall D, et al. The epidemiological impact of the NHS COVID-19 app. Nature. 2021 Jun;594(7863):408–12.

270. Cencetti G, Santin G, Longa A, Pigani E, Barrat A, Cattuto C, et al. Digital proximity tracing on empirical contact networks for pandemic control. Nat Commun. 2021 Mar 12;12(1):1655.

271. Wang J, Yang W, Pan L, Ji JS, Shen J, Zhao K, et al. Prevention and control of COVID-19 in nursing homes, orphanages, and prisons. Environ Pollut. 2020 Nov 1;266:115161.

272. Garibaldi PMM, Ferreira NN, Moraes GR, Moura JC, Espósito DLA, Volpe GJ, et al. Efficacy of COVID-19 outbreak management in a skilled nursing facility based on serial testing for early detection and control. Braz J Infect Dis. 2021 Mar 1;25(2):101570.

273. Bueno de Mesquita PJ, Delp WW, Chan WR, Bahnfleth WP, Singer BC. Control of airborne infectious disease in buildings: Evidence and research priorities. Indoor Air. 2022;32(1):e12965.

274. McCarter YS. Infectious Disease Outbreaks on Cruise Ships. Clin Microbiol Newsl. 2009 Nov 1;31(21):161–8.

275. Cebrian M. The past, present and future of digital contact tracing. Nat Electron. 2021 Jan;4(1):2–4.

276. Measures to contain the COVID-19 outbreak in migrant worker dormitories [Internet]. Ministry of Manpower Singapore. Available from: https://www.mom.gov.sg/newsroom/press-releases/2020/1214-measures-to-contain-the-covid-19-outbreak-in-migrant-worker-dormitories

277. Joint MOM-BCA-EDB Press Release on Enabling Targeted Quarantining through Contact-Tracing Devices for more than 450,000 Workers [Internet]. Ministry of Manpower Singapore. Available from: https://www.mom.gov.sg/newsroom/press-releases/2020/1016-enabling-targeted-quarantining-through-contact-tracing-devices

278. Foreign workforce numbers [Internet]. Ministry of Manpower Singapore. Available from: https://www.mom.gov.sg/documents-and-publications/foreign-workforce-numbers

279. How does MOH determine if someone is a close contact of a confirmed case? - AskGov [Internet]. Available from: https://ask.gov.sg/questions/473

280. Measures to contain the COVID-19 outbreak in migrant worker dormitories [Internet]. Ministry of Manpower Singapore. Available from: https://www.mom.gov.sg/newsroom/press-releases/2020/1214-measures-to-contain-the-covid-19-outbreak-in-migrant-worker-dormitories

281. Lyubchich V, Gel YR, El-Shaarawi A. On detecting non-monotonic trends in environmental time series: a fusion of local regression and bootstrap. Environmetrics. 2013;24(4):209–26.

282. Joffe AR. COVID-19: Rethinking the Lockdown Groupthink. Front Public Health. 2021;9:625778.

283. Ganesan B, Al-Jumaily A, Fong KNK, Prasad P, Meena SK, Tong RKY. Impact of Coronavirus Disease 2019 (COVID-19) Outbreak Quarantine, Isolation, and Lockdown Policies on Mental Health and Suicide. Front Psychiatry [Internet]. 2021;12. Available from: https://www.frontiersin.org/article/10.3389/fpsyt.2021.565190

284. Gloster AT, Lamnisos D, Lubenko J, Presti G, Squatrito V, Constantinou M, et al. Impact of COVID-19 pandemic on mental health: An international study. PloS One. 2020;15(12):e0244809.

285. Altieri M, Santangelo G. The Psychological Impact of COVID-19 Pandemic and Lockdown on Caregivers of People With Dementia. Am J Geriatr Psychiatry Off J Am Assoc Geriatr Psychiatry. 2021 Jan;29(1):27–34.

286. Yuan Z, Xiao Y, Dai Z, Huang J, Zhang Z, Chen Y. Modelling the effects of Wuhan's lockdown during COVID-19, China. Bull World Health Organ. 2020 Jul 1;98(7):484–94.

287. Alfano V, Ercolano S. The Efficacy of Lockdown Against COVID-19: A Cross-Country Panel Analysis. Appl Health Econ Health Policy. 2020 Jun 3;1–9.

288. Koo JR, Cook AR, Lim JT, Tan KW, Dickens BL. Modelling the Impact of Mass Testing to Transition from Pandemic Mitigation to Endemic COVID-19. Viruses. 2022 May;14(5):967.

289. CDC. Health Departments [Internet]. Centers for Disease Control and Prevention. 2020. Available from: https://www.cdc.gov/coronavirus/2019-ncov/php/contact-tracing/contact-tracing-plan/digital-contact-tracing-tools.html

290. Ng OT, Marimuthu K, Koh V, Pang J, Linn KZ, Sun J, et al. SARS-CoV-2 seroprevalence and transmission risk factors among high-risk close contacts: a retrospective cohort study. Lancet Infect Dis. 2021 Mar 1;21(3):333–43.

291. Emanuel EJ, Persad G, Upshur R, Thome B, Parker M, Glickman A, et al. Fair Allocation of Scarce Medical Resources in the Time of Covid-19. N Engl J Med. 2020 May 21;382(21):2049–55.

292. Livingston E, Desai A, Berkwits M. Sourcing Personal Protective Equipment During the COVID-19 Pandemic. JAMA. 2020 May 19;323(19):1912–4.

293. 30 September 2020 Daily Report on COVID-19 [Internet]. Available from: https://www.moh.gov.sg/docs/librariesprovider5/local-situation-report/situation-report---30-sep-2020.pdf

v

Appendix 1. Migrant worker contact survey (English version)

Migrant worker contact survey

This survey is a research study by NUS Saw Swee Hock School of Public Health and Temasek Foundation. We want to know if the safety measures to stop COVID-19 are working, and if they can be changed to be lighter. **This survey is voluntary—you do not have to do it.** The information you provide will be kept confidential. It will not be shared with your company or dormitory. If you take part we ask for two things:

1. Anonymous answers to the six questions on the other side of this paper.
2. To extract anonymous data from your BluePass.

The BluePass will tell us how many people were near you in the last few weeks. We will be able to know what time of the day this was. We will not know who those people are. We cannot link the data from your BluePass to you or the people you met.

If you participate, it will take a few minutes to extract the data from your BluePass and do the questions. Thank you!

Principal investigator: Dr Alex Cook.

Contact email: ephcar@nus.edu.sg

NUS Institutional Review Board

Contact email: irb@nus.edu.sg

Telephone: +65 6516 4311 (during office hours)

Q1: How old are you?

24 or younger	25–29	30–34	35–39	40–44	45 or older

Q2: Where do you come from?

Bangladesh	China	India	Myanmar	Other country

Q3: Which unit do you stay in?

Block	Unit

Q4: How many other men live in your unit?

Q5: Have you tested positive for COVID-19?

Yes	No	I don't know

If Yes:

Q6: Do you feel you have fully recovered from COVID-19?

If you still feel tired or have any other symptoms, please answer No.

Yes	No

Thank you!

For research team

QR												

Q completed	DE completed	QR captured	Drink received